MTH102

Linear Algebra

2025–2026–I Semester

17 October 2025

Preface

This discusses the content covered in the class, along with some examples, exercises, and problem sets. You are encouraged to go through it. Solving the exercises would greatly benefit in understanding the content of the course. The exercises, remarks, marked with an asterisk (*), are supplementary. Further supplementary material has been put in the appendix. The typos/inaccuracies may be reported to jpsaha@iiserb.ac.in.

Timetable

The classes are scheduled on

- Mondays,
- Wednesdays,
- Fridays

from **11:00am to 11:55am** in L5, LHC. The tutorial is scheduled on

- Thursdays

from **5:30pm to 6:30pm** in L5, LHC.

Office hours

The office hours will be held in **Room no. 108, AB1** on

- Mondays, during **5:00pm to 6:00pm**,
- Wednesdays, during **5:00pm to 6:00pm**,
- Fridays, during **5:00pm to 6:00pm**,
- Saturdays, during **10:00am to 11:00am**.

The grading policy is as follows.

Components	Weightage
Quiz	20%
Mid-semester examination	35%
End-semester examination	45%

Calendar

- The academic calendar is available at https://www.iiserb.ac.in/doaa/schedule.
- **Quiz 1** is scheduled on **Saturday, 11th October, 2025** from **3:00pm to 4:00pm**, and will be based on the content covered from **Monday, 1st September, 2025** until **Saturday, 4th October, 2025**.
- **Mid-semester examination** will be based on the content covered from **Monday, 1st September, 2025** until **Saturday, 18th October, 2025**. It will be held as per the **Mid-Semester Examination Schedule**, available at https://acad.iiserb.ac.in/pdf/Mid_Sem_Schedule.pdf.
- **Quiz 2** is scheduled on **Saturday, 22nd November, 2025** from **3:00pm to 4:00pm**, and will be based on the content covered from **Monday, 1st September, 2025** until **Saturday, 15th November, 2025**.

The announcements related to the course will be made on the following Google classroom page.

https://classroom.google.com/u/1/c/ODAwOTg3NDg2Mjg0

The notes can be accessed at the link below (until all the students join the Google classroom).

https://jpsaha.github.io/home/blog/2025/mth102/

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Introduction

Chapter 1. Sets

Definition

A **set** is a well-defined collection of objects. The objects of a set are called its **elements** or **members**.

Examples

- The set of vowels a, e, i, o, u.
- The set consisting of $2, 4, 6, 8, 10$.
- The set consisting of $3, 15, 35, 63, 99$.

Here are further examples.

Examples

- The set of positive integers divisible by 3.
- The set of prime numbers less than 100.
- The set of positive integers which can be expressed as the product of two distinct primes.
- The set of positive integers which can be expressed as the product of two or more distinct primes.
- The set of positive integers which are smaller than 100 and share no common factor with 100.

Remark

Note that the first few sets are described by writing down all its elements, whereas in the latter examples, the sets are described by the rules or properties which determine whether a particular object is an element or not.

Notation

A set is usually denoted by an uppercase letter, for example,

$$A, B, C, X, Y, Z,$$

whereas the lowercase letters are used to denote its elements. For instance, one may use

- a to denote an element of A,
- b to denote an element of B,
- c to denote an element of C,
- x to denote an element of X,
- y to denote an element of Y,
- z to denote an element of Z.

Several special sets are denoted by certain standard symbols. Here are some such examples.

$\mathbb{N}$	the set of positive integers
$\mathbb{Z}$	the set of integers

If x is an element of a set X, then one says that **x belongs to X**, and writes

$$x \in X.$$

Note that 2 belongs to $\mathbb{N}$, but $\frac{1}{2}$ does not belong to $\mathbb{N}$. One writes

$$2 \in \mathbb{N}, \frac{1}{2} \notin \mathbb{N}.$$

Notation

There are two ways to write down a set. For example, consider the set of vowels, which is written as

$$V = \{\text{a, e, i, o, u}\}.$$

Note that the elements are seperated by commas and enclosed in braces $\{\}$.

Another way to write down a set is by stating the *rules* or *properties* which determine whether a particular object is an element or not. For example, the set of even positive integers is written as

$$E = \{x \in \mathbb{Z} : x \text{ is even}, x > 0\}.$$

This reads

"E is the set of elements x in $\mathbb{Z}$ such that x is even and $x > 0$".

Example

Note that the set $A = \{3, 15, 35, 63, 99\}$ is equal to

$$\{x \in \mathbb{Z} : x \text{ is equal to } n^2 - 1 \text{ for some } n \in \{2, 4, 6, 8, 10\}\}.$$

Since $1, 2$ are the roots of the polynomial $x^2 - 3x + 2$, it follows that

$$\{x \in \mathbb{N} : x^2 - 3x + 2 = 0\} = \{1, 2\}.$$

Also note that

$$\mathbb{N} = \{1, 2, 3, \ldots\},$$
$$\mathbb{Z} = \{0, 1, -1, 2, -2, 3, -3, \ldots\}.$$

One uses the symbol $:=$ to indicate that the symbol on the left is being defined by the symbol on the right.

$$\mathbb{Q} := \left\{\frac{m}{n} : m, n \in \mathbb{Z} \text{ and } n \neq 0\right\}.$$

It is called the *set of rational numbers*.

If the number of elements of a set is finite, then it is called a **finite set**. If a set is not finite, it is called an **infinite set**. A set containing exactly one element is called a **singleton set**.

Example

The sets

$$\{\text{a, e, i, o, u}\}, \{3, 15, 35, 63, 99\}, \{x \in \mathbb{N} : x^2 - 3x + 2 = 0\}$$

are finite. The sets $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are infinite.

Exercise 1.1. Determine the elements of the following sets.

- $\{x \in \mathbb{N} : x^2 - 1 = 0\}$.
- $\{x \in \mathbb{Z} : x^2 - 1 = 0\}$.

Solution. Note that 1 is the only element of $\mathbb{N}$ satisfying the equation $x^2 - 1 = 0$. This gives

$$\boxed{\{x \in \mathbb{N} : x^2 - 1 = 0\} = \{1\}}.$$

Since $1, -1$ are precisely all the elements of $\mathbb{Z}$ satisfying $x^2 - 1 = 0$, it follows that

$$\boxed{\{x \in \mathbb{Z} : x^2 - 1 = 0\} = \{1, -1\}}.$$

□

§1.1 Basic terminologies

If A, B are sets, and every element of A also belongs to B, then we say that A is a **subset** of B, and write

$$A \subseteq B.$$

If A is not a subset of B, one writes

$$A \not\subseteq B.$$

Example

- Note that every set is a subset of itself.
- Note that $\mathbb{N}$ is a subset of $\mathbb{Z}$, and $\mathbb{Z}$ is a subset of $\mathbb{Q}$. One writes
$$\mathbb{N} \subseteq \mathbb{Z}, \quad \mathbb{Z} \subseteq \mathbb{Q}.$$
- Also note that
$$\mathbb{Z} \not\subseteq \mathbb{N}, \mathbb{Q} \not\subseteq \mathbb{Z}, \mathbb{Q} \not\subseteq \mathbb{N}.$$

If A is a subset of B and A is not equal to B, then A is said to be a *proper subset of* B, and one writes

$$A \subsetneq B.$$

Note that

$$\mathbb{N} \subsetneq \mathbb{Z}, \mathbb{Z} \subsetneq \mathbb{Q}.$$

Exercise 1.2. What does it mean to say that A is not a subset of B?

Solution. Note that A is a subset of B is equivalent to the statement that each element of A lies in B. Hence, A is not a subset of B is equivalent to saying that some element of A does not belong to B. □

Exercise 1.3. Show that if A is a subset of B and B is a subset of C, then A is a subset of C.

Solution. Let x be an element of A. Since A is a subset of B, it follows that x belongs to B. Using that B is a subset of C, we obtain that x belongs to C. Hence, the element x lies in C. This shows that any element x of A belongs to C. This proves that A is a subset of C. □

Exercise 1.4. Show that no element n of $\mathbb{N}$ satisfies $n^4 - 5n^2 + 6 = 0$.

Tip

Try to show that no natural number n satisfies any of the equations

$$n^2 = 2, n^2 = 3.$$

Solution. Note that the polynomial $x^4 - 5x^2 + 6$ factorizes as

$$x^4 - 5x^2 + 6 = (x^2 - 2)(x^2 - 3).$$

Let n be a natural number. If $n = 1$, then $n^2 \neq 2$ and $n^2 \neq 3$ hold. If $n \geq 2$, then $n^2 \geq 4$, and hence, $n^2 \neq 2$ and $n^2 \neq 3$ hold. This shows that no natural number n satisfies

$$(n^2-2)(n^2-3)=0,$$

that is, the equation

$$n^4-5n^2+6=0.$$

□

Consider the set

$$A=\{n\in\mathbb{N}:n^4-5n^2+6=0\}.$$

Note that the set A has no elements. Any such set is called the **empty set** or **null set**, and is denoted by the symbol

$$\emptyset.$$

Two sets, A and B, are said to be **equal**[1] if any element of A belongs to B, and any element of B also belongs to A, or equivalently,

$$A\subseteq B \text{ and } B\subseteq A$$

holds. If two sets A, B are not equal, one writes

$$A\neq B.$$

Exercise 1.5. What does it mean to say that two sets A, B are not equal?

Tip

Does Exercise 1.2 help?

Solution. Note that two sets A, B are equal is equivalent to saying that A is a subset of B, and B is a subset of C. Hence, the statement $A\neq B$ is equivalent to saying that A is **not** a subset of B, **or** B is **not** a subset of A, which is equivalent to the statement that some element of A does not belong to B, **or** some element of B does not belong to A. □

§1.2 Operations on sets

Exercise 1.6. Consider the sets

$$\{1,2,3,4\},\{3,4,5,6\}.$$

Determine the elements common to these sets.

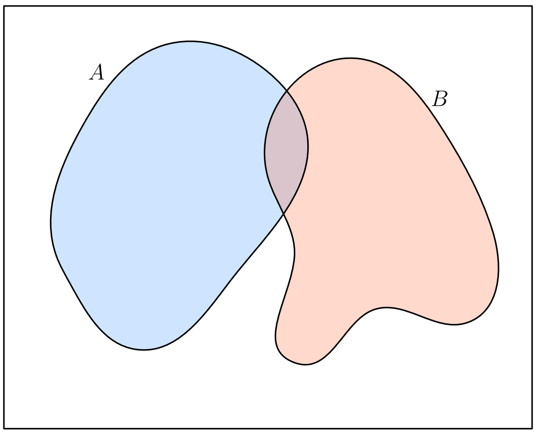

Figure 1: Two sets A and B

[1] *It* may appear obvious! The advantage of putting forth definitions is to set up notations and conventions, so that these are not left to interpretations!

Definition

If A, B are sets, their **union** is denoted by $A \cup B$, which is defined as

$$A \cup B := \{x : x \in A \text{ or } x \in B\}.$$

See Figure 2.

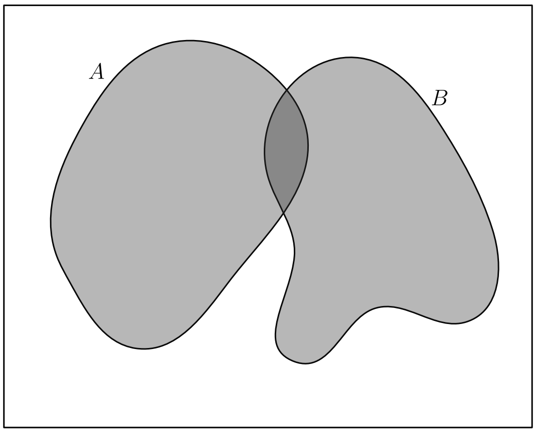

Figure 2: The union of A and B

Exercise 1.7. Determine the union of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

Usage of "or" in the inclusive sense

The word "or" is used in the inclusive sense, allowing two or more of the conditions to be satisfied. In common terminology, this inclusive sense is denoted by "and/or".

Exercise 1.8. If A, B are sets, show that

$$A \subseteq A \cup B, B \subseteq A \cup B.$$

Solution. Let x be an element of A. Then note that x lies $A \cup B$. This shows that A is a subset of $A \cup B$. Similarly, it follows that B is a subset of $A \cup B$. □

Exercise 1.9. If A, B are subsets of a set C, then $A \cup B$ is a subset of C.

Solution. Let x be an element of $A \cup B$. It follows that x belongs to A or x belongs to B. If x belongs to A, then using A is a subset of C, we obtain that x lies in C. Further, if x belongs to B, then using that B is a subset of C, we obtain that x lies in C. Combining the above cases, it follows that x lies in C. Consequently, any element of $A \cup B$ is an element of C, or equivalently, $A \cup B$ is a subset of C. □

Exercise 1.10. If A is a set, show that

$$A \cup A = A.$$

Solution. Note that any element x of $A \cup A$ belongs to A or belongs to A, and hence it lies in A. This shows that $A \cup A$ is a subset of A. Further, for any element y of A, it belongs to $A \cup A$. Hence, A is a subset of $A \cup A$. This proves that

$$\boxed{A \cup A = A}.$$

□

Exercise 1.11 (Commutative property). If A, B are sets, show that

$$A \cup B = B \cup A.$$

Solution. Let P, Q be sets, and let x be an element of $P \cup Q$. It follows that x belongs to P or x belongs to Q. This shows that x belongs to Q or x belongs to P. This implies that x is an element of $Q \cup P$. Hence, $P \cup Q$ is a subset of $Q \cup P$ for any two sets P, Q.

Consequently, the set $A \cup B$ is a subset of $B \cup A$, and $B \cup A$ is a subset of $A \cup B$. This shows that

$$\boxed{A \cup B = B \cup A}.$$

□

Definition

If A, B, C are sets, their **union** is denoted by $A \cup B \cup C$, which is defined as

$$A \cup B \cup C := \{x : x \in A \text{ or } x \in B \text{ or } x \in C\}.$$

Exercise 1.12 (Associative property). If A, B, C are sets, show that

$$A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C).$$

Use Exercise 1.11 and Exercise 1.12, to show that for any three sets A, B, C, the following sets are equal

$$A \cup B \cup C, A \cup C \cup B, B \cup A \cup C, B \cup C \cup A, C \cup A \cup B, C \cup B \cup A.$$

Definition

If A, B are sets, their **intersection** is denoted by $A \cap B$, which is defined as

$$A \cap B := \{x : x \in A \text{ and } x \in B\}.$$

See Figure 3.

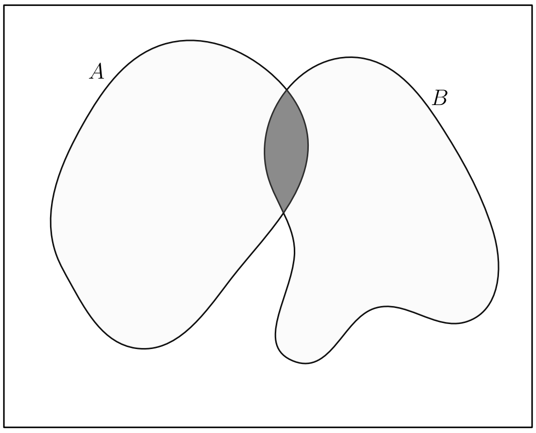

Figure 3: The intersection of A and B

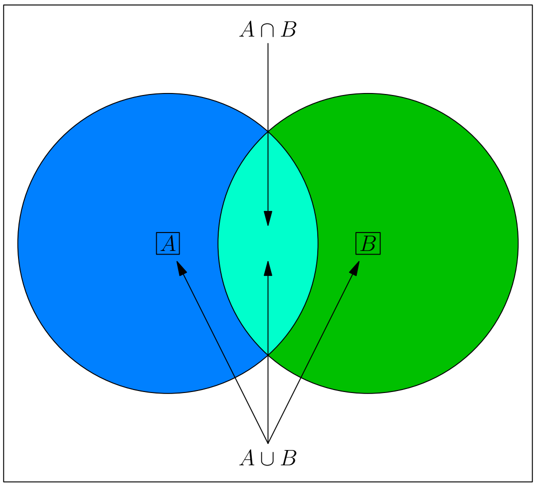

Figure 4: The union and intersection of A and B

Exercise 1.13. Determine the intersection of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

Exercise 1.14. If A, B are sets, show that

$$A \cap B \subseteq A, A \cap B \subseteq B.$$

Solution. For any element x of $A \cap B$, it belongs to A and it belongs to B. Hence, $A \cap B$ is a subset of A, and $A \cap B$ is also a subset of B. $\square$

Exercise 1.15. If A, B, C are sets satisfying

$$C \subseteq A, C \subseteq B,$$

then show that

$$C \subseteq A \cap B$$

holds.

Exercise 1.16. Identify the integers among

$$1, 2, 3, 4, \ldots, 20$$

which are divisible by at least one of 2 and 5.

Solution. In the following, the integers among $1, 2, \ldots, 20$ divisible by 2 are marked.

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10,$$

$$11, 12, 13, 14, 15, 16, 17, 18, 19, 20.$$

Note that there 10 integers among $1, 2, \ldots, 20$, which are divisible by 2. The integers among $1, 2, \ldots, 20$ divisible by 5 are marked below.

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10,$$

$$11, 12, 13, 14, 15, 16, 17, 18, 19, 20.$$

Note that there are 4 integers among $1, 2, \ldots, 20$, which are divisible by 5. $\square$

Remark

Note that some integers are counted/marked twice. This is an instance of *double counting*, which refers to the counting of certain element twice. In the above, the integers 10 and 20 are "counted twice". Thus, the number of integers among $1, 2, \ldots, 20$, divisible by 2 or 5, is equal to $10 + 4 - 2 = \boxed{12}$.

Definition

Two sets A, B are said to be **disjoint** if
$$A \cap B = \emptyset.$$

Lemma 1.17 (Inclusion-exclusion principle). *Let X be a set, and A, B be finite subsets of X. Then*
$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Proof. Note that the set $A \cup B$ is equal to the union of the disjoint sets A and $B \setminus A$. This shows that
$$|A \cup B| = |A| + |B \setminus A|.$$
Also note that the set B is equal to the union of the disjoint subsets $A \cap B$ and $B \setminus A$. This gives
$$|B| = |A \cap B| + |B \setminus A|.$$
It follows that
$$\boxed{|A \cup B| = |A| + |B| - |A \cap B|}\,.$$
□

Exercise 1.18 (Inclusion-exclusion principle). Using Lemma 1.17 or otherwise, show that for finite subsets A, B, C of a set X,
$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.$$

Solution. Note that
$$\begin{aligned}
&|A \cup B \cup C| \\
&= |(A \cup B) \cup C| \\
&= |A \cup B| + |C| - |(A \cup B) \cap C| && \text{(by Lemma 1.17)} \\
&= |A \cup B| + |C| - |(A \cap C) \cup (B \cap C)| && \text{(by Fact 1.23)} \\
&= |A \cup B| + |C| - (|A \cap C| + |B \cap C| - |(A \cap C) \cap (B \cap C)| && \text{(by Lemma 1.17)} \\
&= |A \cup B| + |C| - |A \cap C| - |B \cap C| + |(A \cap C) \cap (B \cap C)| \\
&= |A \cup B| + |C| - |A \cap C| - |B \cap C| + |A \cap C \cap B \cap C| \\
&= |A \cup B| + |C| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \\
&= |A| + |B| - |A \cap B| + |C| - |A \cap C| - |B \cap C| + |A \cap C \cap B \cap C| && \text{(by Lemma 1.17)} \\
&= |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.
\end{aligned}$$
□

Exercise 1.19. Determine the number of integers among
$$1, 2, 3, 4, \ldots, 100,$$
which are divisible by at least one of $6, 10, 15$.

Solution. Let X denote the set

$$\{1, 2, 3, ..., 100\}.$$

Let A (resp. B, C) denote the set of elements X which are divisible by 6 (resp. 10, 15). Note that

$$|A| = 16, |B| = 10, |C| = 6$$

hold. Observe that $A \cap B$ consists of the elements of X which are divisible by 6 and by 10, that is, divisible by their least common multiple, which is equal to 30. Similarly, $B \cap C$ consists of the elements of X which are divisible by the least common multiple of 10, 15, which is equal to 30. Further, it follows that $C \cap A$ also consists of the multiples of 30 lying between 1 and 100. Moreover, the set $A \cap B \cap C$ consists of the integers between 1 and 100, which are divisible by the least common multiple of 6, 10, 15, that is, the integer 30. By the inclusion-exclusion principle (Exercise 1.18), we obtain

$$\begin{aligned}
|A \cup B \cup C| &= |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C| \\
&= |A| + |B| + |C| - 3|A \cap B| + |A \cap B \cap C| \\
&= |A| + |B| + |C| - 2|A \cap B| \\
&= 16 + 10 + 6 - 2 \times 3 \\
&= 26.
\end{aligned}$$

Note that the set $A \cup B \cup C$ consists of the integers between 1 and 100 which are divisible by at least one of 6, 10, 15. This shows that there are precisely $\boxed{26}$ such integers. □

Exercise 1.20. If A is a set, show that

$$A \cap A = A.$$

Tip

Does Exercise 1.10 help? If not, what about its solution?

Exercise 1.21 (Commutative property). If A, B are sets, show that

$$A \cap B = B \cap A.$$

Tip

Does Exercise 1.11 help? If not, what about its solution?

Definition

If A, B, C are sets, their **intersection** is denoted by $A \cap B \cap C$, which is defined as

$$A \cap B \cap C := \{x : x \in A \text{ and } x \in B \text{ and } x \in C\}.$$

Exercise 1.22 (Associative property). If A, B, C are sets, show that

$$A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C).$$

Tip

Does Exercise 1.12 help? If not, what about its solution?

Fact 1.23 (Distributive property). If A, B, C are sets, show that

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

A proof of the above is provided in Chapter 6.

Remark

All sets under consideration in a given context, are assumed to be contained in a 'large' fixed set, called the *universal set*. For example, while studying the positive integers (for instance, some sets consisting of the positive integers), the set $\mathbb{N}$ can be taken as the universal set. While studying the integers, the set $\mathbb{Z}$ can be taken as the universal set.

Definition

Let A, B be subsets of a set X. The **complement of A in B** is denoted by $B \setminus A$, and is defined by

$$B \setminus A := \{x \in X : x \in B, x \notin A\}$$

It is also called the **difference of B and A**.

The complement of A in X, is called the *complement of* A, and is denoted by A^c, and is defined to be

$$A^c := X \setminus A.$$

Exercise 1.24. Show that

$$A^c = \{x \in X : x \notin A\},$$

where X denotes the underlying universal set.

Solution. Since $X \setminus A$ is equal to $\{x \in X : x \notin A\}$, it follows that

$$\boxed{A^c = \{x \in X : x \notin A\}}.$$

□

Exercise 1.25. Determine the complement of $\{1, 2, 3, 4\}$ in $\{1, 3, 5, 7\}$, and the complement of $\{1, 3, 5, 7\}$ in $\{1, 2, 3, 4\}$.

Exercise 1.26. If A, B are subsets of a set X, show that

$$A^c \cap B = B \setminus A.$$

Solution. Note that

$$\begin{aligned} A^c \cap B &= (X \setminus A) \cap B \\ &= \{x \in X : x \in X \setminus A \text{ and } x \in B\} \\ &= \{x \in X : x \notin A \text{ and } x \in B\} \\ &= \{x \in X : x \in B \text{ and } x \notin A\} \\ &= B \setminus A. \end{aligned}$$

□

Exercise 1.27. If A is a subset of a set X, then show that

$$A \cup A^c = X, A \cap A^c = \emptyset, (A^c)^c = A.$$

Fact 1.28. If A, B are subsets of a set X, then show that
$$(A \cup B)^c = A^c \cap B^c, (A \cap B)^c = A^c \cup B^c.$$

A proof of the above is provided in Chapter 6.

Exercise 1.29. Let A, B be subsets of a set X. Show that the following statements are equivalent.
$$A \subseteq B,$$
$$A \cap B = A,$$
$$A \cup B = B,$$
$$B^c \subseteq A^c.$$

Fact 1.30. If A, B, C are sets, show that
1. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$,
2. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.

A proof of the above is provided in Chapter 6.

Exercise 1.31. Let A, B be subsets of a set X. Show that the sets
$$A \setminus B, B \setminus A$$
are disjoint.

Definition

The **symmetric difference** of two sets A, B, denoted by $A \Delta B$, is defined as
$$A \Delta B := (A \cup B) \setminus (A \cap B).$$

Exercise 1.32. Determine the symmetric difference of $\{1, 2, 3, 4\}$ and $\{1, 3, 5, 7\}$.

Exercise 1.33. If A, B are sets, then show that
$$A \Delta B = (A \setminus B) \cup (B \setminus A),$$
$$A \Delta B = B \Delta A,$$
$$A \cup B = (A \Delta B) \cup (A \cap B),$$
$$(A \Delta B) \cap (A \cap B) = \emptyset.$$

Solution. Note that
$$\begin{aligned}
A \Delta B &= (A \cup B) \setminus (A \cap B) \\
&= (A \cup B) \cap (A \cap B)^c \\
&= (A \cap (A \cap B)^c) \cup (B \cap (A \cap B)^c) && \text{(by Fact 1.23)} \\
&= (A \cap (A^c \cup B^c)) \cup (B \cap (A^c \cup B^c)) && \text{(by Fact 1.28)} \\
&= ((A \cap A^c) \cup (A \cap B^c)) \cup ((B \cap A^c) \cup (B \cap B^c)) && \text{(by Fact 1.23)} \\
&= (\emptyset \cup (A \setminus B)) \cup ((B \setminus A) \cup \emptyset) && \text{(by Exercise 1.26)} \\
&= (A \setminus B) \cup (B \setminus A).
\end{aligned}$$

Also note that

$$\begin{aligned} A\Delta B &= (A \cup B) \setminus (A \cap B) \\ &= (B \cup A) \setminus (B \cap A) \\ &= B\Delta A. \end{aligned}$$

Since $A \cap B$ is a subset of $A \cup B$, we obtain

$$\begin{aligned} A \cup B &= ((A \cup B) \setminus (A \cap B)) \cup (A \cap B) \\ &= (A\Delta B) \cup (A \cap B). \end{aligned}$$

Using $A\Delta B = (A \cup B) \setminus (A \cap B)$, it follows that $A\Delta B$ contains no element of $A \cap B$. This implies that

$$(A\Delta B) \cap (A \cap B) = \emptyset.$$

□

§1.3 Family of sets

Definition

Let $n \geq 2$ be an integer, and let $A_1, \ldots, A_n$ be sets. The **union of $A_1, \ldots, A_n$** is denoted by $A_1 \cup \cdots \cup A_n$, and is defined by

$$A_1 \cup \cdots \cup A_n := \{x : x \text{ belongs to } A_i \text{ for some } 1 \leq i \leq n\}.$$

The **intersection of $A_1, \ldots, A_n$** is denoted by $A_1 \cap \cdots \cap A_n$, and is defined by

$$A_1 \cap \cdots \cap A_n := \{x : x \text{ belongs to } A_i \text{ for all } 1 \leq i \leq n\}.$$

§1.4 Power set

Example

Note that the subsets of $\{1, 2, 3\}$ are

$$\begin{aligned} &\{1, 2, 3\}, \\ &\{1, 2\}, \{2, 3\}, \{3, 1\}, \\ &\{1\}, \{2\}, \{3\}, \\ &\emptyset. \end{aligned}$$

Definition

Let A be a set. The set of subsets of A is denoted by $\mathcal{P}(A)$. It is called the **power set of A**.

Example

Note that

$$\begin{aligned}
\mathcal{P}(\{1\}) &= \{\emptyset, \{1\}\},\\
\mathcal{P}(\{x\}) &= \{\emptyset, \{x\}\},\\
\mathcal{P}(\{x,y\}) &= \{\emptyset, \{x\}, \{y\}, \{x,y\}\},\\
\mathcal{P}(\{x,y,z\}) &= \{\emptyset, \{x\}, \{y\}, \{z\}, \{x,y\}, \{y,z\}, \{z,x\}, \{x,y,z\}\},\\
\mathcal{P}(\{1,2,3\}) &= \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, \{1,2,3\}\},\\
\mathcal{P}(\{1,2,3,4\}) &= \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, \{1,2,3\},\\
&\qquad \{4\}, \{1,4\}, \{2,4\}, \{3,4\}, \{1,2,4\}, \{2,3,4\}, \{1,3,4\}, \{1,2,3,4\}\}.
\end{aligned}$$

Remark

Convince yourself that for any integer $n \geq 0$, the power set of a set of size n has size 2^n.

Exercise 1.34. Write down the power set of each of the following sets:

$$\{1,2\}, \{1,2,3\}, \{1,\{2,3\}\}, \{1,2,3,4\}.$$

§1.5 Cartesian product of sets

Definition

If A, B are sets, then the **cartesian product** $A \times B$ of A and B is the set of all ordered pairs of the form (a,b) with $a \in A$ and $b \in B$. That is,

$$A \times B := \{(a,b) : a \in A, b \in B\}.$$

If $A = \{1,2\}$ and $B = \{2,3,4\}$, then

$$A \times B = \{(1,2), (1,3), (1,4), (2,2), (2,3), (2,4)\}.$$

Exercise 1.35. If A, B are finite sets, then show that

$$|A \times B| = |A| \times |B|,$$

where for a set X, its number of elements is denoted by $|X|$.

§1.6 Sets and subsets of real and complex numbers

§1.6.1 Real numbers

$$\begin{aligned}
\mathbb{N} &= \text{the set of positive integers}\\
&= \{1,2,3,\dots\},\\
\mathbb{Z} &= \text{the set of integers}\\
&= \{0,1,-1,2,-2,3,-3,\dots\},\\
\mathbb{Q} &= \text{the set of rational numbers}\\
&= \left\{\frac{m}{n} : m \in \mathbb{Z}, n \in \mathbb{N}\right\},\\
\mathbb{R} &= \text{the set of real numbers.}
\end{aligned}$$

Remark

There are real numbers which are not rational. For instance, the number $\sqrt{2}$ is not rational.

Lemma 1.36. *No rational number x satisfies $x^2 = 2$.*

Proof. On the contrary, let us assume that some rational number x satisfies $x^2 = 2$. Replacing x by $-x$ if necessary, we may and do assume that x is positive. Write $x = \frac{a}{b}$ for some positive integers a, b. Let d denote the greatest common divisor of a, b. Write $a = da_0, b = db_0$ where a_0, b_0 are positive integers. Note that $\left(\frac{a_0}{b_0}\right)^2 = 2$.

Let us prove the two claims below. After establishing them, we will use them to complete the proof of the lemma.

Claim

None of a_0, b_0 is divisible by 3.

Proof of the Claim

On the contrary, let us assume that 3 divides at least one of the integers a_0, b_0.

If 3 divides a_0, then 3 divides a_0^2. Using $a_0^2 = 2b_0^2$, it follows that 3 divides b_0. Further, if 3 divides b_0, then 3 divides $2b_0^2$, and hence 3 divides a_0^2. This shows that a_0 is divisible by 3.

In both the cases, it follows that the greatest common divisor of a_0, b_0 is larger than 1. This shows that the greatest common divisor of da_0, db_0 is larger than d. Using $a = da_0, b = db_0$, we obtain that the greatest common divisor of a, b is larger than d, which is impossible. This contradicts the assumption that some rational number x satisfies $x^2 = 2$. This proves the claim.

Claim

If n is an integer not divisible by 3, then n^2 is equal to $3k + 1$ for some integer k, depending on n.

Proof of the Claim

Let q (resp. r) denote the quotient (resp. remainder) obtained upon dividing n by 3. This gives $n = 3q + r$. Note that

$$\begin{aligned} n^2 &= (3q + r)^2 \\ &= 9q^2 + 6qr + r^2 \end{aligned}$$

Since $n = 3q + r$ and 3 does not divide n, it follows that r is equal to 1 or 2. If $r = 1$, then

$$n^2 = 3(3q^2 + 2qr) + 1.$$

If $r = 2$, then

$$n^2 = 3(3q^2 + 2qr) + 4 = 3(3q^2 + 2qr + 1) + 1.$$

This proves the claim.

Using the claims above, it follows that

$$\begin{aligned} a_0^2 &= 3c + 1, \\ b_0^2 &= 3d + 1 \end{aligned}$$

for some integers c, d. This gives

$$3c + 1 = 2(3d + 1).$$

This yields $1 = 3(c - 2d)$, which is impossible.

This completes the proof of the lemma. $\square$

Definition

Let a, b be real numbers with $a \leq b$. Then the **closed interval** $[a, b]$ is defined by

$$[a, b] := \{x \in \mathbb{R} : a \leq x \leq b\}.$$

The **open interval** (a, b) is defined by

$$(a, b) := \{x \in \mathbb{R} : a < x < b\}.$$

Exercise 1.37. Determine the cartesian product of $[1, 2]$ and $[3, 4] \cup [5, 6]$.

Exercise 1.38. Identify the set

$$\left\{x \in \mathbb{R} \setminus \{0\} : x + \frac{1}{x} \geq 2\right\}.$$

Solution. Let x be an element of $\mathbb{R} \setminus \{0\}$. Note that

$$\begin{aligned} x + \frac{1}{x} - 2 &= \frac{x^2 + 1 - 2x}{x} \\ &= \frac{(x-1)^2}{x} \end{aligned}$$

hold. This shows that the condition

$$x + \frac{1}{x} \geq 2$$

is equivalent to the condition

$$\frac{(x-1)^2}{x} \geq 0. \tag{1}$$

If $x \neq 1$, then we obtain $(x-1)^2 > 0$, and then, Equation 1 yields $\frac{1}{x} > 0$, or equivalently, $x > 0$. This gives that

$$\left\{x \in \mathbb{R} \setminus \{0\} : x + \frac{1}{x} \geq 2\right\} \setminus \{1\} \subseteq \{x \in \mathbb{R} : x > 0\},$$

which implies

$$\left\{x \in \mathbb{R} \setminus \{0\} : x + \frac{1}{x} \geq 2\right\} \subseteq \{x \in \mathbb{R} : x > 0\}.$$

Also note that if $x > 0$, then

$$\frac{(x-1)^2}{x} \geq 0,$$

or equivalently,

$$x + \frac{1}{x} \geq 2.$$

It follows that

$$\boxed{\left\{x \in \mathbb{R} \setminus \{0\} : x + \frac{1}{x} \geq 2\right\} = \{x \in \mathbb{R} : x > 0\}}.$$

$\square$

Exercise 1.39. Let A, B be subsets of $\mathbb{R}$ defined by

$$A = \{x \in \mathbb{R} : x^2 \geq 0\},$$
$$B = \{x \in \mathbb{R} : x^3 \geq 0\}.$$

Determine the sets $A \cup B$, $A \cap B$, $A \setminus B$, $B \setminus A$.

Exercise 1.40. For a positive integer k, let A_k denote the set of integral multiples of k, that is,

$$A_k := \{r \in \mathbb{Z} : r = k\ell \text{ for some } \ell \in \mathbb{Z}\}.$$

Let m, n be positive integers. For each of the following statements, determine the equivalent conditions on the integers m, n.

1. $A_m \subseteq A_n$
2. $A_m \subsetneq A_n$
3. $A_m \not\subseteq A_n$
4. $A_m = A_n$
5. $A_m \cap A_n = \emptyset$
6. $A_m \setminus A_n \neq \emptyset$
7. $A_m \setminus A_n \neq \emptyset$ or $A_n \setminus A_m \neq \emptyset$
8. $A_m \setminus A_n \neq \emptyset$ and $A_n \setminus A_m \neq \emptyset$

§1.6.2 Complex numbers

Note that no real number x satisfies $x^2 = -1$. To "resolve" this, one "introduces"[2] an element i satisfying

$$\boxed{i^2 = -1.}$$

The element i can be multiplied by any real number, and the product can also be added to any real number. This leads to the numbers of the form $a + ib$, where a, b are real numbers. Such numbers are called the **complex numbers**, and the set of such numbers is denoted by $\mathbb{C}$.

Definition

The **set of complex numbers**, is denoted by $\mathbb{C}$, and is defined by

$$\mathbb{C} := \{a + ib : a, b \in \mathbb{R}\}.$$

Exercise 1.41. Show that $\mathbb{R}$ is a subset of $\mathbb{C}$.

Definition

For a complex number $z = a + ib$ with $a, b \in \mathbb{R}$, the **real part** (resp. **imaginary part**) of z, denoted by $\mathfrak{R}(z)$ (resp. $\mathfrak{I}(z)$), is defined by

$$\mathfrak{R}(z) = a,$$
$$\mathfrak{I}(z) = b.$$

[2]To introduce is to adjoin.

Definition

For complex numbers $z = a + ib, w = c + id$ with $a, b, c, d \in \mathbb{R}$, the **sum** $z + w$ and the **product** $z \cdot w$ of z and w are defined by

$$z + w := (a + c) + i(b + d),$$
$$z \cdot w := (ac - bd) + i(ad + bc).$$

Definition

For a complex number $z = a + ib$ with $a, b \in \mathbb{R}$, the **conjugate** of z, denoted by $\overline{z}$, is defined by

$$\overline{z} := a - ib.$$

Definition

For a complex number $z = a + ib$ with $a, b \in \mathbb{R}$, the **absolute value** of z, also called the **modulus** of z, is denoted by $|z|$, and is defined by

$$|z| := \sqrt{a^2 + b^2}.$$

Exercise 1.42. If z, w are complex numbers, then show that

1. $\overline{z + w} = \overline{z} + \overline{w}$,
2. $\overline{z \cdot w} = \overline{z} \cdot \overline{w}$,
3. $|z| \geq 0$, and $|z| = 0$ if and only if $z = 0$,
4. $|z \cdot w| = |z|\ |w|$.

Exercise 1.43. If z is a complex number, then show that

$$z \cdot \overline{z} = |z|^2.$$

Exercise 1.44. For any real number θ, show that

$$|\cos\theta + i\sin\theta| = 1.$$

Fact 1.45. If z is a complex number satisfying $|z| = 1$, then there exists a real number θ such that

$$z = \cos\theta + i\sin\theta.$$

Consequently, if z is a nonzero complex number, then for some real number θ,

$$z = |z|\ (\cos\theta + i\sin\theta)$$

holds.

Chapter 2. Induction principles

Example

Show that
$$1+2+3+\cdots+n=\frac{n(n+1)}{2}$$
holds for any positive integer n.

Solution

For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that
$$1+2+3+\cdots+n=\frac{n(n+1)}{2}.$$
Note that $P(1)$ holds. Let k be an element of $\mathbb{N}$, and assume that $P(k)$ holds. Note that
$$\begin{aligned} 1+2+3+\cdots+k+(k+1) &= \frac{k(k+1)}{2}+(k+1) \quad (\text{using } P(k)) \\ &= (k+1)\left(\frac{k}{2}+1\right) \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$
This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n.

Definition

If A is a nonempty subset of $\mathbb{N}$, then an element a_0 of A is called a **least element** of A if $a_0 \leq a$ for all $a \in A$.

Exercise 2.1. If A is a nonempty subset of $\mathbb{N}$, and a_1, a_2 are least elements of A, then show that $a_1 = a_2$.

Solution. Since a_1, a_2 are least elements of A, they belong to A. Using that a_1 is a least element of A, we obtain that $a_1 \leq a_2$. Further, using that a_2 is a least element of A, we obtain that $a_2 \leq a_1$. Combining these two inequalities, it follows that $a_1 = a_2$. □

Remark

By Exercise 2.1, a nonempty subset of $\mathbb{N}$ has at most one least element, to be called **the** least element, if it exists.

Well-ordering principle

Every nonempty subset of $\mathbb{N}$ has a least element.

Principle of mathematical induction

For each $n \in \mathbb{N}$, let $P(n)$ be a statement. Suppose $P(1)$ is true, and that whenever $P(n)$ holds for some $n \geq 1$, the statement $P(n+1)$ holds. Then $P(n)$ is true for all $n \geq 1$.

Principle of strong induction

For each $n \in \mathbb{N}$, let $P(n)$ be a statement. Suppose $P(1)$ is true, and that whenever $P(1), P(2), ..., P(n)$ hold for some $n \geq 1$, the statement $P(n+1)$ holds. Then $P(n)$ is true for all $n \geq 1$.

Remark

The above three principles are equivalent. A proof of this is provided in Chapter 7 (see Theorem 7.1).

Exercise 2.2. Show that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for all $n \in \mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
&\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} \\
&= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} && \text{(using } P(k)\text{)} \\
&= \frac{k(k+2)+1}{(k+1)(k+2)} \\
&= \frac{(k+1)^2}{(k+1)(k+2)} \\
&= \frac{k+1}{k+2}.
\end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.3. Show that

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

for all $n \in \mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
&\frac{1}{1\cdot 3}+\frac{1}{3\cdot 5}+\cdots+\frac{1}{(2k-1)(2k+1)}+\frac{1}{(2(k+1)-1)(2(k+1)+1)}\\
&=\frac{k}{2k+1}+\frac{1}{(2k+1)(2k+3)} && \text{(using } P(k)\text{)}\\
&=\frac{2k^2+3k+1}{(2k+1)(2k+3)}\\
&=\frac{(k+1)(2k+1)}{(2k+1)(2k+3)}\\
&=\frac{k+1}{2(k+1)+1}.
\end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.4. Show that

$$\frac{1}{1\cdot 2\cdot 3}+\frac{1}{2\cdot 3\cdot 4}+\cdots+\frac{1}{n(n+1)(n+2)}=\frac{n(n+3)}{4(n+1)(n+2)}$$

for all $n\in\mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that

$$\frac{1}{1\cdot 2\cdot 3}+\frac{1}{2\cdot 3\cdot 4}+\cdots+\frac{1}{n(n+1)(n+2)}=\frac{n(n+3)}{4(n+1)(n+2)}.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$
\begin{aligned}
&\frac{1}{1\cdot 2\cdot 3}+\frac{1}{2\cdot 3\cdot 4}+\cdots+\frac{1}{k(k+1)(k+2)}+\frac{1}{(k+1)(k+2)(k+3)}\\
&=\frac{k(k+3)}{4(k+1)(k+2)}+\frac{1}{(k+1)(k+2)(k+3)} && \text{(using } P(k)\text{)}\\
&=\frac{k(k+3)^2+4}{4(k+1)(k+2)(k+3)}\\
&=\frac{(k+1)(k+3)^2-((k+3)^2-4)}{4(k+1)(k+2)(k+3)}\\
&=\frac{(k+1)(k+3)^2-((k+3)^2-2^2)}{4(k+1)(k+2)(k+3)}\\
&=\frac{(k+1)(k+3)^2-(k+1)(k+5)}{4(k+1)(k+2)(k+3)}\\
&=\frac{(k+1)((k+3)^2-(k+5))}{4(k+1)(k+2)(k+3)}\\
&=\frac{k^2+6k+9-(k+5)}{4(k+2)(k+3)}\\
&=\frac{k^2+5k+4}{4(k+2)(k+3)}\\
&=\frac{(k+1)((k+1)+3)}{4((k+1)+1)((k+1)+2)}.
\end{aligned}
$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.5. Show that

$$3+11+\cdots+(8n-5)=4n^2-n$$

for all $n\in\mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that

$$3+11+\cdots+(8n-5)=4n^2-n.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$
\begin{aligned}
&3+11+\cdots+(8k-5)+(8(k+1)-5)\\
&=(4k^2-k)+(8k+3) && \text{(using } P(k)\text{)}\\
&=4k^2+7k+3\\
&=4k^2+8k+4-(k+1)\\
&=4(k+1)^2-(k+1).
\end{aligned}
$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.6. Show that

$$1^2+3^2+\cdots+(2n-1)^2=\frac{4n^3-n}{3}$$

for all $n\in\mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that

$$1^2 + 3^2 + \cdots + (2n-1)^2 = \frac{4n^3 - n}{3}.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
&1^2 + 3^2 + \cdots + (2k-1)^2 + (2(k+1)-1)^2 \\
&= \frac{4k^3 - k}{3} + (2k+1)^2 && \text{(using } P(k)\text{)} \\
&= \frac{4k^3 - k + 3(2k+1)^2}{3} \\
&= \frac{4k^3 - k + 12k^2 + 12k + 3}{3} \\
&= \frac{4k^3 + 12k^2 + 11k + 3}{3} \\
&= \frac{4(k+1)^3 - (k+1)}{3}.
\end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.7. Show that

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1} n^2 = (-1)^{n+1} \frac{n(n+1)}{2}$$

for all $n \in \mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1} n^2 = (-1)^{n+1} \frac{n(n+1)}{2}.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
&1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{k+1} k^2 + (-1)^{(k+1)+1} (k+1)^2 \\
&= (-1)^{k+1} \frac{k(k+1)}{2} + (-1)^{k+2} (k+1)^2 && \text{(using } P(k)\text{)} \\
&= (-1)^{k+1} \left(\frac{k(k+1)}{2} - (k+1)^2 \right) \\
&= (-1)^{k+1} \left(\frac{k(k+1) - 2(k+1)^2}{2} \right) \\
&= (-1)^{k+1} \left(\frac{-(k+1)(k+2)}{2} \right) \\
&= (-1)^{(k+1)+1} \frac{(k+1)((k+1)+1)}{2}.
\end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.8. Show that

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

hold for any positive integer n.

Solution. For a positive integer n, let $P(n)$ denote the statement that

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

Note that $P(1)$ holds. Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
&1^2 + 2^2 + 3^2 + \cdots + k^2 + (k+1)^2 \\
&= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \qquad \text{(using } P(k)\text{)} \\
&= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \\
&= \frac{(k+1)(k(2k+1) + 6(k+1))}{6} \\
&= \frac{(k+1)(2k^2 + k + 6k + 6)}{6} \\
&= \frac{(k+1)(2k^2 + 7k + 6)}{6} \\
&= \frac{(k+1)(k+2)(2k+3)}{6} \\
&= \frac{(k+1)(k+2)(2(k+1)+1)}{6}.
\end{aligned}$$

Also note that

$$\begin{aligned}
&1^3 + 2^3 + 3^3 + \cdots + k^3 + (k+1)^3 \\
&= \left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3 \qquad \text{(using } P(k)\text{)} \\
&= (k+1)^2 \left(\frac{k^2}{4} + (k+1)\right) \\
&= (k+1)^2 \left(\frac{k^2}{4} + k + 1\right) \\
&= (k+1)^2 \left(\frac{(k+2)^2}{4}\right) \\
&= \left(\frac{(k+1)((k+1)+1)}{2}\right)^2.
\end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n.

□

Exercise 2.9. Show that
$$\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{2n}=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots+\frac{1}{2n-1}-\frac{1}{2n}$$
for all $n \in \mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that
$$\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{2n}=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots+\frac{1}{2n-1}-\frac{1}{2n}.$$
Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that
$$\begin{aligned}
&\frac{1}{(k+1)+1}+\cdots+\frac{1}{2k+1}+\frac{1}{2(k+1)}\\
&=\frac{1}{k+1}+\frac{1}{(k+1)+1}+\cdots+\frac{1}{2k}+\frac{1}{2k+1}+\frac{1}{2(k+1)}-\frac{1}{k+1}\\
&=\frac{1}{k+1}+\frac{1}{(k+1)+1}+\cdots+\frac{1}{2k}+\frac{1}{2k+1}-\frac{1}{2(k+1)}\\
&=\left(1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots+\frac{1}{2k-1}-\frac{1}{2k}\right)+\frac{1}{2k+1}-\frac{1}{2(k+1)}\\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\text{(using } P(k))\\
&=\left(1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots+\frac{1}{2k-1}-\frac{1}{2k}\right)+\frac{1}{2k+1}-\frac{1}{2k+2}\\
&=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots+\frac{1}{(2(k+1))-1}-\frac{1}{(2(k+1))}.
\end{aligned}$$
This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Definition

An integer is called **divisible** by a nonzero integer m if it is equal to mk for some integer k.

Exercise 2.10. Show that n^3+5n is divisible by 6 for all $n \in \mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that n^3+5n is divisible by 6.

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that
$$\begin{aligned}
(k+1)^3+5(k+1)&=(k^3+3k^2+3k+1)+(5k+5)\\
&=(k^3+5k)+(3k^2+3k+6).
\end{aligned}$$
By the induction hypothesis, the integer k^3+5k is divisible by 6. Also note that $3k^2+3k+6=3(k^2+k+2)$ is divisible by 3. Further, one of the integers $k, k+1$ is even, and hence k^2+k is even. This gives that k^2+k+2 is even, and hence $3(k^2+k+2)$ is divisible by 2. It follows that $3(k^2+k+2)$ is divisible by 6. This shows that $(k+1)^3+5(k+1)$ is divisible by 6. This proves that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds.

By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.11. Show that $5^{2n}-1$ is divisible by 8 for all $n \in \mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that $5^{2n}-1$ is divisible by 8.

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned} 5^{2(k+1)} - 1 &= 5^{2k+2} - 1 \\ &= 5^2 \cdot 5^{2k} - 1 \\ &= 25 \cdot 5^{2k} - 1 \\ &= (24+1) \cdot 5^{2k} - 1 \\ &= 24 \cdot 5^{2k} + \left(5^{2k} - 1\right). \end{aligned}$$

By the induction hypothesis, the integer $5^{2k} - 1$ is divisible by 8. Also note that $24 \cdot 5^{2k}$ is divisible by 8. This shows that $5^{2(k+1)} - 1$ is divisible by 8. This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds.

By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.12. Show that $5^n - 4n - 1$ is divisible by 16 for all $n \in \mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that $5^n - 4n - 1$ is divisible by 16.

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned} 5^{k+1} - 4(k+1) - 1 &= 5 \cdot 5^k - 4k - 4 - 1 \\ &= 5 \cdot \left(5^k - 4k - 1\right) + 5 \cdot (4k+1) - 4k - 5 \\ &= 5 \cdot \left(5^k - 4k - 1\right) + 16k. \end{aligned}$$

By the induction hypothesis, the integer $5^k - 4k - 1$ is divisible by 16. Using the above, it follows that $5^{k+1} - 4(k+1) - 1$ is divisible by 16. This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds.

By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.13. Show that $6^n - 5n - 1$ is divisible by 25 for all $n \in \mathbb{N}$.

Exercise 2.14. Show that $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9 for all $n \in \mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9.

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned} &(k+1)^3 + (k+2)^3 + (k+3)^3 \\ &= \left(k^3 + (k+1)^3 + (k+2)^3\right) - k^3 + (k+3)^3 \\ &= \left(k^3 + (k+1)^3 + (k+2)^3\right) - k^3 + \left(k^3 + 9k^2 + 27k + 27\right) \\ &= \left(k^3 + (k+1)^3 + (k+2)^3\right) + \left(9k^2 + 27k + 27\right). \end{aligned}$$

By the induction hypothesis, the integer $k^3 + (k+1)^3 + (k+2)^3$ is divisible by 9. Also note that $9k^2 + 27k + 27$ is divisible by 9. This shows that $(k+1)^3 + (k+2)^3 + (k+3)^3$ is divisible by 9. This implies that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds.

By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.15. Show that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

for all $n \in \mathbb{N}$ satisfying $n > 1$.

Solution. For a positive integer n, let $P(n)$ denote the statement that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

holds.

Note that $P(2)$ holds, since

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} = 1 + \frac{\sqrt{2}}{2} > \sqrt{2}.$$

Let k be a positive integer such that $k \geq 2$ and $P(k)$ holds. Note that

$$\begin{aligned} & \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \\ & > \sqrt{k} + \frac{1}{\sqrt{k+1}} && \text{(using } P(k)\text{)} \\ & \geq \sqrt{k+1}, \end{aligned}$$

where the last inequality follows since

$$\begin{aligned} k - \left(\sqrt{k+1} - \frac{1}{\sqrt{k+1}}\right)^2 &= k - (k+1) + 2 - \frac{1}{k+1} \\ &= 1 - \frac{1}{k+1} \\ &> 0. \end{aligned}$$

This shows that if k is a positive integer satisfying $k \geq 2$ and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n satisfying $n \geq 2$.

□

Exercise 2.16. Show that $3^n \geq n^2$ for all $n \in \mathbb{N}$.

Solution. For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that $3^n \geq n^2$. Note that $P(1)$ holds. Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned} (k+1)^2 &= k^2 + 2k + 1 \\ &\leq 3^k + 2k + 1 && \text{(using } P(k)\text{)} \\ &\leq 3^k + 3^k + 3^k && \text{(using } k \geq 1\text{)} \\ &= 3 \cdot 3^k \\ &= 3^{k+1}. \end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.17. Show that $n! > 2^n$ for all $n \in \mathbb{N}$ satisfying $n \geq 4$.

Solution. For a positive integer n, let $P(n)$ denote the statement that $n! > 2^n$.

Note that $P(4)$ holds, since $4! = 24 > 16 = 2^4$.

Let k be a positive integer such that $k \geq 4$ and $P(k)$ holds. Note that

$$\begin{aligned}(k+1)! &= (k+1)k! \\ &\geq (k+1)\cdot 2^k \quad (\text{using } P(k)) \\ &> 2\cdot 2^k \qquad\quad (\text{using } k \geq 4) \\ &= 2^{k+1}.\end{aligned}$$

This shows that if k is a positive integer satisfying $k \geq 4$ and $P(k)$ holds, then $P(k+1)$ also holds.

By the principle of induction, it follows that $P(n)$ holds for any positive integer n satisfying $n \geq 4$.

□

Warning

It is important to verify the base case, that is, verifying that $P(1)$ holds, even if it is trivial. For example, if we define $P(n)$ to be the statement that

$$n = n + 1,$$

then $P(n)$ implies $P(n+1)$ for any $n \in \mathbb{N}$. However, $P(1)$ does not hold. Another example is to take $Q(n)$ to be the statement that $2n+1$ is even, then $Q(n)$ implies $Q(n+1)$ for any $n \in \mathbb{N}$. However, $Q(1)$ does not hold.

Here are a few exercises where strong induction is useful.

Exercise 2.18. Show that every positive integer greater than 1 is a prime or is a product of prime numbers.

Solution. For any integer $n \geq 2$, let $P(n)$ denote the statement that n is a prime or is a product of prime numbers.

Note that $P(2)$ holds, since 2 is a prime.

Let $k \geq 2$ be an integer such that $P(m)$ holds for all integers m satisfying $2 \leq m \leq k$. If $k+1$ is a prime, then $P(k+1)$ holds. If $k+1$ is not a prime, then there exist integers a, b such that $k+1 = ab$ and $1 < a \leq b < k+1$. This gives that $2 \leq a \leq k$ and $2 \leq b \leq k$. By the induction hypothesis, both a and b are either primes or products of primes. It follows that $k+1 = ab$ is a product of primes. This shows that if $k \geq 2$ is an integer such that $P(m)$ holds for all integers m satisfying $2 \leq m \leq k$, then $P(k+1)$ also holds.

By the principle of strong induction, it follows that $P(n)$ holds for any integer $n \geq 2$. □

Exercise 2.19. Let the integers $x_1, x_2, \ldots$ be defined by

$$\begin{aligned}x_1 &:= 1, \\ x_2 &:= 2, \\ x_{n+2} &:= \frac{1}{2}(x_n + x_{n+1})\end{aligned}$$

for all $n \in \mathbb{N}$. Show that $1 \leq x_n \leq 2$ for all $n \in \mathbb{N}$.

Tip

For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that $1 \leq x_m \leq 2$ for all $m \in \mathbb{N}$ satisfying $m \leq n$.

Exercise 2.20. The Fibonacci sequence $F_0, F_1, F_2, \ldots$ is defined by

$$
\begin{aligned}
F_0 &:= 0, \\
F_1 &:= 1, \\
F_2 &:= 1, \\
F_{n+2} &:= F_n + F_{n+1}
\end{aligned}
$$

for all $n \in \mathbb{N}$. Show that

$$F_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)$$

for all $n \in \mathbb{N} \cup \{0\}$.

Exercise 2.21. Show that

$$F_0^2 + F_1^2 + F_2^2 + \cdots + F_n^2 = F_n F_{n+1}$$

for all $n \in \mathbb{N} \cup \{0\}$.

Linear algebra

Chapter 3. Matrices

§3.1 Vectors

§3.1.1 Definitions and examples

A vector is a column of numbers. The numbers in the vector are called its **components** or **coordinates**.

Remark

Throughout this course, most of the vectors to be considered, will have real or complex components.

Recall that for two sets A, B, their cartesian product $A \times B$ is defined as

$$A \times B := \{(a, b) \mid a \in A, b \in B\}.$$

The cartesian product of $\mathbb{R}$ with itself is denoted by $\mathbb{R}^2$. In other words,

$$\boxed{\mathbb{R}^2 := \{(x, y) \mid x \in \mathbb{R}, y \in \mathbb{R}\}}\,.$$

Similarly, one defines $\mathbb{R}^3$ as $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$, that is,

$$\boxed{\mathbb{R}^3 := \{(x, y, z) \mid x \in \mathbb{R}, y \in \mathbb{R}, z \in \mathbb{R}\}}\,.$$

Definition

If $n \geq 2$ is an integer and $A_1, A_2, \ldots, A_n$ are sets, then their **cartesian product** $A_1 \times A_2 \times \cdots \times A_n$ is defined as

$$A_1 \times A_2 \times \cdots \times A_n := \{(a_1, a_2, \ldots, a_n) \mid a_i \in A_i \text{ for all } i = 1, 2, \ldots, n\}.$$

The n-fold cartesian product of $\mathbb{R}$ with itself, that is, the cartesian product of n copies of $\mathbb{R}$, is denoted by $\mathbb{R}^n$. In other words,

$$\boxed{\mathbb{R}^n := \{(x_1, x_2, \ldots, x_n) \mid x_i \in \mathbb{R} \text{ for all } i = 1, 2, \ldots, n\}}\,.$$

Similarly, one defines $\mathbb{C}^n$ as the n-fold cartesian product of $\mathbb{C}$ with itself, that is,

$$\boxed{\mathbb{C}^n := \{(z_1, z_2, \ldots, z_n) \mid z_i \in \mathbb{C} \text{ for all } i = 1, 2, \ldots, n\}}\,.$$

Example

The elements

$$\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix}$$

of $\mathbb{R}^2$ are vectors. The elements

$$\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix}, \begin{pmatrix}10\\20\\-2025\end{pmatrix}$$

of $\mathbb{R}^3$ are vectors too. The elements

$$\begin{pmatrix}1+i\\2-i\\-3i\end{pmatrix}, \begin{pmatrix}0\\\pi\\1+i\end{pmatrix}$$

of $\mathbb{C}^3$ are vectors as well.

§3.1.2 Vector operations

Definition

Two vectors $\begin{pmatrix}x_1\\x_2\\\vdots\\x_n\end{pmatrix}$ and $\begin{pmatrix}y_1\\y_2\\\vdots\\y_n\end{pmatrix}$ in $\mathbb{R}^n$ are added component-wise. That is, the **sum** of two elements $\begin{pmatrix}x_1\\x_2\\\vdots\\x_n\end{pmatrix}$ and $\begin{pmatrix}y_1\\y_2\\\vdots\\y_n\end{pmatrix}$ in $\mathbb{R}^n$ is defined as

$$\begin{pmatrix}x_1\\x_2\\\vdots\\x_n\end{pmatrix} + \begin{pmatrix}y_1\\y_2\\\vdots\\y_n\end{pmatrix} := \begin{pmatrix}x_1+y_1\\x_2+y_2\\\vdots\\x_n+y_n\end{pmatrix}.$$

If $c \in \mathbb{R}$ is a real number and $v = \begin{pmatrix}x_1\\x_2\\\vdots\\x_n\end{pmatrix}$ is a vector in $\mathbb{R}^n$, then the **scalar multiplication** of c with the vector v is defined as

$$c \cdot \begin{pmatrix}x_1\\x_2\\\vdots\\x_n\end{pmatrix} := \begin{pmatrix}c\cdot x_1\\c\cdot x_2\\\vdots\\c\cdot x_n\end{pmatrix}.$$

Note that $1 \cdot v$ is equal to v. Instead of $(-1) \cdot v$, we write $-v$. Further, $c \cdot v$ is often written as cv.

Warning

In the above definition, we have used the same notation "+" for addition of two vectors and addition of two real numbers, and the same notation "·" for scalar multiplication of a vector by a real number and multiplication of two real numbers. The meaning will be clear from the context.

Remark

Note that the sum of two vectors in $\mathbb{R}^n$ is again a vector in $\mathbb{R}^n$. Also, the scalar multiplication of a vector in $\mathbb{R}^n$ by a real number is again a vector in $\mathbb{R}^n$.

Remark

Similar definitions work for vectors in $\mathbb{C}^n$.

Example

Consider the vectors $u = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $v = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$ in $\mathbb{R}^3$. Note that

$$u + v = \begin{pmatrix} 1+4 \\ 2+5 \\ 3+6 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \\ 9 \end{pmatrix},$$

and if $c = 3$, then

$$c \cdot u = 3 \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \cdot 1 \\ 3 \cdot 2 \\ 3 \cdot 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix}.$$

§3.2 Matrix

§3.2.1 Definitions and examples

Definition

A **matrix** is a rectangular array of numbers arranged in rows and columns. A matrix with m rows and n columns is called an $m \times n$ **matrix**. The numbers in the matrix are called its **entries** or **elements**. A matrix is called a **square matrix** if its number of rows, and its number of columns are equal.

Definition

Two matrices are **equal** if they have the same size and their corresponding entries are equal.

Remark

The **size** of a matrix is always given in the form "(the number of rows) $\times$ (the number of columns)".

Example

The following is a 2×3 matrix:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}.$$

The following is a 3×2 matrix:

$$B = \begin{pmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{pmatrix}.$$

Example

Here is an example of two matrices which are not equal.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \neq \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$$

If A is a matrix, its (i, j)-th entry, that is, the entry common to the i-th row and the j-th column is denoted by a_{ij}. Thus an $m \times n$ matrix A can be represented as

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

One also writes $A = (a_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}$ to denote the matrix A, or one writes $A = (a_{ij})$ if the size of A is clear from the context.

Remark

Note that for an $m \times n$ matrix A, one writes $A = (\text{the } (i, j)\text{-th entry of } A)_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}$.

If $A = (a_{ij})$ is an $n \times n$ matrix, then the entries $a_{11}, a_{22}, \ldots, a_{nn}$ are called its **diagonal entries**. A square matrix is called a **diagonal matrix** if its entries, other the diagonal entries, are equal to zero.

§3.2.2 Matrix times a vector

The product

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$$

is defined by “combining” each row of the matrix with the vector as follows:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} = \begin{pmatrix} 1 \times 7 + 2 \times 8 + 3 \times 9 \\ 4 \times 7 + 5 \times 8 + 6 \times 9 \end{pmatrix},$$

which is equal to the vector $\begin{pmatrix} 50 \\ 122 \end{pmatrix}$.

As we see, for this to work, the number of columns of the matrix must be equal to the number of rows of the vector. Thus, when the product of an $m \times n$ matrix with a vector in $\mathbb{R}^n$ is defined similarly, the result is a vector in $\mathbb{R}^m$.

Definition

If

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

is an $m \times n$ matrix with entries in $\mathbb{R}$ and $\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ is a vector in $\mathbb{R}^n$, then one defines

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} := \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{pmatrix}.$$

Note that the result is a vector in $\mathbb{R}^m$.

§3.2.3 Matrix times a matrix

Similar to the case of vectors, one defines addition, subtraction, and scalar multiplication of matrices entry-wise. For addition and subtraction, the two matrices must be of the same size. Indeed, if

$$A = (a_{ij})_{\substack{1 \le i \le m, \\ 1 \le j \le n}}, B = (b_{ij})_{\substack{1 \le i \le m, \\ 1 \le j \le n}}$$

are two $m \times n$ matrices, then their **sum** $A + B$ and **difference** $A - B$ are defined as

$$A + B := (a_{ij} + b_{ij})_{\substack{1 \le i \le m, \\ 1 \le j \le n}},$$

and

$$A - B := (a_{ij} - b_{ij})_{\substack{1 \le i \le m, \\ 1 \le j \le n}}$$

respectively. If c is a real number and $A = (a_{ij})_{\substack{1 \le i \le m, \\ 1 \le j \le n}}$ is an $m \times n$ matrix, then the **scalar multiplication** of A with c is defined as

$$c \cdot A := (c \cdot a_{ij})_{\substack{1 \le i \le m, \\ 1 \le j \le n}}.$$

Note that $1 \cdot A$ is equal to A. Instead of $-1 \cdot A$, we write $-A$. Further, $c \cdot A$ is often written as cA.

Remark

Note that the sum of two matrices with entries in $\mathbb{R}$ is a matrix with entries in $\mathbb{R}$. Also, the scalar multiplication of a matrix with entries in $\mathbb{R}$ by a real number is again a matrix with entries in $\mathbb{R}$.

Remark

Similar definitions work for matrices with entries in $\mathbb{C}$.

Example

If

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

and

$$B = \begin{pmatrix} 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix},$$

then

$$A + B = \begin{pmatrix} 1+7 & 2+8 & 3+9 \\ 4+10 & 5+11 & 6+12 \end{pmatrix} = \begin{pmatrix} 8 & 10 & 12 \\ 14 & 16 & 18 \end{pmatrix},$$

and if $c = 3$, then

$$c \cdot A = 3 \cdot \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 3 \cdot 1 & 3 \cdot 2 & 3 \cdot 3 \\ 3 \cdot 4 & 3 \cdot 5 & 3 \cdot 6 \end{pmatrix} = \begin{pmatrix} 3 & 6 & 9 \\ 12 & 15 & 18 \end{pmatrix}.$$

Exercise 3.1. If A, B, C are matrices of the same size, show that

$$A + (B + C) = (A + B) + C.$$

Exercise 3.2. If A, B are matrices of the same size, show that

$$A + B = B + A.$$

Exercise 3.3. If A is an $m \times n$ matrix, show that

$$A + 0_{m \times n} = 0_{m \times n} + A = A$$

holds, where $0_{m \times n}$ denotes the $m \times n$ zero matrix, that is, the $m \times n$ matrix whose entries are all zero.

Exercise 3.4. Show that if A is a matrix and c, d are real numbers, then

$$(c + d)A = cA + dA$$

and

$$c(dA) = (cd)A.$$

Definition

The **transpose** of an $m \times n$ matrix

$$A = (a_{ij})_{\substack{1 \le i \le m, \\ 1 \le j \le n}}$$

is the $n \times m$ matrix

$$A^T := (a_{ji})_{\substack{1 \le i \le n, \\ 1 \le j \le m}}.$$

That is, if

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix},$$

then its **transpose** is the $n \times m$ matrix

$$A^T := \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}.$$

Example

If

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix},$$

then its transpose is the 3×2 matrix

$$A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

Remark

Note that the transpose of a matrix is obtained by interchanging its rows and columns.

Exercise 3.5. If A is a matrix, what is $\left(A^T\right)^T$?

Exercise 3.6. Show that if A, B are matrices of the same size, then

$$(A + B)^T = A^T + B^T.$$

Solution. Write

$$A = (a_{ij})_{\substack{1 \le i \le m, \\ 1 \le j \le n}}, B = (b_{ij})_{\substack{1 \le i \le m, \\ 1 \le j \le n}}.$$

Note that

$$
\begin{aligned}
(A+B)^T &= \left((a_{ij}+b_{ij})_{\substack{1\le i\le m,\\ 1\le j\le n}} \right)^T \\
&= (a_{ji}+b_{ji})_{\substack{1\le i\le n,\\ 1\le j\le m}} \\
&= (a_{ji})_{\substack{1\le i\le n,\\ 1\le j\le m}} + (b_{ji})_{\substack{1\le i\le n,\\ 1\le j\le m}} \\
&= A^T + B^T.
\end{aligned}
$$

□

Exercise 3.7. Show that if A denotes a matrix, then

$$(A^T)^T = A.$$

Solution. Write

$$A = (a_{ij})_{\substack{1\le i\le m,\\ 1\le j\le n}}.$$

Note that

$$
\begin{aligned}
(A^T)^T &= \left((a_{ji})_{\substack{1\le i\le n,\\ 1\le j\le m}} \right) \\
&= (a_{ij})_{\substack{1\le i\le m,\\ 1\le j\le n}} \\
&= A.
\end{aligned}
$$

□

Definition

The **product** of an $m \times n$ matrix A with an $n \times p$ matrix B is defined as follows:

$$AB := (c_{ij})_{\substack{1 \le i \le m, \\ 1 \le j \le p}},$$

where

$$c_{ij} := a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

for all $1 \le i \le m$ and $1 \le j \le p$. In other words, given an $m \times n$ matrix

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

and an $n \times p$ matrix

$$B = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{pmatrix},$$

we have

$$AB = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1p} \\ c_{21} & c_{22} & \cdots & c_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mp} \end{pmatrix},$$

where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

for all $1 \le i \le m$ and $1 \le j \le p$. Note that the result is an $m \times p$ matrix.

In other words, the (i, j)-th entry of the product AB is obtained by “combining” the i-th row of A with the j-th column of B.

Remark

Note that the product AB is defined only when the number of columns of A is equal to the number of rows of B.

Example

If

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

and

$$B = \begin{pmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{pmatrix},$$

then the product AB is the 2×2 matrix

$$\begin{aligned} AB &= \begin{pmatrix} 1 \times 7 + 2 \times 9 + 3 \times 11 & 1 \times 8 + 2 \times 10 + 3 \times 12 \\ 4 \times 7 + 5 \times 9 + 6 \times 11 & 4 \times 8 + 5 \times 10 + 6 \times 12 \end{pmatrix} \\ &= \begin{pmatrix} 58 & 64 \\ 139 & 154 \end{pmatrix}. \end{aligned}$$

Question

Suppose A, B are matrices.

- Under what conditions are the products AB and BA defined?
- Under what conditions is the product $B^T A^T$ defined?

Exercise 3.8. Show that if A, B, C are matrices such that the products $A(BC)$ and $(AB)C$ are defined, then

$$A(BC) = (AB)C.$$

Exercise 3.9. Provide examples to show that in general, matrix multiplication is not commutative, that is, $AB \neq BA$ for some matrices A, B.

Exercise 3.10 (**)**.** Show that for any $n \in \mathbb{N}$ with $n \geq 2$, there are $n \times n$ matrices A, B such that

$$AB \neq BA.$$

Question

Can induction be used for the above exercise?

Exercise 3.11. If A is an $n \times n$ matrix, show that

$$AI_n = I_n A = A$$

holds, where I_n denotes the $n \times n$ diagonal matrix, with all diagonal entries equal to 1, that is,

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}.$$

Remark

The matrix I_n as above, is called the **identity matrix** of order n, or the $n \times n$ **identity matrix**.

Exercise 3.12. If A is an $m \times n$ matrix, show that

$$I_m A = A, AI_n = A$$

where I_m (resp. I_n) denote the $m \times m$ (resp. $n \times n$) identity matrix.

Exercise 3.13. If A is an $m \times n$ matrix with entries in $\mathbb{R}$ and c is a real number, then

$$cA = (cI_n)A = A(cI_m),$$

where I_n (resp. I_m) denotes the $n \times n$ (resp. $m \times m$) identity matrix.

Exercise 3.14. Show that if A, B are matrices, then

$$(AB)^T = B^T A^T.$$

Remark

Given a square matrix A, that is, a matrix with the same number of rows and columns, the product AA is often denoted by A^2. Similarly, one defines $A^3, A^4, \ldots$.

Definition

Let A be an $n \times n$ matrix, and k be a positive integer. The **k-th power** of A, denoted by A^k, is defined as the product of k copies of A.

Exercise 3.15. Show that if

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

then

$$A^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}$$

for all $n \in \mathbb{N}$ with $n \geq 1$, where F_n denotes the n-th Fibonacci number.

Exercise 3.16. Compute

$$\begin{pmatrix} 0 & \cdots & 0 & 1 \\ 0 & \cdots & 1 & 0 \\ \vdots & ⋰ & \vdots & \vdots \\ 1 & \cdots & 0 & 0 \end{pmatrix}^2.$$

Exercise 3.17. Show that if A, B, C are matrices such that the sum $B + C$, and the products $AB, AC, A(B + C)$ are defined, then

$$A(B + C) = AB + AC.$$

Exercise 3.18. Show that if A, B, C are matrices such that the sum $B+C$, and the products $AB, AC, A(B+C)$ are defined, then

$$(A+B)C = AC + BC.$$

§3.2.4 Few to many more!

Exercise

Suppose A is a 3×2 matrix, and

$$A\begin{pmatrix}1\\4\end{pmatrix} = \begin{pmatrix}2\\8\\14\end{pmatrix},$$

$$A\begin{pmatrix}3\\7\end{pmatrix} = \begin{pmatrix}-4\\8\\20\end{pmatrix}.$$

Determine

$$A\begin{pmatrix}103\\407\end{pmatrix}, A\begin{pmatrix}97\\393\end{pmatrix}.$$

Solution. Observe that

$$\begin{pmatrix}103\\407\end{pmatrix} = 100\begin{pmatrix}1\\4\end{pmatrix} + \begin{pmatrix}3\\7\end{pmatrix}.$$

This yields

$$\begin{aligned}
A\begin{pmatrix}103\\407\end{pmatrix} &= A\left(100\begin{pmatrix}1\\4\end{pmatrix} + \begin{pmatrix}3\\7\end{pmatrix}\right)\\
&= 100A\begin{pmatrix}1\\4\end{pmatrix} + A\begin{pmatrix}3\\7\end{pmatrix}\\
&= 100\begin{pmatrix}2\\8\\14\end{pmatrix} + \begin{pmatrix}-4\\8\\20\end{pmatrix}\\
&= \begin{pmatrix}200-4\\800+8\\1400+20\end{pmatrix}\\
&= \begin{pmatrix}196\\808\\1420\end{pmatrix}.
\end{aligned}$$

Also note that

$$\begin{pmatrix}97\\393\end{pmatrix} = 100\begin{pmatrix}1\\4\end{pmatrix} - \begin{pmatrix}3\\7\end{pmatrix}.$$

Using this we obtain

$$\begin{aligned}
A\begin{pmatrix} 97 \\ 393 \end{pmatrix} &= A\left(100\begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 7 \end{pmatrix}\right) \\
&= 100A\begin{pmatrix} 1 \\ 4 \end{pmatrix} - A\begin{pmatrix} 3 \\ 7 \end{pmatrix} \\
&= 100\begin{pmatrix} 2 \\ 8 \\ 14 \end{pmatrix} - \begin{pmatrix} -4 \\ 8 \\ 20 \end{pmatrix} \\
&= \begin{pmatrix} 200 + 4 \\ 800 - 8 \\ 1400 - 20 \end{pmatrix} \\
&= \begin{pmatrix} 204 \\ 792 \\ 1380 \end{pmatrix}.
\end{aligned}$$

□

Remark

The above exercise indicates that if we know how a matrix acts on a bunch of vectors, then we can determine how it acts on any vector that can be expressed as a *linear combination* of those vectors.

A **linear combination** of vectors $v_1, v_2, ..., v_k$ is a vector of the form

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k,$$

where $c_1, c_2, ..., c_k$ are real numbers[3].

Note that any vector in $\mathbb{R}^2$ is a linear combination of the vectors

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

So, if A is a 3×2 matrix and we know how A acts on the vectors

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

that is, we know the products

$$A\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad A\begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

then we can determine how A acts on any vector in $\mathbb{R}^2$. Indeed, for any $\begin{pmatrix} x \\ y \end{pmatrix}$ in $\mathbb{R}^2$, we have

$$\begin{pmatrix} x \\ y \end{pmatrix} = x\begin{pmatrix} 1 \\ 0 \end{pmatrix} + y\begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

and hence we obtain

$$\begin{aligned}
A\begin{pmatrix} x \\ y \end{pmatrix} &= A\left(x\begin{pmatrix} 1 \\ 0 \end{pmatrix} + y\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) \\
&= xA\begin{pmatrix} 1 \\ 0 \end{pmatrix} + yA\begin{pmatrix} 0 \\ 1 \end{pmatrix}.
\end{aligned}$$

[3]One can also allow the c_i's to be complex numbers, depending on the situation.

Exercise

Show that every vector in $\mathbb{R}^2$ can be expressed as a linear combination of the vectors

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix}.$$

Solution. Note that for any $\begin{pmatrix} x \\ y \end{pmatrix}$ in $\mathbb{R}^2$, we have

$$\begin{pmatrix} x \\ y \end{pmatrix} = x\begin{pmatrix} 1 \\ 0 \end{pmatrix} + y\begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

that is, $\begin{pmatrix} x \\ y \end{pmatrix}$ is a linear combination of the vectors

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

So, it suffices[4] to show that

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

can be expressed as linear combinations of the vectors

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix}.$$

Note that

$$\begin{aligned} 7\begin{pmatrix} 1 \\ 4 \end{pmatrix} - 4\begin{pmatrix} 3 \\ 7 \end{pmatrix} &= \begin{pmatrix} -5 \\ 0 \end{pmatrix}, \\ 3\begin{pmatrix} 1 \\ 4 \end{pmatrix} - 1\begin{pmatrix} 3 \\ 7 \end{pmatrix} &= \begin{pmatrix} 0 \\ 5 \end{pmatrix}, \end{aligned}$$

which yields

$$\begin{aligned} \begin{pmatrix} 1 \\ 0 \end{pmatrix} &= -\frac{7}{5}\begin{pmatrix} 1 \\ 4 \end{pmatrix} + \frac{4}{5}\begin{pmatrix} 3 \\ 7 \end{pmatrix}, \\ \begin{pmatrix} 0 \\ 1 \end{pmatrix} &= \frac{3}{5}\begin{pmatrix} 1 \\ 4 \end{pmatrix} - \frac{1}{5}\begin{pmatrix} 3 \\ 7 \end{pmatrix}. \end{aligned}$$

□

Remark

A careful reading of the above exercises and the remark indicates that if there are a few vectors $v_1, v_2, \ldots, v_k$ in $\mathbb{R}^n$ such that every vector in $\mathbb{R}^n$ can be expressed as a linear combination of those vectors, then knowing how a matrix A acts on the vectors $v_1, v_2, \ldots, v_k$ is sufficient to determine how A acts on any vector in $\mathbb{R}^n$.

[4] How does it suffice?

Example

Note that

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix},$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 8 \end{pmatrix},$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix}$$

hold. The vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

are called the **standard basis vectors** of $\mathbb{R}^3$. The standard basis vectors of $\mathbb{R}^2$ are

$$e_1 := \begin{pmatrix} 1 \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

In general, the vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

lying in $\mathbb{R}^n$ are called the **standard basis vectors** of $\mathbb{R}^n$.

Example

Note that

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix},$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

hold. The vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

are called the **standard basis vectors** of $\mathbb{R}^2$.

Exercise 3.19. If A is an $m \times n$ matrix, then show that for all $i = 1, 2, \dots, n$, the product Ae_i is equal to the i-th column of A, where e_i denotes the i-th standard basis vector of $\mathbb{R}^n$, that is,

$$e_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

with 1 in the i-th position, and 0's elsewhere.

Remark

Exercise 3.19 shows that the columns of a matrix A can be obtained by multiplying A with the standard basis vectors. Further, if A is an $m \times n$ matrix, and we know how A acts on the standard basis vectors of $\mathbb{R}^n$, then we can determine all the entries of A, and hence we can determine how A acts on any vector v in $\mathbb{R}^n$. Note that this is achieved by determining the entries of A.

Moreover, the first remark in Section 3.2.4 indicates that if we know how A acts on the standard basis vectors of $\mathbb{R}^n$, then using that every element of $\mathbb{R}^n$ can be expressed as a linear combination of the standard basis vectors, we can determine how A acts on any vector v in $\mathbb{R}^n$.

Furthermore, if $v_1, v_2, \ldots, v_k$ are elements of $\mathbb{R}^n$, then knowing how a matrix A acts on $v_1, v_2, \ldots, v_k$ is sufficient to determine how A acts on any vector which can be expressed as a linear combination of $v_1, \ldots, v_k$.

§3.3 System of linear equations

Exercise

Solve the system of equations in the variables x, y:

$$\begin{aligned} 2x + 3y &= 8, \\ 5x + 7y &= 19. \end{aligned}$$

Solution. Multiplying the first equation by 7, and the second equation by 3, and taking the difference of the equations obtained, we get

$$7 \cdot 2x - 3 \cdot 5x = 7 \cdot 8 - 3 \cdot 19,$$

which yields

$$x = \frac{7 \cdot 8 - 3 \cdot 19}{7 \cdot 2 - 3 \cdot 5}.$$

Similarly, multiplying the first equation by 5, and the second equation by 2, and taking the difference of the equations obtained, we get

$$5 \cdot 3y - 2 \cdot 7y = 5 \cdot 8 - 2 \cdot 19,$$

which yields

$$y = \frac{5 \cdot 8 - 2 \cdot 19}{5 \cdot 3 - 2 \cdot 7}.$$

□

Alternate solution. Note that the given system of equations can be rewritten as

$$\begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 8 \\ 19 \end{pmatrix}, \tag{2}$$

which can be thought as a concise way of putting the information

$$\begin{pmatrix} 2x+3y \\ 5x+7y \end{pmatrix} = \begin{pmatrix} 8 \\ 19 \end{pmatrix}. \tag{3}$$

Recall that in the previous solution,

- to solve for x, the first equation was multiplied by 7, and the second equation by 3, and next, we considered the difference of the equations obtained,
- to solve for y, the first equation was multiplied by 5, and the second equation was multiplied by 2, and then, we considered the difference of the equations obtained.

The two steps can be put together in the matrix form as

$$\begin{pmatrix} 7 & -3 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} 2x+3y \\ 5x+7y \end{pmatrix}.$$

Using Equation 3, it yields

$$\begin{pmatrix} 7 & -3 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} 2x+3y \\ 5x+7y \end{pmatrix} = \begin{pmatrix} 7 & -3 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} 8 \\ 19 \end{pmatrix}.$$

The advantage of this process is that, first y gets eliminated in the first step and helps to find x, and in the next step, x gets eliminated and leads to finding y. Stated in matrix form, we have that

$$\begin{pmatrix} 7 & -3 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} 2x+3y \\ 5x+7y \end{pmatrix} = \begin{pmatrix} (7 \cdot 2 - 3 \cdot 5)x \\ (5 \cdot 3 - 2 \cdot 7)y \end{pmatrix}.$$

We could have also written

$$\begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 2x+3y \\ 5x+7y \end{pmatrix} = \begin{pmatrix} (7 \cdot 2 - 3 \cdot 5)x \\ (-5 \cdot 3 + 2 \cdot 7)y \end{pmatrix}.$$

Thus, rewriting the given sysem of equations in the matrix form as in Equation 3, and multiplying it from the left by the matrix

$$\begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix},$$

we obtain

$$\begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 2x+3y \\ 5x+7y \end{pmatrix} = \begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 8 \\ 19 \end{pmatrix},$$

which gives

$$\begin{pmatrix} (7 \cdot 2 - 3 \cdot 5)x \\ (-5 \cdot 3 + 2 \cdot 7)y \end{pmatrix} = \begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 8 \\ 19 \end{pmatrix},$$

that is, we have

$$(7 \cdot 2 - 3 \cdot 5) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 8 \\ 19 \end{pmatrix},$$

and hence, the solution to the given system of equations is given by

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{7 \cdot 2 - 3 \cdot 5} \begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 8 \\ 19 \end{pmatrix}.$$

□

Summarizing the above. The given system of equations can be rewritten as

$$\begin{pmatrix} 2x+3y \\ 5x+7y \end{pmatrix} = \begin{pmatrix} 8 \\ 19 \end{pmatrix}, \tag{4}$$

or equivalently,

$$\begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 8 \\ 19 \end{pmatrix}, \tag{5}$$

Multiplying the above from the left by the matrix

$$\frac{1}{7\cdot 2-3\cdot 5}\begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix}, \tag{6}$$

we obtain

$$\frac{1}{7\cdot 2-3\cdot 5}\begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix}\begin{pmatrix} 2x+3y \\ 5x+7y \end{pmatrix} = \frac{1}{7\cdot 2-3\cdot 5}\begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix}\begin{pmatrix} 8 \\ 19 \end{pmatrix},$$

which shows that the solution to the given system of equations is given by

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{7\cdot 2-3\cdot 5}\begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix}\begin{pmatrix} 8 \\ 19 \end{pmatrix}.$$

□

Remark

In the above, Equation 5 was multiplied by the matrix as in Equation 6, since multiplying the matrix

$$\begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$$

as in Equation 5 **from the left** by the matrix

$$\frac{1}{7\cdot 2-3\cdot 5}\begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix},$$

as in Equation 6 yields

$$\begin{aligned}\frac{1}{7\cdot 2-3\cdot 5}\begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix} &= \frac{1}{7\cdot 2-3\cdot 5}\begin{pmatrix} 7\cdot 2-3\cdot 5 & 0 \\ 0 & -5\cdot 3+2\cdot 7 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},\end{aligned}$$

which is the 2×2 identity matrix.

§3.4 Invertible matrices

Definition

Let A be an $n\times n$ matrix. If there is an $n\times n$ matrix B such that

$$AB = BA = I_n,$$

then A is called **invertible** and the matrix B is called an **inverse** of A.

Fact 3.20. Show that if B, C are inverses of an $n\times n$ matrix A, then $B = C$.

A proof of the above is provided in Chapter 8.

Remark

If A is an $n\times n$ matrix, and A admits an inverse, then it is called **the** inverse of A, and is denoted by A^{-1}.

Exercise 3.21. Determine whether the identity matrix I_n is invertible, and if it is, find its inverse.

Exercise 3.22. If A is an invertible square matrix, then show that A^{-1} is also invertible, and that
$$(A^{-1})^{-1} = A.$$

Solution. Suppose A is an invertible square matrix. We have
$$AA^{-1} = A^{-1}A = I.$$
This shows that A^{-1} is also invertible, and that
$$(A^{-1})^{-1} = A.$$
□

Exercise 3.23. Let A, B be invertible $n \times n$ matrices. Show that AB is invertible, and that
$$(AB)^{-1} = B^{-1}A^{-1}.$$

Compare the above exercise with Exercise 3.14.

Solution. Note that
$$\begin{aligned}(AB)(B^{-1}A^{-1}) &= A(BB^{-1})A^{-1} \\ &= AI_nA^{-1} \\ &= AA^{-1} \\ &= I_n,\end{aligned}$$
and
$$\begin{aligned}(B^{-1}A^{-1})(AB) &= B^{-1}(A^{-1}A)B \\ &= B^{-1}I_nB \\ &= B^{-1}B \\ &= I_n.\end{aligned}$$
This shows that AB is invertible, and that
$$(AB)^{-1} = B^{-1}A^{-1}.$$
□

Lemma 3.24. *Let*
$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$
be a 2×2 matrix. The following statements are equivalent.

- *A is invertible,*
- *$ad - bc$ is nonzero,*
- *every vector $\begin{pmatrix} e \\ f \end{pmatrix}$ can be expressed as a linear combination of the columns of A.*

Moreover, if A is invertible, then its inverse is equal to
$$\frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Proof. Note that
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix},$$
$$\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix}$$
hold[5].

Assume that if A is invertible. It follows that

$$
\begin{aligned}
(ad-bc)A^{-1} &= \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix} A^{-1} \\
&= \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} A^{-1} \\
&= \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} AA^{-1} \\
&= \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} I_2 \\
&= \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.
\end{aligned}
$$

Note that if $ad-bc=0$, then a, b, c, d are equal to 0, which implies $A=0$, which is impossible since $AA^{-1}=I_2$. Hence, $ad-bc$ is nonzero, and

$$A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Conversely, note that if $ad-bc \neq 0$, then it follows that the matrix

$$\frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

is the inverse[6] of A.

Also note that if every vector $\begin{pmatrix} e \\ f \end{pmatrix}$ can be expressed as a linear combination of the columns of A, then in particular, the vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ can be expressed as a linear combination of the columns of A. This implies that there are real numbers x_1, y_1, x_2, y_2 such that

$$A \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

and

$$A \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

This shows that

$$A \begin{pmatrix} x_1 & x_2 \\ y_1 & y_2 \end{pmatrix} = I_2,$$

implying that A is invertible[7]. Further, if $ad-bc$ is nonzero, then as observed above, it follows that

$$
\begin{aligned}
\frac{d}{ad-bc} \begin{pmatrix} a \\ c \end{pmatrix} + \frac{-c}{ad-bc} \begin{pmatrix} b \\ d \end{pmatrix} &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \\
\frac{-b}{ad-bc} \begin{pmatrix} a \\ c \end{pmatrix} + \frac{a}{ad-bc} \begin{pmatrix} b \\ d \end{pmatrix} &= \begin{pmatrix} 0 \\ 1 \end{pmatrix}
\end{aligned}
$$

hold, which shows that every vector $\begin{pmatrix} e \\ f \end{pmatrix}$ can be expressed as a linear combination of the columns of A. $\square$

[5] This is an instance of Cramer's rule.
[6] Why?
[7] How?

Definition

If $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is a 2×2 matrix, then its **determinant** is denoted by $\det A$, and is defined as

$$\det A := ad - bc.$$

Remark

The determinant of a 2×2 matrix can be used to determine whether the matrix is invertible or not. Indeed, by Lemma 3.24, the following statements are equivalent.

- The matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is invertible.
- The determinant $\det\left(\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)\right)$ is nonzero.

Moreover, if $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is invertible, then its inverse is given by

$$\frac{1}{\det\left(\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)\right)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Exercise 3.25. If A, B are 2×2 matrices, then show that

$$\det(AB) = \det(A)\det(B).$$

Solution. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

and

$$B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$$

be 2×2 matrices. Note that

$$AB = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}.$$

Using the definition of determinant, we obtain

$$\begin{aligned} \det(AB) &= (ae + bg)(cf + dh) - (af + bh)(ce + dg) \\ &= acef + adeh + bcfg + bdgh - acef - adfg - bceh - bdgh \\ &= adeh - adfg + bcfg - bceh \\ &= (ad - bc)(eh - fg) \\ &= \det(A)\det(B). \end{aligned}$$

□

Exercise 3.26. Solve the system of equations in the variables x, y:

$$\begin{aligned} 5x + 2y &= 11, \\ 3x + 4y &= 8. \end{aligned}$$

Solution. Note that the given system of equations can be rewritten as

$$\begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 11 \\ 8 \end{pmatrix}, \tag{7}$$

Note that

$$\det\left(\begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix}\right) = 5 \cdot 4 - 2 \cdot 3 = 14,$$

which is nonzero. Hence, the matrix $\begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix}$ is invertible, and

$$\begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix}^{-1} = \frac{1}{14}\begin{pmatrix} 4 & -2 \\ -3 & 5 \end{pmatrix}.$$

Multiplying Equation 7 from the left by the inverse of the matrix $\begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix}$, we obtain

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{14}\begin{pmatrix} 4 & -2 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} 11 \\ 8 \end{pmatrix},$$

which yields

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{14}\begin{pmatrix} 44 - 16 \\ -33 + 40 \end{pmatrix} = \begin{pmatrix} 2 \\ \frac{1}{2} \end{pmatrix}.$$

□

Exercise 3.27. Solve the system of equations in the variables x, y:

$$12x - 25y = -47,$$
$$-7x + 30y = 51.$$

Solution. The given system of equations can be put in the matrix form as

$$\begin{pmatrix} 12 & -25 \\ -7 & 30 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -47 \\ 51 \end{pmatrix}, \tag{8}$$

Note that

$$\det\left(\begin{pmatrix} 12 & -25 \\ -7 & 30 \end{pmatrix}\right) = 12 \cdot 30 - (-25) \cdot (-7) = 360 - 175 = 185,$$

which is nonzero. It follows that the matrix $\begin{pmatrix} 12 & -25 \\ -7 & 30 \end{pmatrix}$ is invertible, and

$$\begin{pmatrix} 12 & -25 \\ -7 & 30 \end{pmatrix}^{-1} = \frac{1}{185}\begin{pmatrix} 30 & 25 \\ 7 & 12 \end{pmatrix}.$$

Multiplying Equation 8 from the left by the inverse of the matrix $\begin{pmatrix} 12 & -25 \\ -7 & 30 \end{pmatrix}$, we obtain

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{185}\begin{pmatrix} 30 & 25 \\ 7 & 12 \end{pmatrix}\begin{pmatrix} -47 \\ 51 \end{pmatrix},$$

which yields

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{185}\begin{pmatrix} -1410 + 1275 \\ -329 + 612 \end{pmatrix} = \begin{pmatrix} -\frac{135}{185} \\ \frac{283}{185} \end{pmatrix} = \begin{pmatrix} -\frac{27}{37} \\ \frac{283}{185} \end{pmatrix}.$$

□

Exercise 3.28. Solve the system of equations in the variables x, y, z:

$$2x + 3y + z = 1,$$
$$4x + y + 2z = 2,$$
$$3x + 2y + 3z = 3.$$

Solution. The given system of equations can be rewritten as

$$A\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \tag{9}$$

where

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 4 & 1 & 2 \\ 3 & 2 & 3 \end{pmatrix}.$$

Note that A^{-1} is equal to[8]

$$\frac{1}{15}\begin{pmatrix} 1 & 7 & -5 \\ 6 & -3 & 0 \\ -5 & -5 & 10 \end{pmatrix}.$$

Indeed, observe that

$$\begin{aligned} A \times \frac{1}{15}\begin{pmatrix} 1 & 7 & -5 \\ 6 & -3 & 0 \\ -5 & -5 & 10 \end{pmatrix} &= \frac{1}{15}\begin{pmatrix} 2 & 3 & 1 \\ 4 & 1 & 2 \\ 3 & 2 & 3 \end{pmatrix}\begin{pmatrix} 1 & 7 & -5 \\ 6 & -3 & 0 \\ -5 & -5 & 10 \end{pmatrix} \\ &= \frac{1}{15}\begin{pmatrix} 15 & 0 & 0 \\ 0 & 15 & 0 \\ 0 & 0 & 15 \end{pmatrix} \\ &= I_3, \end{aligned}$$

and

$$\begin{aligned} \frac{1}{15}\begin{pmatrix} 1 & 7 & -5 \\ 6 & -3 & 0 \\ -5 & -5 & 10 \end{pmatrix} \times A &= \frac{1}{15}\begin{pmatrix} 1 & 7 & -5 \\ 6 & -3 & 0 \\ -5 & -5 & 10 \end{pmatrix}\begin{pmatrix} 2 & 3 & 1 \\ 4 & 1 & 2 \\ 3 & 2 & 3 \end{pmatrix} \\ &= \frac{1}{15}\begin{pmatrix} 15 & 0 & 0 \\ 0 & 15 & 0 \\ 0 & 0 & 15 \end{pmatrix} \\ &= I_3. \end{aligned}$$

Multiplying Equation 9 from the left by A^{-1}, we obtain

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1}\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix},$$

which yields

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{15}\begin{pmatrix} 1 & 7 & -5 \\ 6 & -3 & 0 \\ -5 & -5 & 10 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \frac{1}{15}\begin{pmatrix} 1+14-15 \\ 6-6+0 \\ -5-10+30 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

□

Exercise 3.29. Let A, B be 2×2 matrices. Show that if AB is invertible, then both A and B are invertible.

Compare the above with Exercise 3.74, Exercise 3.75, Exercise 3.92.

Exercise 3.30. Let A be an $n \times n$ matrix. If P is an invertible $n \times n$ matrix, then show that

$$\left(PAP^{-1}\right)^k = PA^kP^{-1}$$

holds for any positive integer k.

[8]One may wonder how to find the inverse of an invertible 3×3 matrix. One way is to use the Gaussian elimination method, to be discussed in Section 3.7. Can one also have a formula for the inverse of an invertible 3×3 matrix, similar to the formula for the inverse of a 2×2 matrix as in Lemma 3.24? See Lemma 3.104 for the details.

Solution. We prove the statement by induction. Let P be an invertible $n \times n$ matrix. For a positive integer k, let $P(k)$ denote the statement

$$(PAP^{-1})^k = PA^kP^{-1}.$$

Note that $P(1)$ holds. Assume that $P(k)$ holds for some positive integer k. Then, using the induction hypothesis, we obtain

$$\begin{aligned}(PAP^{-1})^{k+1} &= (PAP^{-1})^k(PAP^{-1}) \\ &= (PA^kP^{-1})(PAP^{-1}) \\ &= PA^k(P^{-1}P)AP^{-1} \\ &= PA^kI_nAP^{-1} \\ &= PA^{k+1}P^{-1}.\end{aligned}$$

This shows that $P(k+1)$ holds. Hence, by the principle of mathematical induction, $P(k)$ holds for all positive integers k. □

§3.5 Systems of linear equations again

§3.5.1 Linear combination of the columns of a matrix

Here are further examples of systems of linear equations.

System of linear equations	Matrix form	Solution(s)	Associated matrix
$x + 2y = 1,$ $3x - 5y = -7$	$\begin{pmatrix} 1 & 2 \\ 3 & -5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -7 \end{pmatrix}$	Unique	Invertible
$x + 2y = 1,$ $2x + 4y = 2$	$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$	Infinitely many[9]	Not invertible
$x + 2y = 1,$ $3x + 6y = 2$	$\begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$	None	Not invertible

? Question

Which of such systems of equations can be solved?

[9] Why?

Remark

Consider the following system of linear equations.

$$12345x + 67890y = 11111,$$
$$54321x + 9876y = 22222.$$

To solve it, we need to find a vector $\begin{pmatrix} x \\ y \end{pmatrix}$ in $\mathbb{R}^2$ satisfying

$$\begin{pmatrix} 12345 & 67890 \\ 54321 & 9876 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 11111 \\ 22222 \end{pmatrix}.$$

In other words, we need to find real numbers x, y such that

$$\begin{aligned}\begin{pmatrix} 12345 & 67890 \\ 54321 & 9876 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 12345 & 67890 \\ 54321 & 9876 \end{pmatrix}(xe_1 + ye_2) \\ &= x\begin{pmatrix} 12345 & 67890 \\ 54321 & 9876 \end{pmatrix}e_1 + y\begin{pmatrix} 12345 & 67890 \\ 54321 & 9876 \end{pmatrix}e_2 \\ &= x\begin{pmatrix} 12345 \\ 54321 \end{pmatrix} + y\begin{pmatrix} 67890 \\ 9876 \end{pmatrix}\end{aligned}$$

is equal to $\begin{pmatrix} 11111 \\ 22222 \end{pmatrix}$, where e_1, e_2 are the standard basis vectors of $\mathbb{R}^2$.

It is left to determining whether some linear combination of the columns of the matrix $\begin{pmatrix} 12345 & 67890 \\ 54321 & 9876 \end{pmatrix}$ is equal to $\begin{pmatrix} 11111 \\ 22222 \end{pmatrix}$. If such a linear combination exists, then the system of equations has a solution, and if such a linear combination does not exist, then the system of equations does not have a solution. If there is more than one such linear combination, then the system of equations has more than one solution, etc.

Remark

The above discussion shows that solving a system of linear equations is equivalent to determining whether **some linear combination** of the columns of a matrix is equal to the given vector. This is the reason why the notions[10] of *column space* and *linear span of vectors* are important in the study of systems of linear equations.

Remark

Consider a matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, and a vector $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ where $a, b, c, d, \alpha, \beta$ are real numbers. If $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is **not** a linear combination of the columns of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, that is, if $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is **not** a linear combination of the vectors $\begin{pmatrix} a \\ c \end{pmatrix}$ and $\begin{pmatrix} b \\ d \end{pmatrix}$, then the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

admits **no** solution.

Lemma 3.31. *Let A be a 2×2 matrix, and let $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ be a vector of $\mathbb{R}^2$. Show that the following statements are equivalent.*

- *The vector $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is a linear combination of the columns of the matrix A.*
- *The system of equations*

[10] The notions of column space and linear span of vectors will be introduced formally at a later stage.

$$A\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

admits at least one solution.

Proof. Assume that $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is a linear combination of the columns of the matrix A. This shows that there are real numbers x_0, y_0 such that

$$A\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Hence, the system of equations

$$A\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

admits at least one solution.

Conversely, assume that the system of equations

$$A\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

admits at least one solution, say $\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$. It follows that

$$A\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

This shows that $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is a linear combination of the columns of the matrix A. □

§3.5.2 Homogeneous system of linear equations

Let us first consider the solutions to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{10}$$

Note that $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is a linear combination of the columns of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, since

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0\begin{pmatrix} a \\ c \end{pmatrix} + 0\begin{pmatrix} b \\ d \end{pmatrix}.$$

This shows that $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is a solution to Equation 10. It is called the **trivial solution** to Equation 10.

Case 1

Let us first consider the case that the matrix A **is** invertible.

Note that under this hypothesis, Equation 10 has no solution other than $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$. Indeed, if $\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ is another solution to Equation 10, then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This shows that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1}\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

which yields

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Hence, if A is invertible, then the only solution to Equation 10 is $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Observation

If Equation 10 has at least one solution other than $\binom{0}{0}$, then Equation 10 has infinitely many solutions. Indeed, if $\binom{x_2}{y_2}$ is a solution to Equation 10 other than $\binom{0}{0}$, then for any real number t, the vector $t\binom{x_2}{y_2}$ is also a solution to Equation 10, and hence, Equation 10 has infinitely many solutions.

Case 2

Now let us consider the case that the matrix A is **not** invertible. By Lemma 3.24, we have $ad - bc = 0$, this implies that

$$d\begin{pmatrix} a \\ c \end{pmatrix} - c\begin{pmatrix} b \\ d \end{pmatrix} = \begin{pmatrix} ad - bc \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$
$$b\begin{pmatrix} a \\ c \end{pmatrix} - a\begin{pmatrix} b \\ d \end{pmatrix} = \begin{pmatrix} 0 \\ bc - ad \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

and consequently,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} d \\ -c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} b \\ -a \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This shows that if A is **not** invertible, then the vectors $\binom{d}{-c}$ and $\binom{b}{-a}$ are solutions to Equation 10.

Subcase 2a

If the vectors $\binom{d}{-c}$ and $\binom{b}{-a}$ are equal to the zero vector, equivalently, if a, b, c, d are all equal to 0, then any vector $\binom{x}{y}$ of $\mathbb{R}^2$ is a solution to Equation 10.

Subcase 2b

If not both of the vectors $\binom{d}{-c}$ and $\binom{b}{-a}$ are equal to the zero vector (equivalently, at least one of them is nonzero) and A is not invertible, then Equation 10 admits a solution other than $\binom{0}{0}$, and moreover, not all vectors of $\mathbb{R}^2$ are solutions to Equation 10 since not all of $\binom{1}{0}$, $\binom{0}{1}$ are solutions to Equation 10.

Lemma 3.32. *Consider the system of linear equations*

$$A\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \tag{11}$$

where A is a 2×2 matrix. The following statements are equivalent.

- *The matrix A is invertible.*
- *The system of equations as in Equation 11 admits a unique solution, that is, it admits no solution other than the trivial solution $\binom{0}{0}$.*
- *The trivial linear combination (that is, the linear combination using zeroes as the coefficients) of the columns of the matrix A is the only linear combination of the columns of A that is equal to $\binom{0}{0}$.*

Furthermore, the following statements are equivalent.

- *Any element of* $\mathbb{R}^2$ *is a solution to* Equation 11.
- *The matrix* A *is the zero matrix.*

Moreover, the following statements are equivalent too.

- *The system of equations as in* Equation 11 *admits a solution other than the trivial solution* $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, *and not all elements of* $\mathbb{R}^2$ *are solutions to* Equation 11.
- *The matrix* A *is not invertible and nonzero.*

A proof of the above is provided in Chapter 8.

Exercise 3.33. Let A be a 2×2 matrix. Let V be the set of solutions to the system of equations

$$A\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

that is,

$$V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : A\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}.$$

Show that for any $u, v \in V$ and $c \in \mathbb{R}$, the element $u + v$ of $\mathbb{R}^2$ lies in V, and the element cu also lies in V.

Solution. Let u, v be elements of V, and let c be a real number. Note that

$$A(u + v) = Au + Av = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

and

$$A(cu) = c(Au) = c\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This shows that both $u + v$ and cu lie in V. □

Remark

What about the difference of two elements of V? If u, v are elements of V and c, d are real numbers, then what can be said about the element $cu + dv$ of $\mathbb{R}^2$?

Exercise 3.34. Let A be a 2×2 matrix. Let V be the set of vectors which can be expressed as a linear combination of the columns of A, that is[11],

$$V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : \begin{pmatrix} x \\ y \end{pmatrix} = A\begin{pmatrix} s \\ t \end{pmatrix} \text{ for some } s, t \in \mathbb{R} \right\}.$$

Show that for any $u, v \in V$ and $c \in \mathbb{R}$, the element $u + v$ of $\mathbb{R}^2$ lies in V, and the element cu also lies in V.

Solution. Let u, v be elements of V, and let c be a real number. By the definition of V, there exist real numbers s_1, t_1, s_2, t_2 such that

[11]Observe that

$$A\begin{pmatrix} s \\ t \end{pmatrix} = sC_1 + tC_2,$$

where C_1, C_2 denote the first and second columns of A respectively.

$$u = A\begin{pmatrix} s_1 \\ t_1 \end{pmatrix},$$

$$v = A\begin{pmatrix} s_2 \\ t_2 \end{pmatrix}.$$

Note that

$$u + v = A\begin{pmatrix} s_1 \\ t_1 \end{pmatrix} + A\begin{pmatrix} s_2 \\ t_2 \end{pmatrix} = A\left(\begin{pmatrix} s_1 \\ t_1 \end{pmatrix} + \begin{pmatrix} s_2 \\ t_2 \end{pmatrix}\right) = A\begin{pmatrix} s_1 + s_2 \\ t_1 + t_2 \end{pmatrix},$$

and

$$cu = c\left(A\begin{pmatrix} s_1 \\ t_1 \end{pmatrix}\right) = A\left(c\begin{pmatrix} s_1 \\ t_1 \end{pmatrix}\right) = A\begin{pmatrix} cs_1 \\ ct_1 \end{pmatrix}.$$

This shows that both $u + v$ and cu lie in V. □

Exercise 3.35. Let A be an $m \times n$ matrix. Let V be the set of solutions to the system of equations

$$Au = 0,$$

that is,

$$V = \{v \in \mathbb{R}^n : Av = 0\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $\mathbb{R}^n$ lies in V.

Solution. Let u, v be elements of V, and let c, d be real numbers. Note that

$$A(cu + dv) = c(Au) + d(Av) = c \cdot 0 + d \cdot 0 = 0.$$

This shows that $cu + dv$ lies in V. □

Compare the following exercise with Exercise 3.65.

Exercise 3.36. Let A be an $m \times n$ matrix. Let V be the set of vectors which can be expressed as a linear combination of the columns of A, that is,

$$V = \{v \in \mathbb{R}^m : v = Au \text{ for some } u \in \mathbb{R}^n\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $\mathbb{R}^m$ lies in V.

Solution. Let u, v be elements of V, and let c, d be real numbers. By the definition of V, there exist vectors $u_1, u_2 \in \mathbb{R}^n$ such that

$$u = Au_1,$$

$$v = Au_2.$$

Note that

$$cu + dv = c(Au_1) + d(Au_2) = A(cu_1) + A(du_2) = A(cu_1 + du_2).$$

This shows that $cu + dv$ lies in V. □

§3.5.3 General system of linear equations

If $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is a linear combination of the columns of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then there exist real numbers x_0, y_0 such that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

This shows that the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

has at least one solution, namely $\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$. If $\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ is another solution to the above system of equations, then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

This shows that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}.$$

Hence,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} - \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

holds. This shows that the difference of any two solutions to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

is a solution to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Conversely, if $\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ is a solution to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

then

$$\begin{aligned} \begin{pmatrix} a & b \\ c & d \end{pmatrix}\left(\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}\right) &= \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \\ &= \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \end{aligned}$$

This shows that if $\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ is a solution to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

then $\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ is also a solution to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

In summary, if $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is a linear combination of the columns of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then the solutions to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

are in one-to-one correspondence with the solutions to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Remark

The above discussion shows that if $\begin{pmatrix}\alpha\\\beta\end{pmatrix}$ is a linear combination of the columns of $\begin{pmatrix}a&b\\c&d\end{pmatrix}$, then the solutions to the system of equations

$$\begin{pmatrix}a&b\\c&d\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\alpha\\\beta\end{pmatrix}$$

are in one-to-one correspondence with the solutions to the system of equations

$$\begin{pmatrix}a&b\\c&d\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}.$$

Hence, to determine the number of solutions to the system of equations

$$\begin{pmatrix}a&b\\c&d\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\alpha\\\beta\end{pmatrix}$$

it suffices to determine the number of solutions to the system of equations

$$\begin{pmatrix}a&b\\c&d\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}.$$

Lemma 3.37. *Let A be a 2×2 matrix, and let $\begin{pmatrix}\alpha\\\beta\end{pmatrix}$ be a vector of $\mathbb{R}^2$. Assume that $\begin{pmatrix}\alpha\\\beta\end{pmatrix}$ is a linear combination of the columns of the matrix A. Then the solutions to the system of equations*

$$A\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\alpha\\\beta\end{pmatrix}$$

are in one-to-one correspondence with the solutions to the system of equations

$$A\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}.$$

More specifically, for any fixed solution $\begin{pmatrix}x_0\\y_0\end{pmatrix}$ to the system of equations

$$A\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\alpha\\\beta\end{pmatrix},$$

the map

$$\begin{pmatrix}x\\y\end{pmatrix}\mapsto\begin{pmatrix}x\\y\end{pmatrix}-\begin{pmatrix}x_0\\y_0\end{pmatrix}$$

is a bijection from the set of solutions to the system of equations

$$A\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\alpha\\\beta\end{pmatrix}$$

to the set of solutions to the system of equations

$$A\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix},$$

having the following map as its inverse:

$$\begin{pmatrix}x\\y\end{pmatrix}\mapsto\begin{pmatrix}x\\y\end{pmatrix}+\begin{pmatrix}x_0\\y_0\end{pmatrix}.$$

Proof. Deferred. □

Remark

The terms “one-to-one correspondence” and “bijection” will be introduced formally at a later stage. The proof of the above lemma is deferred until then.

§3.6 Special types of matrices

§3.6.1 Diagonal, scalar, and triangular matrices

Recall that a matrix is called a **square** matrix if it has the same number of rows and columns.

Definition

A square matrix A is called **diagonal** if all its non-diagonal entries are equal to zero.

Example

The following matrices are diagonal.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \begin{pmatrix} 5 & 0 \\ 0 & -7 \end{pmatrix}, \begin{pmatrix} i & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \pi \end{pmatrix}.$$

Some non-diagonal matrices are as follows.

$$\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Definition

A square matrix A is called **scalar** if it is a diagonal matrix and all its diagonal entries are equal. In other words, an $n \times n$ matrix A is called **scalar** if

$$A = \lambda I_n$$

for some scalar λ.

Example

The following matrices are scalar.

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Some non-scalar matrices are as follows.

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 5 & 0 & 0 \\ 0 & -7 & 0 \\ 0 & 0 & \pi \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Definition

A square matrix A is called **triangular** if all its entries below the main diagonal are equal to zero, or if all its entries above the main diagonal are equal to zero.

A triangular matrix A is called **upper-triangular** if all its entries below the main diagonal are equal to zero.

A triangular matrix A is called **lower-triangular** if all its entries above the main diagonal are equal to zero.

Example

The following matrices are upper-triangular.

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}, \begin{pmatrix} 7 & 8 \\ 0 & 9 \end{pmatrix}.$$

The following matrices are lower-triangular.

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{pmatrix}, \begin{pmatrix} 7 & 0 \\ 8 & 9 \end{pmatrix}.$$

Some non-triangular matrices are as follows.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ 5 & 0 & 6 \end{pmatrix}.$$

§3.6.2 Symmetric, and skew-symmetric matrices

Definition

A square matrix A is called **symmetric** if

$$A^T = A.$$

Example

The following matrices are symmetric.

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}, \begin{pmatrix} 7 & 8 \\ 8 & 9 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Some non-symmetric matrices are as follows.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Definition

A square matrix A is called **skew-symmetric** if

$$A^T = -A.$$

Example

The following matrices are skew-symmetric.

$$\begin{pmatrix} 0 & 2 & 3 \\ -2 & 0 & 5 \\ -3 & -5 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 8 \\ -8 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Here are some matrices that are not skew-symmetric.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Exercise 3.38. Let A be a square matrix. Show that if A is symmetric and skew-symmetric, then A is the zero matrix.

Solution. Assume that A is symmetric and skew-symmetric. It follows that

$$A^T = A,$$
$$A^T = -A.$$

This shows that

$$A = -A,$$

which yields

$$2A = 0,$$

and hence, A is the zero matrix. □

Fact 3.39. Any square matrix can be uniquely expressed as the sum of a symmetric matrix and a skew-symmetric matrix.

A proof of the above is provided in Chapter 8.

§3.6.3 Orthogonal matrices

Definition

An $n \times n$ matrix A with **real entries** is called **orthogonal** if

$$AA^T = A^T A = I_n.$$

Example

The following matrices are orthogonal.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

Some non-orthogonal matrices are as follows.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Remark

There are infinitely many orthogonal matrices. For instance, for any real number θ, the matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

is an orthogonal matrix since

$$\begin{aligned} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}^T \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} &= \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \\ &= \begin{pmatrix} \cos^2\theta + \sin^2\theta & 0 \\ 0 & \cos^2\theta + \sin^2\theta \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \end{aligned}$$

and

$$\begin{aligned} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}^T &= \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \\ &= \begin{pmatrix} \cos^2\theta + \sin^2\theta & 0 \\ 0 & \cos^2\theta + \sin^2\theta \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

Exercise 3.40. Show that for any $n \geq 2$, there are infinitely many $n \times n$ orthogonal matrices.

Exercise 3.41. Let A be an orthogonal matrix. Show that A is invertible and that

$$A^{-1} = A^T.$$

Solution. Let A be an $n \times n$ orthogonal matrix. It follows that

$$AA^T = A^TA = I_n,$$

which shows that A is invertible and that

$$A^{-1} = A^T.$$

□

Exercise 3.42. Let A, B be orthogonal matrices of the same size. Show that the matrix product AB is also an orthogonal matrix.

Solution. Let A, B be $n \times n$ orthogonal matrices. Note that

$$(AB)(AB)^T = ABB^TA^T = AI_nA^T = AA^T = I_n,$$

and similarly,

$$(AB)^T(AB) = B^TA^TAB = B^TI_nB = B^TB = I_n.$$

This shows that the matrix product AB is also an orthogonal matrix. □

§3.6.4 Hermitian and skew-hermitian matrices

Definition

If A is a matrix with complex entries, then the **complex conjugate** of A, denoted by $\overline{A}$, is defined as the matrix of the same size, obtained by taking the complex conjugate of each entry of A.

Example

The complex conjugate of the matrix

$$\begin{pmatrix} 1+2i & 3-4i \\ -i & 5 \end{pmatrix}$$

is the matrix

$$\begin{pmatrix} 1-2i & 3+4i \\ i & 5 \end{pmatrix}.$$

Exercise 3.43. Let A, B be square matrices with complex entries. Show that

$$\overline{AB} = \overline{A}\,\overline{B}.$$

Exercise 3.44. Let A be a square matrix with complex entries. Show that

$$\overline{A^T} = \left(\overline{A}\right)^T.$$

Definition

If A is a matrix with complex entries, then the **conjugate transpose** of A, denoted by A^*, is defined as the transpose of the matrix obtained by taking the complex conjugate of each entry of A, that A^* is equal to $\overline{A}^T$.

Example

The conjugate transpose of the matrix

$$\begin{pmatrix} 1+2i & 3-4i \\ -i & 5 \end{pmatrix}$$

is the matrix

$$\begin{pmatrix} 1-2i & i \\ 3+4i & 5 \end{pmatrix}.$$

Compare the following exercises with Exercise 3.6, Exercise 3.7, Exercise 3.14.

Exercise 3.45. Let A, B be square matrices of the same size with complex entries. Show that

$$(A+B)^* = A^* + B^*.$$

Solution. Note that

$$(A+B)^* = \left(\overline{A+B}\right)^T = \left(\overline{A}+\overline{B}\right)^T = \overline{A}^T + \overline{B}^T = A^* + B^*.$$

□

Exercise 3.46. Let A be a matrix with complex entries. Show that

$$(A^*)^* = A.$$

Solution. Note that

$$(A^*)^* = \left(\overline{A}^T\right)^* = \left(\overline{A^T}\right)^* = \overline{\overline{A^T}}^T = \left(A^T\right)^T = A.$$

□

Exercise 3.47. Let A, B be matrices with complex entries. Show that

$$(AB)^* = B^* A^*.$$

Solution. Note that

$$(AB)^* = \left(\overline{AB}\right)^T = \left(\overline{A}\,\overline{B}\right)^T = \overline{B}^T\,\overline{A}^T = B^* A^*.$$

□

Definition

A square matrix A with complex entries is called **hermitian** if

$$A^* = A,$$

where A^* denotes the conjugate transpose of A.

Example

The following matrices are hermitian.

$$\begin{pmatrix} 1 & 2+i \\ 2-i & 3 \end{pmatrix}, \begin{pmatrix} 5 & i \\ -i & 7 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Some non-hermitian matrices are as follows.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Exercise 3.48. Let A be an $n \times n$ hermitian matrix. Show that all the diagonal entries of A are real numbers. More generally, show that for any $1 \leq i, j \leq n$, the (i, j)-entry and the (j, i)-entry of A are complex conjugates of each other.

Exercise 3.49. Let A, B be hermitian matrices of the same size. Show that the matrix $A + B$ is also a hermitian matrix. Is the matrix AB also a hermitian matrix?

Solution. Let A, B be $n \times n$ hermitian matrices. Note that

$$(A+B)^* = A^* + B^* = A + B.$$

This shows that the matrix $A + B$ is also a hermitian matrix.

The matrix product AB is not necessarily a hermitian matrix. For instance, let

$$A = \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix},$$

$$B = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

Note that both A and B are hermitian matrices. Also note that

$$AB = \begin{pmatrix} 2 & 3i \\ -2i & 3 \end{pmatrix},$$

and hence, the product AB is not a hermitian matrix. □

Definition

A square matrix A with complex entries is called **skew-hermitian** if

$$A^* = -A,$$

where A^* denotes the conjugate transpose of A.

Example

The following matrices are skew-hermitian.

$$\begin{pmatrix} 0 & 2+i \\ -2+i & 0 \end{pmatrix}, \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Some non-skew-hermitian matrices are as follows.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Exercise 3.50. State and prove an analogue of Exercise 3.48 for skew-hermitian matrices.

Exercise 3.51. Let A, B be skew-hermitian matrices of the same size. Show that the matrix $A + B$ is also a skew-hermitian matrix. Is the matrix AB also a skew-hermitian matrix?

Solution. Let A, B be $n \times n$ skew-hermitian matrices. Note that

$$(A+B)^* = A^* + B^* = -A - B = -(A+B).$$

This shows that the matrix $A + B$ is also a skew-hermitian matrix.

The matrix product AB is not necessarily a skew-hermitian matrix. For instance, let

$$A = \begin{pmatrix} 0 & 2+i \\ -2+i & 0 \end{pmatrix},$$

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Note that both A and B are skew-hermitian matrices. Also note that

$$AB = \begin{pmatrix} -2-i & 0 \\ 0 & -2+i \end{pmatrix},$$

which is not a skew-hermitian matrix. □

Exercise 3.52. Let A be a square matrix with complex entries. Show that if A is hermitian and skew-hermitian, then A is the zero matrix.

Solution. Assume that A is hermitian and skew-hermitian. It follows that

$$A^* = A,$$
$$A^* = -A.$$

This shows that

$$A = -A,$$

which yields

$$2A = 0,$$

and hence, A is the zero matrix. □

Fact 3.53. Let A be a square matrix with complex entries. Show that A can be uniquely expressed as the sum of a hermitian matrix and a skew-hermitian matrix.

A proof of the above is provided in Chapter 8.

§3.6.5 Unitary and normal matrices

Definition

An $n \times n$ matrix A with complex entries is called **unitary** if

$$AA^* = A^*A = I_n.$$

Example

The following matrices are unitary.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

Some non-unitary matrices are as follows.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Exercise 3.54. Let A be a unitary matrix. Show that A is invertible and that

$$A^{-1} = A^*.$$

Solution. Let A be a unitary matrix. It follows that

$$AA^* = A^*A = I_n,$$

which shows that A is invertible and that

$$A^{-1} = A^*.$$

□

Exercise 3.55. Let A, B be unitary matrices of the same size. Show that the matrix product AB is also a unitary matrix.

Solution. Let A, B be $n \times n$ unitary matrices. Note that

$$(AB)(AB)^* = ABB^*A^* = AI_nA^* = AA^* = I_n,$$

and similarly,

$$(AB)^*(AB) = B^*A^*AB = B^*I_nB = B^*B = I_n.$$

This shows that the matrix product AB is also a unitary matrix. □

Exercise 3.56. Show that any orthogonal matrix is a unitary matrix. Is the converse true?

Solution. Let A be an orthogonal matrix. It follows that

$$A^T A = I_n,$$

which shows that A is invertible and that

$$A^{-1} = A^T.$$

Since A has real entries, we have $A^* = A^T$, and hence,

$$A^* A = A^T A = I_n.$$

Similarly, we have

$$AA^* = AA^T = I_n.$$

This shows that any orthogonal matrix is a unitary matrix.

The converse is not true in general. For instance, the matrix

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

is a unitary matrix but not an orthogonal matrix. □

Exercise 3.57. Determine whether the following matrices are unitary, hermitian, both, or neither.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Exercise 3.58. Let z, w be complex numbers such that $|z|^2 + |w|^2 = 1$. Show that

$$\begin{pmatrix} z & -\overline{w} \\ w & \overline{z} \end{pmatrix}$$

is a unitary matrix.

Solution. Let z, w be complex numbers such that $|z|^2 + |w|^2 = 1$. Note that

$$\begin{aligned} \begin{pmatrix} z & -\overline{w} \\ w & \overline{z} \end{pmatrix} \begin{pmatrix} z & -\overline{w} \\ w & \overline{z} \end{pmatrix}^* &= \begin{pmatrix} z & -\overline{w} \\ w & \overline{z} \end{pmatrix} \begin{pmatrix} \overline{z} & \overline{w} \\ -w & z \end{pmatrix} \\ &= \begin{pmatrix} z\overline{z} + \overline{w}w & z\overline{w} - \overline{w}z \\ w\overline{z} - \overline{z}w & w\overline{w} + \overline{z}z \end{pmatrix} \\ &= \begin{pmatrix} |z|^2 + |w|^2 & 0 \\ 0 & |z|^2 + |w|^2 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

hold. Also note that

$$
\begin{aligned}
\begin{pmatrix} z & -\overline{w} \\ w & \overline{z} \end{pmatrix}^* \begin{pmatrix} z & -\overline{w} \\ w & \overline{z} \end{pmatrix} &= \begin{pmatrix} \overline{z} & \overline{w} \\ -w & z \end{pmatrix} \begin{pmatrix} z & -\overline{w} \\ w & \overline{z} \end{pmatrix} \\
&= \begin{pmatrix} \overline{z}z + \overline{w}(-w) & \overline{z}(-\overline{w}) + \overline{w}z \\ -wz + zw & -w(-\overline{w}) + z\overline{z} \end{pmatrix} \\
&= \begin{pmatrix} |z|^2 + |w|^2 & 0 \\ 0 & |z|^2 + |w|^2 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}
\end{aligned}
$$

hold. This shows that

$$
\begin{pmatrix} z & -\overline{w} \\ w & \overline{z} \end{pmatrix}
$$

is a unitary matrix. □

Exercise 3.59. Let A be a diagonal matrix with complex entries. Show that A is a unitary matrix if and only if all the diagonal entries of A have absolute value equal to 1.

Definition

A square matrix A with complex entries is called **normal** if

$$AA^* = A^*A,$$

where A^* denotes the conjugate transpose of A.

Example

The following matrices are normal.

$$
\begin{pmatrix} 1 & 2+i \\ 2-i & 3 \end{pmatrix}, \begin{pmatrix} 5 & i \\ -i & 7 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.
$$

Some non-normal matrices are as follows.

$$
\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.
$$

Exercise 3.60. Show that any unitary matrix is a normal matrix. Is the converse true?

Exercise 3.61. Show that any hermitian matrix is a normal matrix. Is the converse true?

Exercise 3.62. Show that any skew-hermitian matrix is a normal matrix. Is the converse true?

Here is a summary of some properties of some of the special types of matrices introduced above.

Matrix	Some property
Diagonal	Entries outside the main diagonal are zero
Scalar	Diagonal matrix with all diagonal entries equal
Triangular	Entries below or above the main diagonal are zero
Upper-triangular	Entries below the main diagonal are zero
Lower-triangular	Entries above the main diagonal are zero
Symmetric	$A^T = A$
Skew-symmetric	$A^T = -A$
Orthogonal	$AA^T = A^T A = I_n$
Hermitian	$A^* = A$
Skew-hermitian	$A^* = -A$
Unitary	$AA^* = A^* A = I_n$
Normal	$AA^* = A^* A$

§3.7 Row reduction

§3.7.1 Matrix units

Definition

The **matrix unit** e_{ij} is the $n \times n$ matrix whose (i, j)-entry is equal to 1 and all other entries are equal to 0.

Example

The matrix unit e_{23} for $n = 4$ is the matrix

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Remark

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we have

$$\begin{aligned} A &= a\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \\ &= ae_{11} + be_{12} + ce_{21} + de_{22}. \end{aligned}$$

More generally, if $A = (a_{ij})$ is an $n \times n$ matrix, then

$$\begin{aligned} A = a_{11}e_{11} &+ a_{12}e_{12} + \cdots + a_{1n}e_{1n} \\ &+ a_{21}e_{21} + a_{22}e_{22} + \cdots + a_{2n}e_{2n} \\ &+ \cdots \\ &+ a_{n1}e_{n1} + a_{n2}e_{n2} + \cdots + a_{nn}e_{nn} \end{aligned}$$

holds.

Example

Let
$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}.$$
Note that
$$\begin{aligned}
e_{12}A &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 4 & 5 & 6 \\ 0 & 0 & 0 \end{pmatrix}, \\
e_{21}A &= \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 2 & 3 \end{pmatrix}, \\
e_{11}A &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{pmatrix}, \\
e_{22}A &= \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 4 & 5 & 6 \end{pmatrix}.
\end{aligned}$$

Remark

More generally, if A is a matrix with n rows, then multiplying A from the left by the matrix unit e_{ij} produces the matrix whose i-th row is equal to the j-th row of A, and all other rows are equal to zero.

Exercise 3.63. Let e_{ij}, e_{kl} be matrix units of the same size. Show that $e_{ij}e_{kl}$ is equal to the matrix unit e_{il} if $j = k$, and is equal to the zero matrix if $j \neq k$, that is,
$$e_{ij}e_{kl} = \begin{cases} e_{il} & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

§3.7.2 Matrix multiplication and linear combinations of rows

Remark

If A is an $m \times n$ matrix and B is an $n \times p$ matrix, then the matrix product AB is an $m \times p$ matrix. Recall that the i-th **column of AB** is a linear combination of the **columns of A** with **coefficients** from the i-th **column of B**.

Remark

Consider the matrices

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix},$$

$$B = \begin{pmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{pmatrix}.$$

Note that

$$AB = \begin{pmatrix} 1 \times 7 + 2 \times 9 + 3 \times 11 & 1 \times 8 + 2 \times 10 + 3 \times 12 \\ 4 \times 7 + 5 \times 9 + 6 \times 11 & 4 \times 8 + 5 \times 10 + 6 \times 12 \end{pmatrix} = \begin{pmatrix} 58 & 64 \\ 139 & 154 \end{pmatrix}.$$

Note that first row of AB is a linear combination of the rows of B with coefficients from the first row of A, and the second row of AB is a linear combination of the rows of B with coefficients from the second row of A. Indeed,

$$\begin{aligned} &(1 \times 7 + 2 \times 9 + 3 \times 11,\ 1 \times 8 + 2 \times 10 + 3 \times 12) \\ &= 1 \times (7,\ 8) + 2 \times (9,\ 10) + 3 \times (11,\ 12), \\ &(4 \times 7 + 5 \times 9 + 6 \times 11,\ 4 \times 8 + 5 \times 10 + 6 \times 12) \\ &= 4 \times (7,\ 8) + 5 \times (9,\ 10) + 6 \times (11,\ 12). \end{aligned}$$

Remark

If A is an $m \times n$ matrix and B is an $n \times p$ matrix, then the matrix product AB is an $m \times p$ matrix, and the i-th **row of AB** is a linear combination of the **rows of B** with **coefficients** from the i-th **row of A**. Thus, left multiplication by a matrix A (in particular, by a square matrix A) can be viewed as a transformation that transforms each row of B into a linear combination of the rows of B, and is often called a **row operation**.

Denote the set of $m \times n$ matrices with entries from $\mathbb{R}$ by $M_{m,n}(\mathbb{R})$.

Compare the following exercise with Exercise 3.35.

Exercise 3.64. Let A be an $m \times n$ matrix. Let V be the set of solutions to the system of equations

$$uA = 0,$$

that is,

$$V = \{v \in M_{1,m}(\mathbb{R}) : vA = 0\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $\mathbb{R}^m$ lies in V.

Compare the following exercise with Exercise 3.36.

Exercise 3.65. Let A be an $m \times n$ matrix. Let V be the set of row vectors which can be expressed as a linear combination of the rows of A, that is,

$$V = \{v \in M_{1,n}(\mathbb{R}) : v = u^T A \text{ for some } u \in \mathbb{R}^m\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $M_{1,n}(\mathbb{R})$ lies in V.

Solution. Let $u, v \in V$ and $c, d \in \mathbb{R}$. By the definition of V, there exist $x, y \in \mathbb{R}^m$ such that

$$u = x^T A,$$
$$v = y^T A.$$

Note that

$$cu + dv = cx^T A + dy^T A = (cx^T + dy^T)A = (cx + dy)^T A.$$

Since $cx + dy \in \mathbb{R}^m$, it follows that $cu + dv \in V$. □

§3.7.3 Elementary row operations and elementary matrices

Definition

An **elementary row operation** on a matrix is one of the following operations:

1. Interchanging two rows.
2. Multiplying a row by a nonzero scalar.
3. Adding a scalar multiple of one row to another row.

To study matrices that perform elementary row operations, we introduce the following special types of matrices, called **elementary matrices**.

Definition

The **elementary matrices** are of three types, and these are obtained by performing the elementary row operations on the identity matrix.

1. The matrix obtained by interchanging the i-th row and the j-th row of the identity matrix, that is, it is given by
$$I_n - e_{ii} - e_{jj} + e_{ij} + e_{ji} \text{ with } i \neq j,$$
2. The matrix obtained by multiplying the i-th row of the identity matrix by a nonzero scalar, that is, it is given by
$$I_n + (\lambda - 1)e_{ii} \text{ with } \lambda \neq 0,$$
3. The matrix obtained by adding a scalar multiple of the j-th row of the identity matrix to its i-th row, that is, it is given by
$$I_n + \lambda e_{ij} \text{ with } i \neq j.$$

Example

The elementary 2×2 matrices are as follows.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} \lambda & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix}, \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix}, \text{where } \lambda \neq 0.$$

Example

The elementary 3×3 matrices are as follows.

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$\begin{pmatrix} \lambda & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \lambda \end{pmatrix} \text{ where } \lambda \neq 0,$$

$$\begin{pmatrix} 1 & 0 & 0 \\ \lambda & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \lambda & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & \lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \lambda & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & \lambda \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \lambda \\ 0 & 0 & 1 \end{pmatrix} \text{ where } \lambda \neq 0.$$

Exercise 3.66. Write down the six elementary 4×4 matrices corresponding to the elementary row operations of interchanging two rows of 4×4 matrices.

Solution. The 6 elementary 4×4 matrices corresponding to the elementary row operations of interchanging two rows of 4×4 matrices are as follows.

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Note that these matrices are the elementary matrices corresponding to the elementary row operations of interchanging

- the first and second rows,
- the first and third rows,
- the first and fourth rows,
- the second and third rows,
- the second and fourth rows,
- the third and fourth rows,

respectively. □

Lemma 3.67. *Left multiplication by an elementary matrix on a matrix A performs the corresponding elementary row operation on the matrix A. More precisely, if E is an elementary matrix obtained by performing an elementary row operation on the identity matrix, then the matrix product EA is the matrix obtained by performing the same elementary row operation on the matrix A.*

A proof of the above is provided in Chapter 8.

Lemma 3.68. *Any elementary matrix is invertible, and its inverse is also an elementary matrix.*

A proof of the above is provided in Chapter 8.

Remark

Note that the elementary row operations on a matrix A are precisely the operations of left multiplying A by the elementary matrices. Thus, if $E_1, E_2, \dots, E_k$ are elementary matrices, then the matrix

$$E_k \dots E_2 E_1 A$$

is obtained by performing a sequence of elementary row operations on the matrix A. Moreover, any matrix that can be obtained from A by performing a sequence of elementary row operations can be expressed in the form

$$EA,$$

where E is an invertible matrix which is a product of elementary matrices.

§3.7.4 Row reduction of a matrix

Definition

Multiplying a matrix A by a matrix E from the left is called **row reduction** of the matrix A if E is a product of elementary matrices. In other words, row reduction of a matrix A is the process of performing a sequence of elementary row operations on the matrix A. It is also known as **Gaussian elimination**.

Example

Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & -1 & -4 \\ 2 & 3 & -1 & -11 \\ -2 & 0 & -3 & 22 \end{pmatrix}.$$

We perform row reduction on the matrix A as follows.

1. First, we add -2 times the first row to the second row, and add 2 times the first row to the third row, to obtain the matrix

$$\begin{pmatrix} 1 & 2 & -1 & -4 \\ 0 & -1 & 1 & -3 \\ 0 & 4 & -5 & 14 \end{pmatrix}.$$

2. Next, we add 4 times the second row to the third row, to obtain the matrix

$$\begin{pmatrix} 1 & 2 & -1 & -4 \\ 0 & -1 & 1 & -3 \\ 0 & 0 & -1 & 2 \end{pmatrix}.$$

3. Finally, we multiply the second row by -1, and multiply the third row by -1, to obtain the matrix

$$\begin{pmatrix} 1 & 2 & -1 & -4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -2 \end{pmatrix}.$$

We now illustrate how row reduction can be used to solve a system of linear equations.

Exercise 3.69. Solve the system of linear equations in five variables x_1, x_2, x_3, x_4, x_5:

$$\begin{aligned} x_1 + 2x_2 - x_3 + 4x_4 + x_5 &= 7, \\ 2x_1 + 3x_2 + x_3 + 5x_4 + 2x_5 &= 14, \\ -x_1 + 4x_2 - 2x_3 - 3x_4 + x_5 &= -10. \end{aligned} \tag{12}$$

Solution. The above system of equations can be expressed in matrix form as

$$\begin{pmatrix} 1 & 2 & -1 & 4 & 1 \\ 2 & 3 & 1 & 5 & 2 \\ -1 & 4 & -2 & -3 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 7 \\ 14 \\ -10 \end{pmatrix}.$$

We perform row reduction on the augmented matrix

$$(A \mid b) = \left(\begin{array}{ccccc|c} 1 & 2 & -1 & 4 & 1 & 7 \\ 2 & 3 & 1 & 5 & 2 & 14 \\ -1 & 4 & -2 & -3 & 1 & -10 \end{array}\right).$$

1. First, we add -2 times the first row to the second row, and add 1 times the first row to the third row, to obtain the matrix
$$\left(\begin{array}{ccccc|c} 1 & 2 & -1 & 4 & 1 & 7 \\ 0 & -1 & 3 & -3 & 0 & 0 \\ 0 & 6 & -3 & 1 & 2 & -3 \end{array}\right).$$
2. Next, we add 6 times the second row to the third row, to obtain the matrix
$$\left(\begin{array}{ccccc|c} 1 & 2 & -1 & 4 & 1 & 7 \\ 0 & -1 & 3 & -3 & 0 & 0 \\ 0 & 0 & 15 & -17 & 2 & -3 \end{array}\right).$$
3. Finally, we multiply the second row by -1, and multiply the third row by $\frac{1}{15}$, to obtain the matrix
$$\left(\begin{array}{ccccc|c} 1 & 2 & -1 & 4 & 1 & 7 \\ 0 & 1 & -3 & 3 & 0 & 0 \\ 0 & 0 & 1 & -\frac{17}{15} & \frac{2}{15} & -\frac{1}{5} \end{array}\right).$$

The solutions to the system of equations Equation 12 are precisely the solutions to the system of equations

$$\begin{pmatrix} 1 & 2 & -1 & 4 & 1 \\ 0 & 1 & -3 & 3 & 0 \\ 0 & 0 & 1 & -\frac{17}{15} & \frac{2}{15} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 7 \\ 0 \\ -\frac{1}{5} \end{pmatrix}.$$

Hence, the solutions of the given system of equations are precisely the elements $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix}$ of $\mathbb{R}^5$ satisfying

$$\begin{aligned} x_1 + 2x_2 - x_3 + 4x_4 + x_5 &= 7, \\ x_2 - 3x_3 + 3x_4 &= 0, \\ x_3 - \frac{17}{15}x_4 + \frac{2}{15}x_5 &= -\frac{1}{5}, \end{aligned}$$

or equivalently,

$$\begin{aligned} x_1 &= 7 - 2x_2 + x_3 - 4x_4 - x_5, \\ x_2 &= 3x_3 - 3x_4, \\ x_3 &= -\frac{1}{5} + \frac{17}{15}x_4 - \frac{2}{15}x_5, \end{aligned}$$

which is equivalent to

$$\begin{aligned} x_3 &= -\frac{1}{5} + \frac{17}{15}s - \frac{2}{15}t, \\ x_2 &= 3\left(-\frac{1}{5} + \frac{17}{15}s - \frac{2}{15}t\right) - 3s \\ &= -\frac{3}{5} + \frac{51}{15}s - \frac{2}{5}t - 3s \\ &= -\frac{3}{5} + \frac{2}{5}s - \frac{2}{5}t \\ x_1 &= 7 - 2\left(-\frac{3}{5} + \frac{2}{5}s - \frac{2}{5}t\right) + \left(-\frac{1}{5} + \frac{17}{15}s - \frac{2}{15}t\right) - 4s - t \\ &= 7 + \frac{6}{5} - \frac{4}{5}s + \frac{4}{5}t - \frac{1}{5} + \frac{17}{15}s - \frac{2}{15}t - 4s - t \\ &= 8 - \frac{11}{3}s - \frac{1}{3}t. \end{aligned}$$

In other words, the set of solutions of the given system of equations is equal to

$$\boxed{\left\{\left(8 - \frac{11}{3}s - \frac{1}{3}t, -\frac{3}{5} + \frac{2}{5}s - \frac{2}{5}t, -\frac{1}{5} + \frac{17}{15}s - \frac{2}{15}t, s, t\right)^T : s, t \in \mathbb{R}\right\}}.$$

□

Theorem 3.70. *Let b denote the vector $\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$, and let A be an $m \times n$ matrix. Let E be a product of elementary matrices. Write $X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$. The solutions to the system of equations*

$$AX = b$$

are precisely the solutions to the system of equations

$$A'X = b',$$

where

$$b' = Eb, \quad A' = EA.$$

A proof of the above is provided in Chapter 8.

Remark

Note that while solving Equation 12, we performed row reduction on the augmented matrix

$$(A \mid b) = \left(\begin{array}{ccccc|c} 1 & 2 & -1 & 4 & 1 & 7 \\ 2 & 3 & 1 & 5 & 2 & 14 \\ -1 & 4 & -2 & -3 & 1 & -10 \end{array}\right),$$

and obtained the matrix

$$\left(\begin{array}{ccccc|c} 1 & 2 & -1 & 4 & 1 & 7 \\ 0 & 1 & -3 & 3 & 0 & 0 \\ 0 & 0 & 1 & -\frac{17}{15} & \frac{2}{15} & -\frac{1}{5} \end{array}\right).$$

Note that one can perform further row reductions. Adding the third row to the first row, and adding 3 times the third row to the second row, we obtain the matrix

$$\left(\begin{array}{ccccc|c} 1 & 2 & 0 & \frac{43}{15} & \frac{17}{15} & \frac{34}{5} \\ 0 & 1 & 0 & -\frac{2}{5} & \frac{2}{5} & -\frac{3}{5} \\ 0 & 0 & 1 & -\frac{17}{15} & \frac{2}{15} & -\frac{1}{5} \end{array}\right).$$

Next, we add -2 times the second row to the first row, to obtain the matrix

$$\left(\begin{array}{ccccc|c} 1 & 0 & 0 & \frac{11}{3} & \frac{1}{3} & 8 \\ 0 & 1 & 0 & -\frac{2}{5} & \frac{2}{5} & -\frac{3}{5} \\ 0 & 0 & 1 & -\frac{17}{15} & \frac{2}{15} & -\frac{1}{5} \end{array}\right).$$

This shows that the augmented matrix corresponding to the system of equations Equation 12 can be row reduced to the matrix

$$\left(\begin{array}{ccccc|c} 1 & 0 & 0 & \frac{11}{3} & \frac{1}{3} & 8 \\ 0 & 1 & 0 & -\frac{2}{5} & \frac{2}{5} & -\frac{3}{5} \\ 0 & 0 & 1 & -\frac{17}{15} & \frac{2}{15} & -\frac{1}{5} \end{array}\right),$$

which shows that the solutions of the system of equations Equation 12 are precisely the elements of the set

$$\boxed{\left\{\left(8 - \frac{11}{3}s - \frac{1}{3}t, -\frac{3}{5} + \frac{2}{5}s - \frac{2}{5}t, -\frac{1}{5} + \frac{17}{15}s - \frac{2}{15}t, s, t\right)^T : s, t \in \mathbb{R}\right\}}.$$

Exercise 3.71. Solve the system of equations in the variables x_1, x_2, x_3:

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 1, \\ 2x_1 + 3x_2 + x_3 &= 2, \\ -x_1 + 4x_2 - 2x_3 &= -3, \\ 3x_1 - x_2 + 4x_3 &= 4, \\ 5x_1 + 2x_2 + 3x_3 &= 5. \end{aligned}$$

Solution. The above system of equations can be expressed in matrix form as

$$\begin{pmatrix} 1 & 2 & -1 \\ 2 & 3 & 1 \\ -1 & 4 & -2 \\ 3 & -1 & 4 \\ 5 & 2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -3 \\ 4 \\ 5 \end{pmatrix}.$$

We perform row reduction on the augmented matrix

$$(A \mid b) = \left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 3 & 1 & 2 \\ -1 & 4 & -2 & -3 \\ 3 & -1 & 4 & 4 \\ 5 & 2 & 3 & 5 \end{array}\right).$$

1. First, we add -2 times the first row to the second row, add 1 times the first row to the third row, add -3 times the first row to the fourth row, and add -5 times the first row to the fifth row, to obtain the matrix
$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & -1 & 3 & 0 \\ 0 & 6 & -3 & -2 \\ 0 & -7 & 7 & 1 \\ 0 & -8 & 8 & 0 \end{array}\right).$$
2. Next, we add 6 times the second row to the third row, add -7 times the second row to the fourth row, and add -8 times the second row to the fifth row, to obtain the matrix
$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & -1 & 3 & 0 \\ 0 & 0 & 15 & -2 \\ 0 & 0 & -14 & 1 \\ 0 & 0 & -16 & 0 \end{array}\right).$$
3. Finally, we multiply the second row by -1, multiply the third row by $\frac{1}{15}$, multiply the fourth row by $-\frac{1}{14}$, and multiply the fifth row by $-\frac{1}{16}$, to obtain the matrix
$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & -\frac{2}{15} \\ 0 & 0 & 1 & -\frac{1}{14} \\ 0 & 0 & 1 & 0 \end{array}\right).$$

Note that the above matrix corresponds to the system of equations

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -\frac{2}{15} \\ \frac{1}{14} \\ 0 \end{pmatrix},$$

which has no solutions. Hence, the given system of equations has no solutions. □

Exercise 3.72. Solve the system of equations in the variables x_1, x_2, x_3, x_4:

$$\begin{aligned} x_1 + 2x_2 - x_3 + 4x_4 &= 5, \\ 2x_1 + 3x_2 + x_3 + 5x_4 &= 8, \\ -x_1 + 4x_2 - 2x_3 - 3x_4 &= -4, \\ 3x_1 - x_2 + 4x_3 + 2x_4 &= 7. \end{aligned}$$

Solution. The above system of equations can be expressed in matrix form as

$$\begin{pmatrix} 1 & 2 & -1 & 4 \\ 2 & 3 & 1 & 5 \\ -1 & 4 & -2 & -3 \\ 3 & -1 & 4 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \\ -4 \\ 7 \end{pmatrix}.$$

We perform row reduction on the augmented matrix

$$(A \mid b) = \left(\begin{array}{cccc|c} 1 & 2 & -1 & 4 & 5 \\ 2 & 3 & 1 & 5 & 8 \\ -1 & 4 & -2 & -3 & -4 \\ 3 & -1 & 4 & 2 & 7 \end{array}\right).$$

1. First, we add -2 times the first row to the second row, add 1 times the first row to the third row, and add -3 times the first row to the fourth row, to obtain the matrix
$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 4 & 5 \\ 0 & -1 & 3 & -3 & -2 \\ 0 & 6 & -3 & 1 & 1 \\ 0 & -7 & 7 & -10 & -8 \end{array}\right).$$

2. Next, we add 6 times the second row to the third row, and add -7 times the second row to the fourth row, to obtain the matrix
$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 4 & 5 \\ 0 & -1 & 3 & -3 & -2 \\ 0 & 0 & 15 & -17 & -11 \\ 0 & 0 & -14 & 11 & 6 \end{array}\right).$$

3. Multiplying the second row by -1, the third row by $\frac{1}{15}$, and the fourth row by $-\frac{1}{14}$, we obtain the matrix
$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 4 & 5 \\ 0 & 1 & -3 & 3 & 2 \\ 0 & 0 & 1 & -\frac{17}{15} & -\frac{11}{15} \\ 0 & 0 & 1 & -\frac{11}{14} & -\frac{3}{7} \end{array}\right).$$

4. Adding -1 times the third row to the fourth row (that is, subtracting the third row from the fourth row), we obtain the matrix
$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 4 & 5 \\ 0 & 1 & -3 & 3 & 2 \\ 0 & 0 & 1 & -\frac{17}{15} & -\frac{11}{15} \\ 0 & 0 & 0 & \frac{73}{210} & \frac{32}{105} \end{array}\right).$$

5. Multiplying the fourth row by $\frac{210}{73}$, we obtain the matrix

$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 4 & 5 \\ 0 & 1 & -3 & 3 & 2 \\ 0 & 0 & 1 & -\frac{17}{15} & -\frac{11}{15} \\ 0 & 0 & 0 & 1 & \frac{64}{73} \end{array}\right).$$

6. Adding -4 times the fourth row to the first row, and adding -3 times the fourth row to the second row, and adding $\frac{17}{15}$ times the fourth row to the third row, we obtain the matrix

$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & \frac{109}{73} \\ 0 & 1 & -3 & 0 & -\frac{46}{73} \\ 0 & 0 & 1 & 0 & \frac{19}{73} \\ 0 & 0 & 0 & 1 & \frac{64}{73} \end{array}\right)$$

7. Adding 3 times the third row to the second row, and adding 1 times the third row to the first row, we obtain the matrix

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & 0 & \frac{128}{73} \\ 0 & 1 & 0 & 0 & \frac{11}{73} \\ 0 & 0 & 1 & 0 & \frac{19}{73} \\ 0 & 0 & 0 & 1 & \frac{64}{73} \end{array}\right)$$

8. Adding -2 times the second row to the first row, we obtain the matrix

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & \frac{106}{73} \\ 0 & 1 & 0 & 0 & \frac{11}{73} \\ 0 & 0 & 1 & 0 & \frac{19}{73} \\ 0 & 0 & 0 & 1 & \frac{64}{73} \end{array}\right).$$

Hence, the given system of equations admits the unique solution

$$\boxed{\begin{pmatrix} \frac{106}{73} \\ \frac{11}{73} \\ \frac{19}{73} \\ \frac{64}{73} \end{pmatrix}}.$$

□

Exercise 3.73. Solve the system of equations in the variables x_1, x_2, x_3:

$$x_1 + 2x_2 - 5x_3 = 20,$$
$$2x_1 + 5x_2 - 7x_3 = 33,$$
$$-x_1 - 2x_2 + 4x_3 = -17.$$

Solution. The above system of equations can be expressed in matrix form as

$$\begin{pmatrix} 1 & 2 & -5 \\ 2 & 5 & -7 \\ -1 & -2 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 20 \\ 33 \\ -17 \end{pmatrix}.$$

We perform row reduction on the augmented matrix

$$(A \mid b) = \left(\begin{array}{ccc|c} 1 & 2 & -5 & 20 \\ 2 & 5 & -7 & 33 \\ -1 & -2 & 4 & -17 \end{array}\right).$$

1. First, we add -2 times the first row to the second row, to obtain the matrix
$$\left(\begin{array}{ccc|c} 1 & 2 & -5 & 20 \\ 0 & 1 & 3 & -7 \\ -1 & -2 & 4 & -17 \end{array}\right).$$
2. Adding 1 times the first row to the third row, we obtain the matrix
$$\left(\begin{array}{ccc|c} 1 & 2 & -5 & 20 \\ 0 & 1 & 3 & -7 \\ 0 & 0 & -1 & 3 \end{array}\right).$$
3. Next, we multiply the third row by -1, to obtain the matrix
$$\left(\begin{array}{ccc|c} 1 & 2 & -5 & 20 \\ 0 & 1 & 3 & -7 \\ 0 & 0 & 1 & -3 \end{array}\right).$$
4. Adding -3 times the third row to the second row, we obtain the matrix
$$\left(\begin{array}{ccc|c} 1 & 2 & -5 & 20 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{array}\right).$$
5. Adding 5 times the third row to the first row, we obtain the matrix
$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{array}\right).$$
6. Finally, adding -2 times the second row to the first row, we obtain the matrix
$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{array}\right).$$

Hence, the given system of equations admits the unique solution
$$\boxed{\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}}.$$

□

Exercise 3.74. Let A, B be square matrices of the same size. Suppose that A is invertible and $AB = I$ holds. Show that B is invertible and $B = A^{-1}$.

Exercise 3.75. Let A, B be square matrices of the same size. Suppose that A is invertible and $BA = I$ holds. Show that B is invertible and $B = A^{-1}$.

Compare the above two exercises with Exercise 3.29, Exercise 3.92.

Exercise 3.76. Let A, B be matrices, suppose that the number of columns of A is equal to the number of rows of B. Suppose AB has at least two columns. Let C be the matrix obtained from AB by removing the last column of AB. Show that there exists a matrix B' such that $C = AB'$.

Exercise 3.77. Let A, B be matrices, suppose that the number of columns of A is equal to the number of rows of B. Suppose AB has at least two rows. Let C be the matrix obtained from AB by removing the last row of AB. Show that there exists a matrix A' such that $C = A'B$.

Remark 3.78. Note that while solving the above system of equations, we performed row reduction on the augmented matrix

$$(A \mid b) = \left(\begin{array}{ccc|c} 1 & 2 & -5 & 20 \\ 2 & 5 & -7 & 33 \\ -1 & -2 & 4 & -17 \end{array}\right),$$

and obtained the matrix

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{array}\right).$$

This shows that multiplying the matrix

$$A = \begin{pmatrix} 1 & 2 & -5 \\ 2 & 5 & -7 \\ -1 & -2 & 4 \end{pmatrix}$$

from the left by the product of some elementary matrices yields the identity matrix I_3, which implies that the matrix A is invertible. Indeed, if $EA = I_3$ where E is the product of some elementary matrices, then using that E is invertible, it follows that $A = E^{-1}EA = E^{-1}I_3 = E^{-1}$, and hence $AE = I_3$, which shows that A is invertible, and A^{-1} is equal to E.

The elementary matrices used to row reduce the matrix A to I_3 are

$$E_1 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

$$E_4 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix}, E_5 = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, E_6 = \begin{pmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Note that $EA = I_3$ holds for

$$\begin{aligned}
E &= E_6E_5E_4E_3E_2E_1\\
&= \begin{pmatrix}1&-2&0\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&5\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&0\\0&1&-3\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}\begin{pmatrix}1&0&0\\0&1&0\\1&0&1\end{pmatrix}\begin{pmatrix}1&0&0\\-2&1&0\\0&0&1\end{pmatrix}\\
&= \begin{pmatrix}1&-2&0\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&5\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&0\\0&1&-3\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}\begin{pmatrix}1&0&0\\-2&1&0\\1&0&1\end{pmatrix}\\
&= \begin{pmatrix}1&-2&0\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&5\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&0\\0&1&-3\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&0\\-2&1&0\\-1&0&-1\end{pmatrix}\\
&= \begin{pmatrix}1&-2&0\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&5\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&0\\1&1&3\\-1&0&-1\end{pmatrix}\\
&= \begin{pmatrix}1&-2&0\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}-4&0&-5\\1&1&3\\-1&0&-1\end{pmatrix}\\
&= \begin{pmatrix}-6&-2&-11\\1&1&3\\-1&0&-1\end{pmatrix}.
\end{aligned}$$

This yields

$$A^{-1} = E = \begin{pmatrix}-6&-2&-11\\1&1&3\\-1&0&-1\end{pmatrix}.$$

Exercise 3.79. Solve the system of equations in the variables x_1, x_2:

$$\begin{aligned} 2x_1 - 3x_2 &= 7,\\ -4x_1 + 5x_2 &= -11. \end{aligned}$$

Solution. The above system of equations can be expressed in matrix form as

$$\begin{pmatrix}2&-3\\-4&5\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix} = \begin{pmatrix}7\\-11\end{pmatrix}.$$

We perform row reduction on the augmented matrix

$$(A \mid b) = \left(\begin{array}{cc|c}2&-3&7\\-4&5&-11\end{array}\right).$$

1. First, we add 2 times the first row to the second row, to obtain the matrix
$$\left(\begin{array}{cc|c}2&-3&7\\0&-1&3\end{array}\right).$$
2. Next, we multiply the second row by -1, to obtain the matrix
$$\left(\begin{array}{cc|c}2&-3&7\\0&1&-3\end{array}\right).$$
3. Add 3 times the second row to the first row to obtain the matrix
$$\left(\begin{array}{cc|c}2&0&-2\\0&1&-3\end{array}\right).$$

4. Finally, we multiply the first row by $\frac{1}{2}$, to obtain the matrix

$$\left(\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & -3 \end{array}\right).$$

Hence, the given system of equations admits the unique solution

$$\boxed{\begin{pmatrix} -1 \\ -3 \end{pmatrix}}.$$

□

Remark

Let A denote the matrix

$$\begin{pmatrix} 2 & -3 \\ -4 & 5 \end{pmatrix}.$$

Note that $EA = I_2$ holds where E is the product of some elementary matrices. More specifically, $EA = I_2$ holds for

$$E = E_4 E_3 E_2 E_1,$$

where

$$E_1 = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, E_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, E_3 = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}, E_4 = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}.$$

One can argue that A is invertible, and that

$$\begin{aligned} A^{-1} &= E \\ &= E_4 E_3 E_2 E_1 \\ &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2} & \frac{3}{2} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2 & -1 \end{pmatrix} \\ &= \begin{pmatrix} -\frac{5}{2} & -\frac{3}{2} \\ -2 & -1 \end{pmatrix}. \end{aligned}$$

§3.7.5 Row echelon form

Definition

A matrix is said to be in **row echelon form** if the following conditions hold.

1. The first nonzero entry in each nonzero row is 1 (called a **leading 1** or a **pivot**).
2. The leading 1 in each nonzero row is to the right of the leading 1 in the previous row (if any).
3. All zero rows (if any) are at the bottom of the matrix.
4. If a column contains a pivot, then all its entries above the pivot are zero.

Example

The following matrices are in row echelon form:

$$\begin{pmatrix} 0 & 1 & -5 & 0 & 0 & 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The following matrices are not in row echelon form:

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Fact 3.80. Through a sequence of elementary row operations, any matrix can be transformed into a matrix in row echelon form.

Definition

If A is a matrix, then the matrix A' obtained from A by performing a sequence of elementary row operations such that A' is in row echelon form is called the **row echelon form** of A.

Exercise 3.81. Transform the matrix

$$\begin{pmatrix} 0 & 2 & -3 & 0 & 0 \\ 1 & -1 & 4 & 0 & 5 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 3 & -5 & 0 & 1 \\ 0 & 3 & -5 & -1 & 2 \end{pmatrix}$$

into a matrix in row echelon form using elementary row operations.

Solution. We perform row reduction on the matrix

$$A = \begin{pmatrix} 0 & 2 & -3 & 0 & 0 \\ 1 & -1 & 4 & 0 & 5 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 3 & -5 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

1. Interchanging the first row and the second row, we obtain the matrix

$$\begin{pmatrix} 1 & -1 & 4 & 0 & 5 \\ 0 & 2 & -3 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 3 & -5 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

2. Next, multiplying the second row by $\frac{1}{2}$, we obtain the matrix

$$\begin{pmatrix} 1 & -1 & 4 & 0 & 5 \\ 0 & 1 & -\frac{3}{2} & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 3 & -5 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

3. Adding -3 times the second row to the fourth row, we obtain the matrix

$$\begin{pmatrix} 1 & -1 & 4 & 0 & 5 \\ 0 & 1 & -\frac{3}{2} & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & -\frac{1}{2} & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

4. Interchanging the third row and the fourth row, we obtain the matrix

$$\begin{pmatrix} 1 & -1 & 4 & 0 & 5 \\ 0 & 1 & -\frac{3}{2} & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 1 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

5. Multiplying the third row by -2, we obtain the matrix

$$\begin{pmatrix} 1 & -1 & 4 & 0 & 5 \\ 0 & 1 & -\frac{3}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

6. Adding -4 times the third row to the first row, and adding $\frac{3}{2}$ times the third row to the second row, we obtain the matrix

$$\begin{pmatrix} 1 & -1 & 0 & 0 & 13 \\ 0 & 1 & 0 & 0 & -3 \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

7. Finally, adding the second row to the first row, we obtain the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 10 \\ 0 & 1 & 0 & 0 & -3 \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

   which is in row echelon form.

□

Exercise 3.82. Solve the system of equations in the variables x_1, x_2, x_3, x_4:

$$\begin{aligned} 2x_2 - 3x_3 &= 0, \\ x_1 - x_2 + 4x_3 &= 5, \\ x_4 &= -2, \\ 3x_2 - 5x_3 &= 1, \\ 3x_2 - 5x_3 - x_4 &= 2. \end{aligned}$$

Solution. The above system of equations can be expressed in matrix form as

$$\begin{pmatrix} 0 & 2 & -3 & 0 \\ 1 & -1 & 4 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 3 & -5 & 0 \\ 0 & 3 & -5 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \\ -2 \\ 1 \\ 2 \end{pmatrix}.$$

We perform row reduction on the augmented matrix. Adding the third row to the fifth row, we obtain the matrix

$$\left(\begin{array}{cccc|c} 0 & 2 & -3 & 0 & 0 \\ 1 & -1 & 4 & 0 & 5 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 3 & -5 & 0 & 1 \\ 0 & 3 & -5 & 0 & 0 \end{array}\right).$$

Adding -1 times the fourth row to the fifth row, we obtain the matrix

$$\left(\begin{array}{cccc|c} 0 & 2 & -3 & 0 & 0 \\ 1 & -1 & 4 & 0 & 5 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 3 & -5 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 \end{array}\right).$$

Multiplying the fifth row by -1, we obtain the matrix

$$\left(\begin{array}{cccc|c} 0 & 2 & -3 & 0 & 0 \\ 1 & -1 & 4 & 0 & 5 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 3 & -5 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right).$$

Performing further elementary row operations, as done in the previous exercise, yields the matrix

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 10 \\ 0 & 1 & 0 & 0 & -3 \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right).$$

The corresponding system of equations has no solution, since the last column of the above matrix contains a pivot. ☐

Fact 3.83. Let

$$M = (A \,|\, b)$$

be the matrix obtained by augmenting a matrix A with a column vector b. If M is in its **row echelon form**, then the following statements are equivalent.

1. The system of equations $AX = b$ admits a solution.

2. The last column of the augmented matrix M contains no pivot.

If one (and hence, both) of the above statements holds, then the solutions to the system of equations $AX = b$ can be obtained by assigning arbitrary values to the variables corresponding to the non-pivot columns of the augmented matrix M, and then solving for the variables corresponding to the pivot columns of M.

A **pivot column** of a matrix in row echelon form is a column that contains a pivot.

Definition

If $M = (A \mid b)$ denotes the augmented matrix of a system of equations $AX = b$, and $M' = (A' \mid b')$ is the matrix in row echelon form obtained by performing a sequence of elementary row operations on M, then a variable corresponding to a non-pivot column of A' is called a **free variable**.

Recall that we obtained the following matrices earlier while reducing some augmented matrices. The first and the third matrices are in row echelon form. The second matrix is **not** in row echelon form.

$$\left(\begin{array}{ccccc|c} 1 & 0 & 0 & \frac{11}{3} & \frac{1}{3} & 8 \\ 0 & 1 & 0 & -\frac{2}{5} & \frac{2}{5} & -\frac{3}{5} \\ 0 & 0 & 1 & -\frac{17}{15} & \frac{2}{15} & -\frac{1}{5} \end{array}\right), \quad \left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & -\frac{2}{15} \\ 0 & 0 & 1 & -\frac{1}{14} \\ 0 & 0 & 1 & 0 \end{array}\right), \quad \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{array}\right).$$

Fact 3.84. Every homogeneous system of linear equations with more variables than equations has at least one free variable, and hence, admits infinitely many solutions. That is, if A is an $m \times n$ matrix with $m < n$, then the homogeneous system of equations $AX = 0$ admits a nonzero solution, and hence, infinitely many solutions.

A proof of the above is provided in Chapter 8.

Fact 3.85. Let A be an $n \times n$ matrix. The following statements are equivalent.

1. The matrix A can be transformed into the identity matrix I_n by performing a sequence of elementary row operations.
2. The matrix A is a product of elementary matrices.
3. The matrix A is invertible.
4. The system of equations $AX = b$ admits a unique solution for every column vector b having n entries.
5. The zero solution is the only solution to the homogeneous system of equations $AX = 0$.

A proof of the above is provided in Chapter 8.

Exercise 3.86. Let A be a square matrix. If a sequence of elementary row operations can be performed on A to obtain the identity matrix, then A is invertible, and the same sequence of elementary row operations, when performed on the identity matrix, yields the inverse of A.

Solution. Assume that a sequence of elementary row operations can be performed on A to obtain the identity matrix. Let $E_1, E_2, \dots, E_k$ be the corresponding elementary matrices. Then we have

$$E_k \dots E_2 E_1 A = I.$$

Since each E_i is invertible, the product

$$E_k \dots E_2 E_1$$

is also invertible. This shows that A is invertible, and that

$$A^{-1} = E_k ... E_2 E_1 I.$$

This shows that performing the same sequence of elementary row operations on the identity matrix yields the inverse of A. □

Definition

The **rank** of a matrix A is the number of pivots in the row echelon form of A. The rank of a matrix A is denoted by $\mathrm{rk}(A)$.

Exercise 3.87. Find the rank of the matrix

$$\begin{pmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & -\frac{2}{15} \\ 0 & 0 & 1 & -\frac{1}{14} \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Exercise 3.88. If A is an $m \times n$ matrix, then show that

$$\mathrm{rk}(A) \le \min\{m, n\}.$$

Exercise 3.89. If A is an $n \times n$ matrix, then show that the following statements are equivalent.

1. The matrix A is invertible.
2. The rank of A is n.

Solution. Using Fact 3.85, it follows that following statements are equivalent.

1. The matrix A is invertible.
2. The matrix A can be transformed into the identity matrix by performing a sequence of elementary row operations.
3. The row echelon form of A has n pivots.
4. The rank of A is equal to n.

□

Exercise 3.90. Let A be an $m \times n$ matrix, and E be an $m \times m$ elementary matrix. Show that

$$\mathrm{rk}(EA) = \mathrm{rk}(A).$$

More generally, if B is an $m \times m$ invertible matrix, then show that

$$\mathrm{rk}(BA) = \mathrm{rk}(A).$$

Exercise 3.91. If A is an invertible square matrix, then show that A^{-1} is also invertible, and that

$$\left(A^{-1}\right)^{-1} = A.$$

Solution. Suppose A is an invertible square matrix. We have

$$AA^{-1} = A^{-1}A = I.$$

This shows that A^{-1} is also invertible, and that

$$\left(A^{-1}\right)^{-1} = A.$$

□

Example

The matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

is **not** in row echelon form, since the second row is a zero row, but is not at the bottom of the matrix. The rank of this matrix is 2, since its row echelon form is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Exercise 3.92. If A, B are square matrices of the same size satisfying $BA = I$, then show that the matrices A, B are invertible, and are the inverses of each other, and in particular, $AB = I$ holds.

Compare the above with Exercise 3.74, Exercise 3.75, Exercise 3.29.

Remark

Does considering the row echelon form of B help?

Solution. Let E be a product of elementary matrices such that EB is in row echelon form. Note that

$$(EB)A = E$$

holds. Since EB is in row echelon form, it has a pivot in every nonzero row. If EB has a zero row, then the corresponding row of E is also zero, and hence E is not invertible, which is not possible since E being a product of elementary matrices is invertible. This shows that EB has no zero row, and hence, has a pivot in every row. This implies that EB is the identity matrix[12]. This shows that B is invertible, and that $B^{-1} = E$. Moreover, we obtain $A = E$, which shows that A is invertible, and that

$$A^{-1} = E^{-1} = \left(B^{-1}\right)^{-1} = B$$

holds. ☐

Exercise 3.93. Let A, B be square matrices of the same size. Show that

$$\mathrm{rk}(AB) \leq \mathrm{rk}(A).$$

Solution. Let E be a product of elementary matrices such that EA is in row echelon form. Note that

$$E(AB) = (EA)B$$

holds. Since EA is in row echelon form, it has a pivot in every nonzero row. If a row of EA is zero, then the corresponding row of $E(AB)$ is also zero, and the corresponding row of $E'(E(AB))$ is also zero for any product E' of elementary matrices. This shows that the number of pivots in the row echelon form of $E(AB)$ is at most the number of pivots in EA. This shows that

$$\mathrm{rk}(AB) = \mathrm{rk}(E(AB)) \leq \mathrm{rk}(EA) = \mathrm{rk}(A).$$

☐

[12] Why?

§3.8 Systems of linear equations in three variables and determinants of 3×3 matrices

§3.8.1 An example of a system of linear equations in three variables

Exercise 3.94. Solve the system of equations in the variables x, y, z:

$$\begin{aligned} x + 2y + 3z &= 1, \\ 4x + 5y + 6z &= 2, \\ 7x + 8y + 10z &= 3. \end{aligned}$$

Remark

If the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{pmatrix}$$

is invertible, then the above system of equations admits a unique solution, which can be found by multiplying A^{-1} from the left to the matrix form of the system of equations, that is, multiplying

$$A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

from the left by A^{-1}, **provided A^{-1} exists**. Note that

$$\frac{1}{3}\begin{pmatrix} -2 & -4 & 3 \\ -2 & 11 & -6 \\ 3 & -6 & 3 \end{pmatrix} A = \frac{1}{3}\begin{pmatrix} -2 & -4 & 3 \\ -2 & 11 & -6 \\ 3 & -6 & 3 \end{pmatrix}\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{pmatrix} = I_3,$$

and

$$A \cdot \frac{1}{3}\begin{pmatrix} -2 & -4 & 3 \\ -2 & 11 & -6 \\ 3 & -6 & 3 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{pmatrix}\begin{pmatrix} -2 & -4 & 3 \\ -2 & 11 & -6 \\ 3 & -6 & 3 \end{pmatrix} = I_3.$$

This shows that A is invertible, and that

$$A^{-1} = \frac{1}{3}\begin{pmatrix} -2 & -4 & 3 \\ -2 & 11 & -6 \\ 3 & -6 & 3 \end{pmatrix}.$$

§3.8.2 Determinants and adjoints of 3×3 matrices

Definition

For a 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

its **adjoint** is denoted by $\operatorname{adj}(A)$, and is defined to be

$$\operatorname{adj}(A) := \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Exercise 3.95. Let

$$A = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$$

where a, b are scalars. Show that $\det(A) = ab$, and that

$$\operatorname{adj}(A) = \begin{pmatrix} b & 0 \\ 0 & a \end{pmatrix}.$$

Exercise 3.96. (Cramer's rule for 2×2 matrices) Prove that for a 2×2 matrix A, the equalities

$$A \cdot \operatorname{adj}(A) = \operatorname{adj}(A) \cdot A = \det(A) I_2$$

hold (cf. Lemma 3.24).

Moreover, if A, B are 2×2 matrices, then

$$\det(AB) = \det(A)\det(B)$$

holds, as shown in Exercise 3.25.

Definition

Let A be a 3×3 matrix. Write $A = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}$. The **determinant** of A is denoted by $\det(A)$, and is defined to be

$$\det(A) := a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2).$$

The **adjoint** of A is denoted by $\operatorname{adj}(A)$, and is defined to be the matrix

$$\operatorname{adj}(A) := \begin{pmatrix} b_2c_3 - b_3c_2 & -(b_1c_3 - b_3c_1) & b_1c_2 - b_2c_1 \\ -(c_3a_2 - c_2a_3) & c_3a_1 - c_1a_3 & -(c_2a_1 - c_1a_2) \\ a_2b_3 - a_3b_2 & -(a_1b_3 - a_3b_1) & a_1b_2 - a_2b_1 \end{pmatrix}.$$

Exercise 3.97. Show that the determinant of the identity matrix I_3 is 1, and that

$$\operatorname{adj}(I_3) = I_3.$$

Exercise 3.98. Let

$$A = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix},$$

where a, b, c are scalars. Show that $\det(A) = abc$, and that

$$\operatorname{adj}(A) = \begin{pmatrix} bc & 0 & 0 \\ 0 & ca & 0 \\ 0 & 0 & ab \end{pmatrix}.$$

Exercise 3.99. Let

$$A = \begin{pmatrix} 1 & -5 & 6 \\ -4 & 8 & -9 \\ 7 & 2 & 10 \end{pmatrix}.$$

Find $\det(A)$ and $\operatorname{adj}(A)$.

Exercise 3.100. Let

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 3 & 2 & 1 \\ 4 & 1 & 2 \end{pmatrix}.$$

Find $\det(A)$ and $\operatorname{adj}(A)$.

Fact 3.101. (Cramer's rule for 3×3 matrices) For any 3×3 matrix A, the following holds.

$$A \cdot \operatorname{adj}(A) = \operatorname{adj}(A) \cdot A = \det(A) I_3.$$

Exercise 3.102. Let A be a 3×3 matrix. If two rows of A are equal, then $\det(A) = 0$.

Lemma 3.103. *If A, B are 3×3 matrices, then*

$$\det(AB) = \det(A)\det(B).$$

A proof of the above is provided in Chapter 8.

The following is an analogue of Lemma 3.24 for 3×3 matrices.

Lemma 3.104. *Let A be a 3×3 matrix. The following statements are equivalent.*

1. *The matrix A is invertible.*
2. *The determinant $\det(A)$ of A is nonzero.*
3. *Every vector*

$$\begin{pmatrix} e \\ f \\ g \end{pmatrix}$$

can be expressed as a linear combination of the columns of A. Moreover, if A is invertible, then

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

holds.

A proof of the above is provided in Chapter 8.

Exercise 3.105. Compute the determinant of the matrix

$$A = \begin{pmatrix} 1 & -5 & 6 \\ 2 & 0 & 9 \\ -1 & 2 & -1 \end{pmatrix}$$

to determine whether it is invertible. If A is invertible, then find its inverse using its adjoint.

Solution. Note that

$$\det(A) = 1 \times (-18) - (-5) \times 7 + 6 \times 4 = -18 + 35 + 24 = 41.$$

Since $\det(A)$ is nonzero, it follows that the matrix A is invertible.

Note that

$$\operatorname{adj}(A) = \begin{pmatrix} -18 & 7 & -45 \\ -7 & 5 & 3 \\ 4 & 3 & 10 \end{pmatrix}.$$

Hence, we obtain

$$A^{-1} = \frac{1}{\det(A)}\operatorname{adj}(A) = \frac{1}{41}\begin{pmatrix} -18 & 7 & -45 \\ -7 & 5 & 3 \\ 4 & 3 & 10 \end{pmatrix}.$$

□

Exercise 3.106. Use Gaussian elimination to determine the row echelon form of the matrix

$$A = \begin{pmatrix} 1 & -5 & 6 \\ 2 & 0 & 9 \\ -1 & 2 & -1 \end{pmatrix}.$$

Use the row echelon form to determine whether A is invertible. If A is invertible, then find its inverse using elementary matrices, corresponding to the elementary row operations used to transform A into its row echelon form.

Solution. Adding -2 times the first row to the second row, we obtain the matrix

$$\begin{pmatrix} 1 & -5 & 6 \\ 0 & 10 & -3 \\ -1 & 2 & -1 \end{pmatrix}.$$

Adding the first row to the third row, we obtain the matrix

$$\begin{pmatrix} 1 & -5 & 6 \\ 0 & 10 & -3 \\ 0 & -3 & 5 \end{pmatrix}.$$

Next, multiplying the second row by $\frac{1}{10}$, we obtain the matrix

$$\begin{pmatrix} 1 & -5 & 6 \\ 0 & 1 & -\frac{3}{10} \\ 0 & -3 & 5 \end{pmatrix}.$$

Adding 3 times the second row to the third row, we obtain the matrix

$$\begin{pmatrix} 1 & -5 & 6 \\ 0 & 1 & -\frac{3}{10} \\ 0 & 0 & \frac{41}{10} \end{pmatrix}.$$

Multiplying the third row by $\frac{10}{41}$, we obtain the matrix

$$\begin{pmatrix} 1 & -5 & 6 \\ 0 & 1 & -\frac{3}{10} \\ 0 & 0 & 1 \end{pmatrix},$$

Adding $\frac{3}{10}$ times the third row to the second row, we obtain the matrix

$$\begin{pmatrix} 1 & -5 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Adding -6 times the third row to the first row, we obtain the matrix

$$\begin{pmatrix} 1 & -5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Finally, adding 5 times the second row to the first row, we obtain the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3.$$

The above is the row echelon form of A. Since the row echelon form of A is the identity matrix I_3, it follows that A is invertible.

The elementary matrices corresponding to the above elementary row operations are

$$E_1 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{10} & 0 \\ 0 & 0 & 1 \end{pmatrix}, E_4 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix},$$

$$E_5 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{10}{41} \end{pmatrix}, E_6 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{3}{10} \\ 0 & 0 & 1 \end{pmatrix}, E_7 = \begin{pmatrix} 1 & 0 & -6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, E_8 = \begin{pmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

It follows that

$$\begin{aligned}
A^{-1} &= E_8 E_7 E_6 E_5 E_4 E_3 E_2 E_1 \\
&= \begin{pmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{3}{10} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{10}{41} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{10} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 5 & -6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{3}{41} \\ 0 & 0 & \frac{10}{41} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{10} & 0 \\ 0 & \frac{3}{10} & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 5 & -\frac{45}{41} \\ 0 & 1 & \frac{3}{41} \\ 0 & 0 & \frac{10}{41} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -\frac{2}{10} & \frac{1}{10} & 0 \\ \frac{4}{10} & \frac{3}{10} & 1 \end{pmatrix} \\
&= \frac{1}{41} \begin{pmatrix} -18 & 7 & -45 \\ -7 & 5 & 3 \\ 4 & 3 & 10 \end{pmatrix}.
\end{aligned}$$

□

Exercise 3.107. Solve the following system of equations in the variables x, y, z.

$$\begin{aligned} x + 2y + 3z &= 1, \\ 4x - 5y + 6z &= 2, \\ 7x - y + 10z &= 3. \end{aligned}$$

The following exercises, marked with asterisk, are supplementary (for the moment). These will be straightforward once we have developed the theory of eigenvalues and eigenvectors. These are included here to illustrate the power (and the ease) of matrix multiplication. This may help motivate the study of eigenvalues and eigenvectors, and appreciate their importance.

Exercise 3.108 (*). Calculate

$$\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}^{123}.$$

Solution. Write

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}.$$

Note that

$$A\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad A\begin{pmatrix} 1 \\ 1 \end{pmatrix} = A\begin{pmatrix} 1 \\ 0 \end{pmatrix} + A\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

It follows that

$$A^{123}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, A^{123}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 3^{123}\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Hence, we obtain

$$A^{123}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = A^{123}\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = 3^{123}\begin{pmatrix} 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3^{123} - 1 \\ 3^{123} \end{pmatrix}.$$

This yields

$$A^{123} = \begin{pmatrix} 1 & 3^{123} - 1 \\ 0 & 3^{123} \end{pmatrix}.$$

□

Exercise 3.109 (*). Calculate

$$\begin{pmatrix} 20 & 25 \\ 0 & 2025 \end{pmatrix}^{2025}.$$

Solution. Note that

$$\begin{pmatrix} 20 & 25 \\ 0 & 2025 \end{pmatrix} = \begin{pmatrix} 1 & \frac{5}{401} \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 20 & 0 \\ 0 & 2025 \end{pmatrix}\begin{pmatrix} 1 & \frac{-5}{401} \\ 0 & 1 \end{pmatrix}.$$

It follows that

$$\begin{aligned}
\begin{pmatrix} 20 & 25 \\ 0 & 2025 \end{pmatrix}^{2025} &= \begin{pmatrix} 1 & \frac{5}{401} \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 20 & 0 \\ 0 & 2025 \end{pmatrix}^{2025}\begin{pmatrix} 1 & \frac{-5}{401} \\ 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 1 & \frac{5}{401} \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 20^{2025} & 0 \\ 0 & 2025^{2025} \end{pmatrix}\begin{pmatrix} 1 & \frac{-5}{401} \\ 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 20^{2025} & \frac{5}{401}(2025^{2025} - 20^{2025}) \\ 0 & 2025^{2025} \end{pmatrix}.
\end{aligned}$$

□

Chapter 4. The space $\mathbb{R}^n$

§4.1 Vectors in $\mathbb{R}^n$

Definition

A vector in $\mathbb{R}^n$ is an ordered n-tuple of real numbers. The set of all such vectors is denoted by $\mathbb{R}^n$. Formally,

$$\mathbb{R}^n = \{(x_1, x_2, ..., x_n) : x_i \in \mathbb{R} \text{ for all } i = 1, 2, ..., n\}.$$

Here, each x_i represents the i-th component of the vector.

It is customary to consider these vectors as column vectors, i.e., as $n \times 1$ matrices. Thus, a vector $x \in \mathbb{R}^n$ is often written as

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

The vector $\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ is called the **zero vector** and is denoted by 0.

Definition

A **linear combination** of vectors $v_1, v_2, ..., v_k \in \mathbb{R}^n$ **with coefficients** $\alpha_1, \alpha_2, ..., \alpha_k \in \mathbb{R}$ is a vector of $\mathbb{R}^n$ of the form

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k.$$

A linear combination is called **trivial** if all its coefficients are zero, that is, if $\alpha_1 = \alpha_2 = \cdots = \alpha_k = 0$. A linear combination of $v_1, v_2, ..., v_k$ is also called a **linear combination** of the set of vectors $\{v_1, v_2, ..., v_k\}$.

Example

Consider the elements $v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ of $\mathbb{R}^2$. The linear combination of v_1 and v_2 with coefficients 2 and -1 is

$$2v_1 - 1v_2 = 2\begin{pmatrix} 1 \\ 2 \end{pmatrix} - 1\begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}.$$

Example

Consider the elements $e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and $e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ of $\mathbb{R}^3$. The linear combination of e_1, e_2, and e_3 with coefficients 3, -2, and 1 is

$$3e_1 - 2e_2 + 1e_3 = 3\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - 2\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 1\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}.$$

Moreover, any vector $v = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3$ can be expressed as a linear combination of e_1, e_2, and e_3:

$$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = x\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + y\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + z\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = xe_1 + ye_2 + ze_3.$$

§4.2 Linear independence and dependence

Definition

Let k be a positive integer. A set of vectors $v_1, v_2, \dots, v_k \in \mathbb{R}^n$ is said to be **linearly dependent** if there exist elements $\alpha_1, \alpha_2, \dots, \alpha_k \in \mathbb{R}$, not all zero, such that

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = 0.$$

A set of vectors $v_1, v_2, \dots, v_k \in \mathbb{R}^n$ is said to be **linearly independent** if it is not linearly dependent, that is, if $\alpha_1, \alpha_2, \dots, \alpha_k \in \mathbb{R}$ satisfy

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = 0,$$

then $\alpha_1 = \alpha_2 = \dots = \alpha_k = 0$ holds. If $v_1, v_2, \dots, v_k$ are elements of $\mathbb{R}^n$, such that the set $\{v_1, v_2, \dots, v_k\}$ is linearly dependent (resp. linearly independent), then the vectors $v_1, v_2, \dots, v_k$ are said to be linearly dependent (resp. linearly independent). The empty subset $\emptyset$ of $\mathbb{R}^n$ is considered to be linearly independent by convention.

Exercise 4.1. Let S be a subset of $\mathbb{R}^n$, containing precisely one element v. Show that S is linearly independent if and only if $v \neq 0$.

Solution. Note that the set $\{v\}$ is linearly dependent if and only if there exists a nonzero scalar $\alpha \in \mathbb{R}$ such that

$$\alpha v = 0.$$

This is equivalent to saying that v is the zero vector. Hence, the set $\{v\}$ is linearly independent if and only if $v \neq 0$. □

Lemma 4.2. *Let k be a positive integer. A set of vectors $v_1, v_2, \dots, v_k \in \mathbb{R}^n$ is linearly dependent if and only if at least one of the vectors can be expressed as a linear combination of the others.*

A proof of the above is provided in Chapter 9.

Example

The vectors $v_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $v_2 = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$, and $v_3 = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$ in $\mathbb{R}^3$ are linearly dependent because

$$v_3 = -1v_1 + 2v_2.$$

Example

The vectors $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and $v_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ in $\mathbb{R}^3$ are linearly independent. Indeed, if

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = 0$$

holds for some $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$, then we obtain

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = 0,$$

implying $\alpha_1 = \alpha_2 = \alpha_3 = 0$.

Example

The vectors $v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$ in $\mathbb{R}^2$ are linearly dependent because

$$v_2 = 2v_1.$$

Example

The vectors $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ in $\mathbb{R}^2$ are linearly independent. Indeed, let $\alpha_1, \alpha_2 \in \mathbb{R}$ be such that

$$\alpha_1 \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ 3 \end{pmatrix} = 0$$

holds. This equation translates to the system of equations:

$$\alpha_1 + 2\alpha_2 = 0,$$
$$4\alpha_1 + 3\alpha_2 = 0,$$

which can be expressed in matrix form as

$$\begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Since the matrix $\begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$ is invertible (its determinant is $1 \times 3 - 2 \times 4 = -5 \neq 0$), the trivial solution $\alpha_1 = \alpha_2 = 0$ is the only solution to the above homogeneous system. This confirms that the vectors $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ are linearly independent.

Revisit this problem in the light of Lemma 3.24, and Lemma 3.32.

Example

The vectors $\begin{pmatrix}1\\2\\3\end{pmatrix}$, $\begin{pmatrix}4\\5\\6\end{pmatrix}$, and $\begin{pmatrix}7\\8\\10\end{pmatrix}$ in $\mathbb{R}^3$ are linearly independent. Indeed, let $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ be such that

$$\alpha_1 \begin{pmatrix}1\\2\\3\end{pmatrix} + \alpha_2 \begin{pmatrix}4\\5\\6\end{pmatrix} + \alpha_3 \begin{pmatrix}7\\8\\10\end{pmatrix} = 0$$

holds. This equation translates to the system of equations:

$$\begin{aligned} \alpha_1 + 4\alpha_2 + 7\alpha_3 &= 0, \\ 2\alpha_1 + 5\alpha_2 + 8\alpha_3 &= 0, \\ 3\alpha_1 + 6\alpha_2 + 10\alpha_3 &= 0, \end{aligned}$$

which can be expressed in matrix form as

$$\begin{pmatrix}1 & 4 & 7\\2 & 5 & 8\\3 & 6 & 10\end{pmatrix}\begin{pmatrix}\alpha_1\\\alpha_2\\\alpha_3\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}.$$

Since the matrix $\begin{pmatrix}1 & 4 & 7\\2 & 5 & 8\\3 & 6 & 10\end{pmatrix}$ is invertible (its determinant is $1 \times (5 \times 10 - 6 \times 8) - 4 \times (2 \times 10 - 3 \times 8) + 7 \times (2 \times 6 - 3 \times 5) = -3 \neq 0$), the only solution to the above homogeneous system is the trivial solution $\alpha_1 = \alpha_2 = \alpha_3 = 0$. This confirms that the vectors $\begin{pmatrix}1\\2\\3\end{pmatrix}$, $\begin{pmatrix}4\\5\\6\end{pmatrix}$, and $\begin{pmatrix}7\\8\\10\end{pmatrix}$ are linearly independent.

Revisit this problem in the light of Lemma 3.104.

Exercise 4.3. In each of the following, determine whether the given vectors are linearly independent or dependent. Justify your answers.

1. $\begin{pmatrix}1\\2\\3\end{pmatrix}$, $\begin{pmatrix}4\\5\\6\end{pmatrix}$, and $\begin{pmatrix}7\\8\\9\end{pmatrix}$ in $\mathbb{R}^3$.
2. $\begin{pmatrix}1\\0\\0\end{pmatrix}$, $\begin{pmatrix}0\\1\\0\end{pmatrix}$, and $\begin{pmatrix}0\\0\\1\end{pmatrix}$ in $\mathbb{R}^3$.
3. $\begin{pmatrix}1\\2\end{pmatrix}$ and $\begin{pmatrix}2\\4\end{pmatrix}$ in $\mathbb{R}^2$.
4. $\begin{pmatrix}1\\4\end{pmatrix}$ and $\begin{pmatrix}2\\3\end{pmatrix}$ in $\mathbb{R}^2$.
5. $\begin{pmatrix}1\\2\\3\end{pmatrix}$, $\begin{pmatrix}4\\5\\6\end{pmatrix}$, and $\begin{pmatrix}7\\8\\10\end{pmatrix}$ in $\mathbb{R}^3$.

Exercise 4.4. Let $v_1 = \begin{pmatrix}1\\-2\\3\end{pmatrix}$, $v_2 = \begin{pmatrix}4\\0\\-1\end{pmatrix}$, $v_3 = \begin{pmatrix}-1\\5\\2\end{pmatrix}$. Determine whether the vectors v_1, v_2, v_3 are linearly independent or dependent. Justify your answer.

Solution. Let x, y, z be real numbers, to be determined, such that

$$xv_1 + yv_2 + zv_3 = 0$$

holds. This equation translates to the system of equations:

$$\begin{pmatrix} 1 & 4 & -1 \\ -2 & 0 & 5 \\ 3 & -1 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

The determinant of the matrix $\begin{pmatrix} 1 & 4 & -1 \\ -2 & 0 & 5 \\ 3 & -1 & 2 \end{pmatrix}$ is

$$5 + 76 - 2 = 79 \neq 0.$$

Since the matrix is invertible, the only solution to the above homogeneous system is the trivial solution $x = y = z = 0$. This shows that the vectors v_1, v_2, v_3 are linearly independent. □

Exercise 4.5. Determine whether the vectors $\begin{pmatrix} 1 \\ 2 \\ -1 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 0 \\ 5 \\ -2 \end{pmatrix}$, $\begin{pmatrix} -2 \\ 1 \\ 3 \\ 4 \end{pmatrix}$, and $\begin{pmatrix} 3 \\ -1 \\ 2 \\ 1 \end{pmatrix}$ in $\mathbb{R}^4$ are linearly independent or dependent. Justify your answer.

Solution. Let $\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathbb{R}$ be such that

$$\alpha_1 \begin{pmatrix} 1 \\ 2 \\ -1 \\ 3 \end{pmatrix} + \alpha_2 \begin{pmatrix} 4 \\ 0 \\ 5 \\ -2 \end{pmatrix} + \alpha_3 \begin{pmatrix} -2 \\ 1 \\ 3 \\ 4 \end{pmatrix} + \alpha_4 \begin{pmatrix} 3 \\ -1 \\ 2 \\ 1 \end{pmatrix} = 0$$

holds. This equation translates to the system of equations:

$$\begin{pmatrix} 1 & 4 & -2 & 3 \\ 2 & 0 & 1 & -1 \\ -1 & 5 & 3 & 2 \\ 3 & -2 & 4 & 1 \end{pmatrix}\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

Let us perform elementary row operations on the matrix

$$A = \begin{pmatrix} 1 & 4 & -2 & 3 \\ 2 & 0 & 1 & -1 \\ -1 & 5 & 3 & 2 \\ 3 & -2 & 4 & 1 \end{pmatrix}.$$

Adding -2 times the first row to the second row yields

$$\begin{pmatrix} 1 & 4 & -2 & 3 \\ 0 & -8 & 5 & -7 \\ -1 & 5 & 3 & 2 \\ 3 & -2 & 4 & 1 \end{pmatrix}.$$

Adding 1 times the first row to the third row yields

$$\begin{pmatrix} 1 & 4 & -2 & 3 \\ 0 & -8 & 5 & -7 \\ 0 & 9 & 1 & 5 \\ 3 & -2 & 4 & 1 \end{pmatrix}.$$

Next, adding -3 times the first row to the fourth row yields

$$\begin{pmatrix} 1 & 4 & -2 & 3 \\ 0 & -8 & 5 & -7 \\ 0 & 9 & 1 & 5 \\ 0 & -14 & 10 & -8 \end{pmatrix}.$$

Note that

$$\begin{pmatrix} 1 & 4 & -2 & 3 \\ 0 & -8 & 5 & -7 \\ 0 & 9 & 1 & 5 \\ 0 & -14 & 10 & -8 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

holds. Since the matrix

$$\begin{pmatrix} -8 & 5 & -7 \\ 9 & 1 & 5 \\ -14 & 10 & -8 \end{pmatrix}$$

is invertible[13], it follows that $\alpha_2 = \alpha_3 = \alpha_4 = 0$. This in turn implies that $\alpha_1 = 0$. This implies that the given vectors are linearly independent. □

Exercise 4.6. Show that the standard basis vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \ldots, e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

of $\mathbb{R}^n$ are linearly independent.

Exercise 4.7. Show that any finite subset of $\mathbb{R}^n$ containing the zero vector is linearly dependent.

Exercise 4.8. Show that any subset of a linearly independent set of vectors is also linearly independent.

Exercise 4.9. Let $v_1, v_2, \ldots, v_k$ be elements of $\mathbb{R}^n$. If $k > n$, then the vectors $v_1, v_2, \ldots, v_k$ are linearly dependent, that is, at least one of the vectors can be expressed as a linear combination of the others.

Solution. Assume that $k > n$. Consider the $n \times k$ matrix A formed by taking $v_1, v_2, \ldots, v_k$ as its columns, that is, A is the $n \times k$ matrix whose j-th column is the vector v_j for each $j = 1, 2, \ldots, k$. Since A is an $n \times k$ matrix with $k > n$, the homogeneous system of equations

$$AX = 0$$

has[14] a non-trivial solution, that is, a solution other than the solution $X = 0$. This shows that there exist coefficients $\alpha_1, \alpha_2, \ldots, \alpha_k$, not all zero, such that

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k = 0.$$

This implies that the vectors $v_1, v_2, \ldots, v_k$ are linearly dependent. □

§4.3 Linear subspaces

Note that $\mathbb{R}^n$ has the following properties:

1. The sum of any two vectors in $\mathbb{R}^n$ is also in $\mathbb{R}^n$.
2. The zero vector is in $\mathbb{R}^n$.
3. For every vector $v \in \mathbb{R}^n$, the vector $-v$ is also in $\mathbb{R}^n$.
4. Any scalar multiple of any vector in $\mathbb{R}^n$ is also in $\mathbb{R}^n$.

These properties motivate the following definition.

[13] Why? Compute its determinant.

[14] Why? Does Fact 3.84 help?

Definition

A subset V of $\mathbb{R}^n$ is called a **linear subspace** of $\mathbb{R}^n$ if it satisfies the following conditions:

1. For any two elements u, v of V, the element $u + v$ of $\mathbb{R}^n$ lies in V.
2. The zero vector of $\mathbb{R}^n$ lies in V.
3. For any $v \in V$, then the vector $-v$ also lies in V.
4. For any $v \in V$ and any scalar $\alpha \in \mathbb{R}$, the scalar multiple αv lies in V.

A linear subspace is also called a **vector subspace** or a **subspace** for brevity.

Exercise 4.10. Determine the linear subspaces of $\mathbb{R}$.

Solution. The only linear subspaces of $\mathbb{R}$ are the subspace containing only the zero vector, that is, the subspace $\{0\}$, and the entire space $\mathbb{R}$ itself. Indeed, let V be a linear subspace of $\mathbb{R}$. If V contains only the zero vector, then $V = \{0\}$. Henceforth, let us assume that V contains a nonzero element x. Since V is a linear subspace of $\mathbb{R}$, for any scalar $\alpha \in \mathbb{R}$, the scalar multiple αx lies in V. This shows that $V = \mathbb{R}$. □

Exercise 4.11. Let V denote the set of all vectors of $\mathbb{R}^n$ whose first component is equal to zero, that is,

$$V = \left\{ \begin{pmatrix} 0 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} : x_i \in \mathbb{R} \text{ for all } i = 2, 3, \ldots, n \right\}.$$

Show that V is a linear subspace of $\mathbb{R}^n$.

Exercise 4.12. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}$ is a linear subspace of $\mathbb{R}^3$.

Solution. Let $u = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$ and $v = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$ be elements of V. Note that

$$u + v = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \\ z_1 + z_2 \end{pmatrix}.$$

Since $x_1 + y_1 + z_1 = 0$ and $x_2 + y_2 + z_2 = 0$, we obtain

$$(x_1 + x_2) + (y_1 + y_2) + (z_1 + z_2) = (x_1 + y_1 + z_1) + (x_2 + y_2 + z_2) = 0 + 0 = 0.$$

This shows that $u + v \in V$.

Note that the zero vector $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ lies in V.

Let $v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ be an element of V. Note that

$$-v = -\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -x \\ -y \\ -z \end{pmatrix}.$$

Since $x + y + z = 0$, we have

$$(-x) + (-y) + (-z) = -(x + y + z) = -0 = 0.$$

This shows that $-v \in V$.

Let $v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ be an element of V and let $\alpha \in \mathbb{R}$ be a scalar. Note that

$$\alpha v = \alpha \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \alpha x \\ \alpha y \\ \alpha z \end{pmatrix}.$$

Since $x + y + z = 0$, we have

$$(\alpha x) + (\alpha y) + (\alpha z) = \alpha(x + y + z) = \alpha \times 0 = 0.$$

$\square$

Exercise 4.13. Let (a, b, c) be a nonzero element of $\mathbb{R}^3$. Show that the set

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : ax + by + cz = 0 \right\}$$

is a linear subspace of $\mathbb{R}^3$.

Exercise 4.14. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x \geq 0, y \geq 0 \right\}$ is not a linear subspace of $\mathbb{R}^2$.

Solution. Note that the vector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ lies in V, and the vector $-1 \times \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$ does not lie in V. This shows that V is not a linear subspace of $\mathbb{R}^2$. $\square$

Exercise 4.15. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x = 1 \right\}$ is not a linear subspace of $\mathbb{R}^2$.

Solution. Note that the zero vector $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ does not lie in V. This shows that V is not a linear subspace of $\mathbb{R}^2$. $\square$

Exercise 4.16. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x = y \right\}$ is a linear subspace of $\mathbb{R}^2$.

Exercise 4.17. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : z = 0 \right\}$ is a linear subspace of $\mathbb{R}^3$.

Exercise 4.18. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 1 \right\}$ is not a linear subspace of $\mathbb{R}^3$.

Solution. Note that the zero vector $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ does not lie in V. This shows that V is not a linear subspace of $\mathbb{R}^3$. $\square$

Exercise 4.19. Show that $\mathbb{R}^n$ is a linear subspace of itself.

Exercise 4.20. Show that the set containing only the zero vector, that is, the set $\{0\}$, is a linear subspace of $\mathbb{R}^n$.

***i* Remark**

Note that a subspace of $\mathbb{R}^n$ is necessarily nonempty since it contains the zero vector of $\mathbb{R}^n$.

Exercise 4.21. Determine the subspaces of $\mathbb{R}^2$.

Exercise 4.22. Determine the subspaces of $\mathbb{R}^3$.

Lemma 4.23. *A subset V of $\mathbb{R}^n$ is a linear subspace if and only if it satisfies the following three conditions:*

1. *The zero vector of $\mathbb{R}^n$ lies in V.*
2. *For any $u, v \in V$, the element $u + v$ of $\mathbb{R}^n$ lies in V.*
3. *For any $v \in V$ and for any $\alpha \in \mathbb{R}$, the element αv of $\mathbb{R}^n$ lies in V.*

Proof. The "only if" part follows from the definition of a linear subspace.

To establish the "if" part, let us assume that V is a subset of $\mathbb{R}^n$ satisfying the three conditions mentioned above. Note that for any $v \in V$, the vector $-v$ can be expressed as $(-1) \cdot v$. Since $v \in V$ and $-1 \in \mathbb{R}$, by the third condition, we have $-v \in V$. $\square$

Lemma 4.24. *A subset V of $\mathbb{R}^n$ is a linear subspace if and only if it satisfies the following two conditions:*

1. *The set V is nonempty.*
2. *For any $u, v \in V$ and any $\alpha, \beta \in \mathbb{R}$, the linear combination $\alpha u + \beta v$ lies in V.*

Compare the following two exercises with Exercise 3.35, Exercise 3.36, Exercise 3.64, Exercise 3.65.

Exercise 4.25. Let A be an $m \times n$ matrix. Let V be the set of solutions to the system of equations

$$Au = 0,$$

that is,

$$V = \{v \in \mathbb{R}^n : Av = 0\}.$$

Show that V is a linear subspace of $\mathbb{R}^n$.

Solution. Let $u, v \in V$ and $\alpha, \beta \in \mathbb{R}$. Note that

$$A(\alpha u + \beta v) = \alpha A u + \beta A v = \alpha 0 + \beta 0 = 0.$$

This shows that $\alpha u + \beta v \in V$. Since the zero vector lies in V, it follows that V is nonempty. This proves that V is a linear subspace of $\mathbb{R}^n$. $\square$

Exercise 4.26. Let A be an $m \times n$ matrix. Let V be the set of vectors which can be expressed as a linear combination of the columns of A, that is,

$$V = \{v \in \mathbb{R}^m : v = Au \text{ for some } u \in \mathbb{R}^n\}.$$

Show that V is a linear subspace of $\mathbb{R}^m$.

Definition

Let A be an $m \times n$ matrix with real entries. The **null space** of A, denoted by $\mathrm{null}(A)$, is the set of solutions to the homogeneous system of equations

$$Au = 0,$$

that is,

$$\mathrm{null}(A) := \{v \in \mathbb{R}^n : Av = 0\}.$$

The **column space** of A, denoted by $\mathrm{col}(A)$, is the set of vectors which can be expressed as a linear combination of the columns of A, that is,

$$\mathrm{col}(A) := \{v \in \mathbb{R}^m : v = Au \text{ for some } u \in \mathbb{R}^n\}.$$

Exercise 4.27. Let A be an $m \times n$ matrix with real entries. Show that for any invertible $m \times m$ matrix E, the null space $\text{null}(A)$ is equal to the null space $\text{null}(EA)$.

Solution. Let $v \in \mathbb{R}^n$. Note that if

$$Av = 0$$

holds, then for any matrix E, one obtains

$$EAv = E0 = 0.$$

Moreover, if E is an invertible matrix and

$$EAv = 0$$

holds, then we obtain

$$Av = E^{-1}(EAv) = E^{-1}0 = 0.$$

This shows that

$$\text{null}(A) = \text{null}(EA)$$

holds for any invertible matrix E. □

Remark

Let A denote the matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Note that the column space $\text{col}(A)$ of A is equal to the span of the vector $\binom{1}{0}$. Let E denote the invertible matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Note that

$$EA = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Hence, the column space $\text{col}(EA)$ of EA is equal to the span of the vector $\binom{0}{1}$. Since $\binom{1}{0}$ is not a scalar multiple of $\binom{0}{1}$, the column spaces $\text{col}(A)$ and $\text{col}(EA)$ are not equal.

Exercise 4.28. Let m, n be positive integers with $m \geq 2$. Provide an example of an $m \times n$ matrix A with real entries and an invertible $m \times m$ matrix E such that the column space $\text{col}(A)$ of A is not equal to the column space $\text{col}(EA)$ of EA.

Exercise 4.29. Let U, V be the subsets of $\mathbb{R}^3$ defined as

$$U = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2y = 0 \right\},$$

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}.$$

Show that both U and V are linear subspaces of $\mathbb{R}^3$. Determine the intersection $U \cap V$ of U and V. Is the intersection $U \cap V$ a linear subspace of $\mathbb{R}^3$?

Solution. Let us first show that U is a linear subspace of $\mathbb{R}^3$. Note that the zero vector lies in U since $0 + 2 \times 0 = 0$. Let $u = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$ and $v = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$ be elements of U. This means that $x_1 + 2y_1 = 0$ and $x_2 + 2y_2 = 0$. Let $\alpha, \beta \in \mathbb{R}$. Note that

$$\alpha u + \beta v = \begin{pmatrix} \alpha x_1 + \beta x_2 \\ \alpha y_1 + \beta y_2 \\ \alpha z_1 + \beta z_2 \end{pmatrix}.$$

Since

$$(\alpha x_1 + \beta x_2) + 2(\alpha y_1 + \beta y_2) = \alpha(x_1 + 2y_1) + \beta(x_2 + 2y_2) = \alpha \times 0 + \beta \times 0 = 0,$$

it follows that $\alpha u + \beta v \in U$. This shows that U is a linear subspace of $\mathbb{R}^3$.

Next, let us show that V is a linear subspace of $\mathbb{R}^3$. Note that the zero vector lies in V since $0 + 0 + 0 = 0$. Let $u = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$ and $v = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$ be elements of V. This means that $x_1 + y_1 + z_1 = 0$ and $x_2 + y_2 + z_2 = 0$. Let $\alpha, \beta \in \mathbb{R}$. Note that

$$\alpha u + \beta v = \begin{pmatrix} \alpha x_1 + \beta x_2 \\ \alpha y_1 + \beta y_2 \\ \alpha z_1 + \beta z_2 \end{pmatrix}.$$

Since

$$\begin{aligned}(\alpha x_1 + \beta x_2) + (\alpha y_1 + \beta y_2) + (\alpha z_1 + \beta z_2) &= \alpha(x_1 + y_1 + z_1) + \beta(x_2 + y_2 + z_2) \\ &= \alpha \times 0 + \beta \times 0 \\ &= 0,\end{aligned}$$

it follows that $\alpha u + \beta v \in V$. This shows that V is a linear subspace of $\mathbb{R}^3$.

Note that the intersection $U \cap V$ of U and V is equal to

$$\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2y = 0 \text{ and } x + y + z = 0 \right\}.$$

In other words, the intersection $U \cap V$ of U and V is the set of solutions to the system of equations

$$\begin{pmatrix} 1 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Adding -1 times the first row to the second row of the matrix

$$\begin{pmatrix} 1 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

yields

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \end{pmatrix}.$$

Multiplying the second row of the above matrix by -1 yields

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \end{pmatrix}.$$

Adding -2 times the second row of the above matrix to the first row of the above matrix yields

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix}.$$

This shows that

$$U \cap V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2y = 0 \text{ and } x + y + z = 0 \right\}$$
$$= \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2z = 0 \text{ and } y - z = 0 \right\},$$

and hence,

$$\boxed{U \cap V = \left\{ \begin{pmatrix} -2t \\ t \\ t \end{pmatrix} : t \in \mathbb{R} \right\}}.$$

Let $u = \begin{pmatrix} -2t_1 \\ t_1 \\ t_1 \end{pmatrix}$ and $v = \begin{pmatrix} -2t_2 \\ t_2 \\ t_2 \end{pmatrix}$ be elements of $U \cap V$. Let $\alpha, \beta \in \mathbb{R}$. Note that

$$\alpha u + \beta v = \begin{pmatrix} -2(\alpha t_1 + \beta t_2) \\ \alpha t_1 + \beta t_2 \\ \alpha t_1 + \beta t_2 \end{pmatrix},$$

which is an element of $U \cap V$. Also note that the zero vector lies in $U \cap V$. This shows that $U \cap V$ is a linear subspace of $\mathbb{R}^3$. □

Exercise 4.30. Show that the intersection of two linear subspaces of $\mathbb{R}^n$ is also a linear subspace of $\mathbb{R}^n$.

Solution. Let U, V be linear subspaces of $\mathbb{R}^n$. Since the zero vector lies in both U and V, it follows that the zero vector lies in $U \cap V$, implying that $U \cap V$ is nonempty. Let $w_1, w_2 \in U \cap V$ and $\alpha, \beta \in \mathbb{R}$. Since $w_1, w_2 \in U$ and U is a linear subspace of $\mathbb{R}^n$, we have $\alpha w_1 + \beta w_2 \in U$. Similarly, since $w_1, w_2 \in V$ and V is a linear subspace of $\mathbb{R}^n$, we have $\alpha w_1 + \beta w_2 \in V$. This shows that $\alpha w_1 + \beta w_2 \in U \cap V$. We conclude that $U \cap V$ is a linear subspace of $\mathbb{R}^n$. □

§4.4 Linear span

Definition

Let k be a positive integer. Let $v_1, v_2, \ldots, v_k$ be elements of $\mathbb{R}^n$. Let S denote the set $\{v_1, v_2, \ldots, v_k\}$. The **linear span** of S is denoted by $\operatorname{span}(S)$, and is defined as the set of all linear combinations of the vectors $v_1, v_2, \ldots, v_k$, that is,

$$\operatorname{span}(S) := \{\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k : \alpha_i \in \mathbb{R} \text{ for all } i = 1, 2, \ldots, k\}.$$

The set $\operatorname{span}(S)$ is also called the **linear span** of the vectors $v_1, v_2, \ldots, v_k$. The **linear span** of the empty set is defined to be the set containing only the zero vector, that is,

$$\operatorname{span}(\emptyset) := \{0\}.$$

Exercise 4.31. Show that the linear span of any finite set of vectors in $\mathbb{R}^n$ is a linear subspace of $\mathbb{R}^n$.

Solution. If S is the empty set, then $\operatorname{span}(S) = \{0\}$. Since the set containing only the zero vector is a linear subspace of $\mathbb{R}^n$, it follows that $\operatorname{span}(S)$ is a linear subspace of $\mathbb{R}^n$ when S is the empty set. Henceforth, let us assume that S is nonempty. Write $S = \{v_1, v_2, \ldots, v_k\}$ where k is a positive integer. Let $u, v \in \operatorname{span}(S)$ and $\alpha, \beta \in \mathbb{R}$. By definition of linear span, there exist elements $\alpha_1, \alpha_2, \ldots, \alpha_k \in \mathbb{R}$ such that

$$u = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k,$$

and there exist elements $\beta_1, \beta_2, ..., \beta_k \in \mathbb{R}$ such that
$$v = \beta_1 v_1 + \beta_2 v_2 + \cdots + \beta_k v_k.$$
Note that
$$\begin{aligned}\alpha u + \beta v &= \alpha(\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k) + \beta(\beta_1 v_1 + \beta_2 v_2 + \cdots + \beta_k v_k) \\ &= (\alpha\alpha_1 + \beta\beta_1)v_1 + (\alpha\alpha_2 + \beta\beta_2)v_2 + \cdots + (\alpha\alpha_k + \beta\beta_k)v_k.\end{aligned}$$
Since $\alpha\alpha_i + \beta\beta_i \in \mathbb{R}$ for each $i = 1, 2, ..., k$, it follows that $\alpha u + \beta v$ is an element of $\operatorname{span}(S)$. Since the zero vector lies in $\operatorname{span}(S)$, it follows that $\operatorname{span}(S)$ is nonempty. This proves that $\operatorname{span}(S)$ is a linear subspace of $\mathbb{R}^n$. □

Exercise 4.32. Let V be a linear subspace of $\mathbb{R}^n$ and S be a finite subset of V. Show that the linear span of S is contained in V, that is,
$$\operatorname{span}(S) \subseteq V.$$

Solution. If S is the empty set, then $\operatorname{span}(S) = \{0\}$. Since V is a linear subspace of $\mathbb{R}^n$, it contains the zero vector. This shows that $\operatorname{span}(S) \subseteq V$ holds when S is the empty set. Henceforth, let us assume that S is nonempty. Let $v_1, v_2, ..., v_k$ be the elements of S, where k is a positive integer. Let $u \in \operatorname{span}(S)$. By definition of linear span, there exist elements $\alpha_1, \alpha_2, ..., \alpha_k \in \mathbb{R}$ such that
$$u = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k.$$
Since $v_1, v_2, ..., v_k \in V$ and V is a linear subspace of $\mathbb{R}^n$, we have $u \in V$. This shows that $\operatorname{span}(S)$ is contained in V. □

Exercise 4.33. Let k be a positive integer, and $v_1, v_2, ..., v_k \in \mathbb{R}^n$ be linearly independent vectors in $\mathbb{R}^n$. Show that if $u \in \mathbb{R}^n$ is not in the span of $\{v_1, v_2, ..., v_k\}$, then the set of vectors $\{v_1, v_2, ..., v_k, u\}$ is linearly independent.

Solution. Let $\alpha_1, \alpha_2, ..., \alpha_k, \beta \in \mathbb{R}$ be such that
$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k + \beta u = 0$$
holds. We need to show that $\alpha_1 = \alpha_2 = \cdots = \alpha_k = \beta = 0$. If $\beta = 0$, then since $v_1, v_2, ..., v_k$ are linearly independent, it follows that $\alpha_1 = \alpha_2 = \cdots = \alpha_k = 0$. Hence, let us assume that $\beta \neq 0$. This implies that
$$u = -\frac{\alpha_1}{\beta}v_1 - \frac{\alpha_2}{\beta}v_2 - \cdots - \frac{\alpha_k}{\beta}v_k,$$
which shows that u lies in $\operatorname{span}(\{v_1, v_2, ..., v_k\})$. This contradicts the assumption that $u \notin \operatorname{span}(\{v_1, v_2, ..., v_k\})$. We conclude that $\beta = 0$, and hence, $\alpha_1 = \alpha_2 = \cdots = \alpha_k = 0$. This shows that the set of vectors $\{v_1, v_2, ..., v_k, u\}$ is linearly independent. □

Exercise 4.34. Let $v_1, v_2, ..., v_k$ be elements of $\mathbb{R}^n$. Show that if $u \in \mathbb{R}^n$ is in the span of $\{v_1, v_2, ..., v_k\}$, then the set of vectors $\{v_1, v_2, ..., v_k, u\}$ is linearly dependent.

Solution. Since $u \in \operatorname{span}(\{v_1, v_2, ..., v_k\})$, there exist elements $\alpha_1, \alpha_2, ..., \alpha_k \in \mathbb{R}$ such that
$$u = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k.$$
This implies that
$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k - 1 \times u = 0.$$
Since at least one of the coefficients $\alpha_1, \alpha_2, ..., \alpha_k, -1$ is nonzero, it follows that the set of vectors $\{v_1, v_2, ..., v_k, u\}$ is linearly dependent. □

Definition

Let U, V be linear subspaces of $\mathbb{R}^n$. The **sum** of U and V, denoted by $U+V$, is defined as
$$U+V := \{w \in \mathbb{R}^n : w = u+v \text{ for some } u \in U, v \in V\}.$$

Exercise 4.35. Let U, V be linear subspaces of $\mathbb{R}^n$. Show that the sum $U+V$ is equal to the sum $V+U$.

Solution. Let $w \in U+V$. By definition of sum of subspaces, there exist elements $u \in U$ and $v \in V$ such that
$$w = u+v.$$
Note that
$$w = u+v = v+u.$$
Since $v \in V$ and $u \in U$, it follows that $w \in V+U$. This shows that $U+V \subseteq V+U$. Similarly, it follows that $V+U \subseteq U+V$. We conclude that $U+V = V+U$. □

Exercise 4.36. Let U, V be linear subspaces of $\mathbb{R}^n$. Show that $U \subseteq U+V$ and $V \subseteq U+V$.

Solution. Let $u \in U$. Note that $u = u+0$, where 0 is the zero vector of $\mathbb{R}^n$. Since the zero vector lies in V, it follows that $u \in U+V$. This shows that $U \subseteq U+V$. Similarly, it follows that $V \subseteq U+V$. □

Exercise 4.37. Let V be a linear subspace of $\mathbb{R}^n$. Show that $V+V = V$.

Solution. Since $V \subseteq V+V$, it suffices to show that $V+V \subseteq V$. Let $w \in V+V$. By definition of sum of subspaces, there exist elements $u, v \in V$ such that
$$w = u+v.$$
Since V is a linear subspace of $\mathbb{R}^n$, we have $w \in V$. This shows that $V+V \subseteq V$. We conclude that $V+V = V$. □

Exercise 4.38. Let U, V be the subsets of $\mathbb{R}^3$ defined as
$$U = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x+2y = 0 \right\},$$
$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x+y+z = 0 \right\}.$$
Determine the sum of U and V.

Exercise 4.39. Show that the sum of two linear subspaces of $\mathbb{R}^n$ is also a linear subspace of $\mathbb{R}^n$.

Solution. Let U, V be linear subspaces of $\mathbb{R}^n$. Since the zero vector lies in both U and V, it follows that the zero vector lies in $U+V$, implying that $U+V$ is nonempty. Let $w_1, w_2 \in U+V$ and $\alpha, \beta \in \mathbb{R}$. By definition of sum of subspaces, there exist elements $u_1, u_2 \in U$ and $v_1, v_2 \in V$ such that
$$w_1 = u_1+v_1, w_2 = u_2+v_2.$$
Note that
$$\begin{aligned} \alpha w_1 + \beta w_2 &= \alpha(u_1+v_1) + \beta(u_2+v_2) \\ &= (\alpha u_1 + \beta u_2) + (\alpha v_1 + \beta v_2). \end{aligned}$$

Since U and V are linear subspaces of $\mathbb{R}^n$, we have $\alpha u_1 + \beta u_2 \in U$ and $\alpha v_1 + \beta v_2 \in V$. This shows that $\alpha w_1 + \beta w_2 \in U + V$. We conclude that $U + V$ is a linear subspace of $\mathbb{R}^n$. $\square$

§4.5 Generation of subspaces

Exercise 4.40. Provide examples of linear subspaces V_1, V_2, V_3 of $\mathbb{R}^3$ satisfying

$$\{0\} \subsetneq V_1 \subsetneq V_2 \subsetneq V_3 \subseteq \mathbb{R}^3.$$

Definition

A linear subspace V of $\mathbb{R}^n$ is said to be **generated** by a finite set of vectors S if V is equal to the linear span of S, that is,

$$V = \operatorname{span}(S).$$

If the above holds, then the set S is called a **generating set** for V.

Remark

Note that if a linear subspace V of $\mathbb{R}^n$ admits a generating set S, then S is a subset of V.

Exercise 4.41. Let e_1, e_2 denote the standard basis vectors of $\mathbb{R}^2$, that is,

$$e_1 := \begin{pmatrix} 1 \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Show that each of the following sets of vectors generates $\mathbb{R}^2$:

1. $\{e_1, e_2\}$,
2. $\{e_1, e_1 + e_2\}$,
3. $\{e_2, e_1 + e_2\}$,
4. $\{e_1 - e_2, e_1 + e_2\}$,
5. $\{e_1, e_2, e_1 + e_2\}$.

Prove that

$$\{ae_1 + be_2, ce_1 + de_2\}$$

generates $\mathbb{R}^2$ if and only if $ad - bc \neq 0$.

Exercise 4.42. Show that the linear subspace $\{0\}$ of $\mathbb{R}^n$ is generated by the set $\{0\}$.

Exercise 4.43. Determine a generating set for the subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}.$$

Solution. Let $v_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$. Note that any vector $\begin{pmatrix} x \\ y \\ z \end{pmatrix} \in V$ can be expressed as

$$\begin{aligned}
\begin{pmatrix} x \\ y \\ z \end{pmatrix} &= \begin{pmatrix} x \\ -x-z \\ z \end{pmatrix} \\
&= x\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + (-z)\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \\
&= xv_1 + (-z)v_2.
\end{aligned}$$

This shows that V is contained in the span of $\{v_1, v_2\}$. Conversely, let $u \in \operatorname{span}(\{v_1, v_2\})$. By definition of linear span, there exist elements $\alpha_1, \alpha_2 \in \mathbb{R}$ such that

$$u = \alpha_1 v_1 + \alpha_2 v_2.$$

Note that

$$\begin{aligned}
u &= \alpha_1\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \alpha_2\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \\
&= \begin{pmatrix} \alpha_1 \\ -\alpha_1 + \alpha_2 \\ -\alpha_2 \end{pmatrix}.
\end{aligned}$$

Since the sum of the components of u is equal to

$$\alpha_1 + (-\alpha_1 + \alpha_2) + (-\alpha_2) = 0,$$

it follows that $u \in V$. This shows that the span of $\{v_1, v_2\}$ is contained in V. We conclude that V is equal to the span of $\{v_1, v_2\}$, and hence $\{v_1, v_2\}$ is a generating set for V. $\square$

Exercise 4.44. Determine a generating set for each of the subspaces mentioned in the exercises in Section 4.3.

Exercise 4.45. Show that $\mathbb{R}^n$ is generated by the standard basis vectors $\{e_1, e_2, \ldots, e_n\}$, where

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \ldots, e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

The following exercise shows that

$$\operatorname{span}(S) + \operatorname{span}(T) = \operatorname{span}(S \cup T)$$

for any finite sets of vectors S and T in $\mathbb{R}^n$.

Exercise 4.46. Let U, V be linear subspaces of $\mathbb{R}^n$. Suppose U is generated by the set of vectors $\{u_1, u_2, \ldots, u_s\}$ and V is generated by the set of vectors $\{v_1, v_2, \ldots, v_t\}$. Show that the sum $U + V$ is generated by the set of vectors $\{u_1, u_2, \ldots, u_s\} \cup \{v_1, v_2, \ldots, v_t\}$.

One can prove the above lemma using Exercise 4.33 and Exercise 4.9. A proof of the above is provided in Chapter 9.

Lemma 4.47. *Every linear subspace of $\mathbb{R}^n$ is generated by a finite set of vectors.*

A proof of the above is provided in Chapter 9.

Lemma 4.48. *Every generating set of a linear subspace V of $\mathbb{R}^n$ contains a generating set consisting of at most n vectors.*

A proof of the above is provided in Chapter 9.

Corollary 4.49. *Every subspace of $\mathbb{R}^n$ is generated by a set of vectors containing at most n vectors.*

Proof. This follows from Lemma 4.47 and Lemma 4.48. □

§4.6 Bases and dimensions

Definition

A **basis** of a linear subspace V of $\mathbb{R}^n$ is a linearly independent set of vectors in V that generates V.

Exercise 4.50. Show that the standard basis vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

form a basis of $\mathbb{R}^n$.

Exercise 4.51. Provide an example of a linear subspace of $\mathbb{R}^n$ admitting more than one basis.

Exercise 4.52. Determine whether the subsets of $\mathbb{R}^2$ as in Exercise 4.41 are bases of $\mathbb{R}^2$.

Exercise 4.53. Determine a basis of the linear subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x - 2y + 6z = 0 \right\}.$$

of $\mathbb{R}^3$.

Solution. Consider the elements

$$v_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix}$$

of $\mathbb{R}^3$. Note that these vectors belong to V. It follows that the span of $\{v_1, v_2\}$ is contained in V. Let us show that $\{v_1, v_2\}$ is a generating set for V. Let $v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ be an element of V. By definition of V, we have

$$x - 2y + 6z = 0.$$

This implies that

$$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$
$$= \begin{pmatrix} x \\ \frac{x+6z}{2} \\ z \end{pmatrix}$$
$$= \frac{x}{2}\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + z\begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix}.$$

This shows that v belongs to the span of $\{v_1, v_2\}$. Consequently, V is equal to the span of $\{v_1, v_2\}$. It remains to show that $\{v_1, v_2\}$ is a linearly independent set of vectors. Let $\alpha_1, \alpha_2 \in \mathbb{R}$ be such that

$$\alpha_1 v_1 + \alpha_2 v_2 = 0.$$

This implies that

$$\alpha_1 \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Equating the first and the third components of the vectors on both sides, we obtain

$$\alpha_1 = 0, \alpha_2 = 0.$$

This shows that the vectors v_1, v_2 are linearly independent. We conclude that $\{v_1, v_2\}$ is a basis of V.

□

Exercise 4.54. Determine a basis of the linear subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y = 0 \right\}.$$

of $\mathbb{R}^3$.

Solution. Consider the elements

$$v_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

of $\mathbb{R}^3$. Note that these vectors belong to V. It follows that the span of $\{v_1, v_2\}$ is contained in V. Let us show that $\{v_1, v_2\}$ is a generating set for V. Let $v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ be an element of V. By definition of V, we have

$$x + y = 0.$$

This implies that

$$\begin{aligned} v &= \begin{pmatrix} x \\ y \\ z \end{pmatrix} \\ &= \begin{pmatrix} x \\ -x \\ z \end{pmatrix} \\ &= x \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + z \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \end{aligned}$$

This shows that v belongs to the span of $\{v_1, v_2\}$. Consequently, V is equal to the span of $\{v_1, v_2\}$. It remains to show that $\{v_1, v_2\}$ is a linearly independent set of vectors. Let $\alpha_1, \alpha_2 \in \mathbb{R}$ be such that

$$\alpha_1 v_1 + \alpha_2 v_2 = 0.$$

This implies that

$$\alpha_1 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Equating the first and the third components of the vectors on both sides, we obtain

$$\alpha_1 = 0, \alpha_2 = 0.$$

This shows that the vectors v_1, v_2 are linearly independent. We conclude that $\{v_1, v_2\}$ is a basis of V. □

Exercise 4.55. Determine a basis of each of the subspaces mentioned in the exercises in Section 4.3.

Lemma 4.56. *Every generating subset of a subspace V of $\mathbb{R}^n$ contains a basis of V.*

A proof of the above is provided in Chapter 9.

Corollary 4.57. *Every subspace of $\mathbb{R}^n$ has a basis containing at most n vectors.*

Proof of Corollary 4.57. By Corollary 4.49, V is generated by a set $\mathcal{G}$ containing at most n vectors. By Lemma 4.56, $\mathcal{G}$ contains a basis $\mathcal{B}$ of V. Since $\mathcal{G}$ contains at most n vectors, it follows that $\mathcal{B}$ contains at most n vectors. □

Lemma 4.58. *Any two bases of a subspace of $\mathbb{R}^n$ contain the same number of vectors.*

A proof of the above is provided in Chapter 9.

Definition

Let V be a linear subspace of $\mathbb{R}^n$. If a basis of V contains precisely k vectors, then one says that the **dimension** of V is k, and writes $\dim(V) = k$.

Exercise 4.59. Show that $\dim(\mathbb{R}^n) = n$.

Lemma 4.60. *Let V be a subspace of $\mathbb{R}^n$. Any linearly independent subset of V can be extended to a basis of V.*

A proof of the above is provided in Chapter 9.

Lemma 4.61. *Let U, V be subspaces of $\mathbb{R}^n$. If U is contained in V, then $\dim(U) \leq \dim(V)$. Moreover, if U is contained in V and $\dim(U) = \dim(V)$, then U is equal to V.*

A proof of the above is provided in Chapter 9.

Lemma 4.62. *Let V be a linear subspace of $\mathbb{R}^n$, and S be a finite subset of V. The following statements are equivalent.*

1. *The set S is a basis of V.*
2. *The set S is a linearly independent set and the number of elements of S is equal to $\dim(V)$.*
3. *The set S is a generating set for V and the number of elements of S is equal to $\dim(V)$.*

A proof of the above is provided in Chapter 9.

Lemma 4.63. *Let S be a finite subset of $\mathbb{R}^n$. The following statements are equivalent.*

1. *The set S is a basis of $\operatorname{span}(S)$.*
2. *The set S is linearly independent.*
3. *The number of elements of S is equal to $\dim(\operatorname{span}(S))$.*

A proof of the above is provided in Chapter 9.

Lemma 4.64. *Let S be a finite subset of $\mathbb{R}^n$. The following statements are equivalent.*

1. *The set S is a basis of $\mathbb{R}^n$.*
2. *The set S is linearly independent and the number of elements of S is equal to n.*
3. *The set S is a generating set for $\mathbb{R}^n$ and the number of elements of S is equal to n.*

Proof of Lemma 4.64. The equivalence of the first three statements follows from Lemma 4.62, taking $V = \mathbb{R}^n$ and using that $\dim(\mathbb{R}^n) = n$. □

Lemma 4.65. *Let S be a subset of $\mathbb{R}^n$ containing precisely n vectors $v_1, v_2, \dots, v_n$. The following statements are equivalent.*

1. *The set S is a basis of $\mathbb{R}^n$.*
2. *The set S is linearly independent.*
3. *The set S is a generating set for $\mathbb{R}^n$.*
4. *The $n \times n$ matrix whose j-th column is the vector v_j for each $j = 1, 2, \dots, n$ is invertible.*

Proof of Lemma 4.65. The equivalence of the first three statements follows from Lemma 4.62, taking $V = \mathbb{R}^n$ and using that $\dim(\mathbb{R}^n) = n$. The equivalence of the second and the fourth statements follows from Fact 3.85. □

Exercise 4.66. Determine a basis and the dimension of each of the subspaces mentioned in the exercises in Section 4.3.

Exercise 4.67. Let A be an $m \times n$ matrix with real entries. Let $\mathcal{B} = \{v_1, \dots, v_n\}$ be a basis of $\mathbb{R}^n$. Show that the set of vectors $\{Av : v \in \mathcal{B}\}$ is a generating set for the column space $\operatorname{col}(A)$ of A.

Solution. Let $w \in \operatorname{col}(A)$. By definition of column space, there exists an element $x \in \mathbb{R}^n$ such that

$$w = Ax.$$

Since $\mathcal{B}$ is a basis of $\mathbb{R}^n$, there exist elements $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ such that

$$x = \alpha_1 v_1 + \dots + \alpha_n v_n.$$

Note that

$$\begin{aligned} w &= Ax \\ &= A(\alpha_1 v_1 + \cdots + \alpha_n v_n) \\ &= \alpha_1 A v_1 + \cdots + \alpha_n A v_n. \end{aligned}$$

This shows that w belongs to the span of $\{Av : v \in \mathcal{B}\}$. We conclude that the column space $\mathrm{col}(A)$ of A is contained in the span of $\{Av : v \in \mathcal{B}\}$.

Let $v \in \mathcal{B}$. Note that $Av \in \mathrm{col}(A)$. This shows that the span of $\{Av : v \in \mathcal{B}\}$ is contained in the column space $\mathrm{col}(A)$ of A.

We conclude that the column space $\mathrm{col}(A)$ of A is equal to the span of $\{Av : v \in \mathcal{B}\}$, and hence $\{Av : v \in \mathcal{B}\}$ is a generating set for the column space $\mathrm{col}(A)$ of A. □

§4.7 Rank-nullity theorem

Exercise 4.68. Determine a basis and the dimension of the null space of

$$A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{pmatrix}.$$

Solution. Note that the null space $\mathrm{null}(A)$ of A is equal to

$$\left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathbb{R}^4 : x_1 + x_4 = 0, x_2 + 2x_4 = 0, x_3 - 3x_4 = 0 \right\},$$

which is equal to

$$\left\{ \begin{pmatrix} -x_4 \\ -2x_4 \\ 3x_4 \\ x_4 \end{pmatrix} \in \mathbb{R}^4 : x_4 \in \mathbb{R} \right\},$$

which is equal to

$$\left\{ \alpha \begin{pmatrix} -1 \\ -2 \\ 3 \\ 1 \end{pmatrix} \in \mathbb{R}^4 : \alpha \in \mathbb{R} \right\}.$$

Hence, the null space $\mathrm{null}(A)$ of A is generated by the nonzero vector

$$e_4 - e_1 - 2e_2 + 3e_3,$$

where e_1, e_2, e_3, e_4 are the standard basis vectors of $\mathbb{R}^4$. It follows that $\mathrm{null}(A)$ is a one-dimensional subspace of $\mathbb{R}^4$, with basis $\{e_4 - e_1 - 2e_2 + 3e_3\}$. □

Exercise 4.69. Determine a basis and the dimension of the column space of

$$A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{pmatrix}.$$

Solution. Note that the column space $\mathrm{col}(A)$ of A is generated by the columns of A, that is, by the set of vectors

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} \right\}.$$

Since the vector $\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$ is equal to the linear combination

$$\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - 3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

it follows that the column space $\mathrm{col}(A)$ of A is generated by the set of vectors

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Since the vectors $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ are linearly independent, it follows that the column space $\mathrm{col}(A)$ of A has basis

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\},$$

and hence is three-dimensional. □

i **Remark**

Note that for the above matrix A in row echelon form, we have

$$\dim(\mathrm{null}(A)) + \dim(\mathrm{col}(A)) = 1 + 3 = 4,$$

which is equal to the number of columns of A.

i **Remark**

Note that the dimension of the null space of the above matrix A in row echelon form is equal to the number of non-pivot columns of A. Also note that the column space $\mathrm{col}(A)$ of A has dimension same as the number of pivot columns of A, which is equal to the rank of A. Hene, for the above matrix A in row echelon form, without determining a basis of the null space and a basis of the column space of A, one can deduce that

$$\dim(\mathrm{null}(A)) + \dim(\mathrm{col}(A)) = \text{the number of columns of } A = 4.$$

Example

Note that the matrix

$$A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{pmatrix}$$

is the row echelon form of the matrix

$$B = \begin{pmatrix} 1 & 2 & -5 & 20 \\ 2 & 5 & -7 & 33 \\ -1 & -2 & 4 & -17 \end{pmatrix}.$$

Since the null spaces of A and B are equal, it follows that the null space $\text{null}(B)$ of B is also a one-dimensional subspace of $\mathbb{R}^4$, with basis $\{e_4 - e_1 - 2e_2 + 3e_3\}$.

Moreover, the column space of B contains the span of the first three columns of B, which are linearly independent, since the matrix formed by these columns is invertible. This shows that the column space $\text{col}(B)$ of B is at least three-dimensional. Since the column space $\text{col}(B)$ of B is a subspace of $\mathbb{R}^3$, it follows that the column space $\text{col}(B)$ of B is three-dimensional. Consequently, we conclude that

$$\dim(\text{null}(B)) + \dim(\text{col}(B)) = 1 + 3 = 4.$$

Remark

The above example illustrates the **rank-nullity theorem** in a special case.

Theorem 4.70. *(Rank-nullity theorem) Let A be an $m \times n$ matrix with real entries. Then, the null space $\text{null}(A)$ of A is a linear subspace of $\mathbb{R}^n$, and the column space $\text{col}(A)$ of A is a linear subspace of $\mathbb{R}^m$. Moreover,*

$$\dim(\text{null}(A)) + \dim(\text{col}(A)) = n$$

holds.

A proof of the above is provided in Chapter 9.

Proof of Theorem 4.70. The fact that the null space $\text{null}(A)$ of A is a linear subspace of $\mathbb{R}^n$ and that the column space $\text{col}(A)$ of A is a linear subspace of $\mathbb{R}^m$

A proof of the equation

$$\dim(\text{null}(A)) + \dim(\text{col}(A)) = n$$

is provided in Chapter 9. □

Definition

Let A be an $m \times n$ matrix. The dimension of the column space $\text{col}(A)$ of A is called the **rank** of A. The dimension of the null space $\text{null}(A)$ of A is called the **nullity** of A.

Remark

If a matrix A is in row echelon form, then the dimension of the column space of A is equal to the number of pivot columns of A. Hence, for the matrices in row echelon form, the above definition of rank agrees with the definition of rank of a matrix given in Section 3.7.4.

Remark

Note that for any matrix A and a product E of elementary matrices, the null space $\mathrm{null}(EA)$ of EA is equal to the null space $\mathrm{null}(A)$ of A. Hence the nullity of EA is equal to the nullity of A. This shows that the nullity of a matrix is invariant under row operations.

It turns out that the dimension of the column space of a matrix is also invariant under row operations, but the proof is more involved than that for nullity. That is, for any matrix A and a product E of elementary matrices, the column space $\mathrm{col}(EA)$ of EA and the column space $\mathrm{col}(A)$ of A have the same dimensions. Hence, the dimension of the column space $\mathrm{col}(A)$ of a matrix A is equal to the number of pivot columns of the row echelon form of A. As a consequence, the above definition of rank agrees with the definition of rank of a matrix given in Section 3.7.4, for any matrix, not necessarily in row echelon form.

Chapter 5. Functions

§5.1 Basic definitions, injections, surjections and bijections

Definition

Let A, B be nonempty sets. A **function** f from a set A to a set B is a rule that assigns to each element $a \in A$, a unique element $b \in B$. We write this as $f : A \to B$, and denote the element b assigned to a by $f(a)$. A function is also called a **mapping** or a **map**.

The set A is called the **domain** or **source** of f, and the set B is called the **codomain** or **target** of f.

If a is an element of A, then the element $f(a)$ of B is called the **image** of a under f.

The **image** or **range** of f, is denoted by $\operatorname{im}(f)$, and is defined as

$$\operatorname{im}(f) := \{b \in B : b = f(a) \text{ for some } a \in A\}.$$

Definition

Let A, B, C, D be nonempty sets, and let $f : A \to B$ and $g : C \to D$ be functions. The functions f and g are said to be **equal**, denoted by $f = g$, if the following conditions hold:

- The domains of f and g are equal, that is, $A = C$.
- The codomains of f and g are equal, that is, $B = D$.
- For every $a \in A$, the images of a under f and g are equal, that is, $f(a) = g(a)$.

Example

Let $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$. The following is a function $f : A \to B$

$$f(1) = x, f(2) = y, f(3) = y.$$

Here, the image of 1 under f is x, the image of 2 under f is y, and the image of 3 under f is also y. The image (or range) of f is equal to $\operatorname{im}(f) = \{x, y\}$.

Example

Consider the function $f : \mathbb{R} \to \mathbb{R}$, defined by

$$f(x) = x^2 \quad \text{for all } x \in \mathbb{R},$$

Note that the image of -2 under f is 4, the image of 0 under f is 0, and the image of 3 under f is 9 etc. The image (or range) of f is

$$\operatorname{im}(f) = \{y \in \mathbb{R} : y \geq 0\}.$$

Example

Let A be an $m \times n$ matrix with real entries. Then, the function $T : \mathbb{R}^n \to \mathbb{R}^m$ defined by

$$T(v) = Av \text{ for all } v \in \mathbb{R}^n$$

is a function from $\mathbb{R}^n$ to $\mathbb{R}^m$. Here, the image of a vector $v \in \mathbb{R}^n$ under T is the vector $Av \in \mathbb{R}^m$. The image of f is

$$\operatorname{im}(f) = \{Av : v \in \mathbb{R}^n\},$$

which is the column space $\operatorname{col}(A)$ of A.

Definition

- A function $f : A \to B$ is said to be **injective** if for any two distinct elements $a_1, a_2 \in A$, we have $f(a_1) \neq f(a_2)$. An injective function is also called an **injection**.
- A function $f : A \to B$ is said to be **surjective** if $\operatorname{im}(f) = B$, that is, for every $b \in B$, there exists an $a \in A$ such that $f(a) = b$. A surjective function is also called a **surjection**.
- A function $f : A \to B$ is said to be **bijective** if it is both injective and surjective. A bijective function is also called a **bijection**.

Remark

Let $f : A \to B$ be a function. Note that the following statements are equivalent:

- The function f is injective.
- For any $a_1, a_2 \in A$, the equality $f(a_1) = f(a_2)$ implies that $a_1 = a_2$.

To show that a function $f : A \to B$ is not injective, it suffices to find two distinct elements $a_1, a_2 \in A$ such that $f(a_1) = f(a_2)$.

Remark

To show that a function $f : A \to B$ is not surjective, it amounts to finding an element $b \in B$ such that there is no element $a \in A$ satisfying $f(a) = b$.

Example

The function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = x^2 \text{ for all } x \in \mathbb{R}$$

is neither injective nor surjective. It is not injective because $f(-2) = f(2) = 4$. It is not surjective because there is no $x \in \mathbb{R}$ such that $f(x) = -1$.

Exercise 5.1. Consider the function $f : (0, 1) \to \mathbb{R}$ defined by

$$f(x) = \frac{1}{x} \text{ for all } x \in (0, 1).$$

Show that f is injective but not surjective.

Solution. To show that f is injective, let $x_1, x_2 \in (0, 1)$ be such that $f(x_1) = f(x_2)$. This implies that

$$\frac{1}{x_1} = \frac{1}{x_2}.$$

Cross-multiplying, we get $x_2 = x_1$, which shows that f is injective.

To show that f is not surjective, note that there is no $x \in (0,1)$ such that $f(x) = 0$. Hence, f is not surjective. $\square$

Example

The function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = 2x + 3 \text{ for all } x \in \mathbb{R}$$

is bijective.

It is injective because if x_1, x_2 are elements of $\mathbb{R}$ such that $f(x_1) = f(x_2)$, then $2x_1 + 3 = 2x_2 + 3$, which implies that $x_1 = x_2$.

It is surjective because for any $y \in \mathbb{R}$, there exists an $x \in \mathbb{R}$ such that $f(x) = y$ holds. In fact, we can take $x = \frac{y-3}{2}$.

Exercise 5.2. Let $M_n(\mathbb{R})$ denote the set of all $n \times n$ matrices with real entries. Show that the function $\det : M_3(\mathbb{R}) \to \mathbb{R}$, defined by

$$A \mapsto \det(A) \text{ for all } A \in M_3(\mathbb{R}),$$

is not injective, but surjective.

Solution. To show that det is not injective, consider the matrices

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and

$$B = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Both matrices have determinant 0, so det is not injective.

To show that det is surjective, let $y \in \mathbb{R}$ be arbitrary. We need to find a matrix $A \in M_3(\mathbb{R})$ such that $\det(A) = y$. Consider the matrix

$$A = \begin{pmatrix} y & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The determinant of A is y, so det is surjective. $\square$

Exercise 5.3. Let $\mathrm{GL}_2(\mathbb{R})$ denote the set of all invertible 2×2 matrices with real entries. Show that the function $\det : \mathrm{GL}_2(\mathbb{R}) \to \mathbb{R} \setminus \{0\}$ defined by

$$A \mapsto \det(A) \text{ for all } A \in \mathrm{GL}_2(\mathbb{R})$$

is surjective but not injective.

Solution. To show that det is not injective, consider the matrices

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and

$$B = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$$

Both matrices have determinant 1, so det is not injective.

To show that det is surjective, let $y \in \mathbb{R} \setminus \{0\}$ be arbitrary. We need to find a matrix $A \in \mathrm{GL}_2(\mathbb{R})$ such that $\det(A) = y$. Consider the matrix

$$A = \begin{pmatrix} y & 0 \\ 0 & 1 \end{pmatrix}$$

The determinant of A is y, which is nonzero. This shows that A is an element of $\mathrm{GL}_2(\mathbb{R})$, and hence, the map det is surjective. □

Remark

If A is a finite set and B is a proper subset of A, then there is no bijection between A and B. However, if A is an infinite set, then there may exist a bijection between A and some proper subset B of A.

Exercise 5.4. Show that the map

$$f : \{..., -6, -4, -2, 0, 2, 4, 6, ...\} \to \{..., -3, -2, -1, 0, 1, 2, 3, ...\}$$

defined by

$$f(n) = \frac{n}{2} \text{ for all } n \in \{..., -6, -4, -2, 0, 2, 4, 6, ...\}$$

is bijection between the set of all even integers and the set of all integers.

Solution. To show that f is injective, let n_1, n_2 be even integers such that $f(n_1) = f(n_2)$. This implies that

$$\frac{n_1}{2} = \frac{n_2}{2}.$$

Cross-multiplying, we get $n_1 = n_2$, which shows that f is injective.

To show that f is surjective, let m be an arbitrary integer. We need to find an even integer n such that $f(n) = m$. Consider the even integer $n = 2m$. Then, we have

$$f(n) = f(2m) = \frac{2m}{2} = m.$$

This shows that f is surjective.

Since f is both injective and surjective, it is bijective. □

Remark

If A if a finite set containing at least two elements, then there is no bijection between A and $A \times A$. However, if A is an infinite set, then there may exist a bijection between A and $A \times A$.

Exercise 5.5. Show that the map

$$f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$$

defined by

$$f(m, n) = 2^{m-1}(2n - 1) \text{ for all } (m, n) \in \mathbb{N} \times \mathbb{N}$$

is a bijection between $\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$.

Solution. To show that f is injective, let $(m_1, n_1), (m_2, n_2) \in \mathbb{N} \times \mathbb{N}$ be such that $f(m_1, n_1) = f(m_2, n_2)$. This implies that

$$2^{m_1-1}(2n_1 - 1) = 2^{m_2-1}(2n_2 - 1).$$

Let us first consider the case that $m_1 \le m_2$. Then, we can rewrite the above equality as

$$(2n_1 - 1) = 2^{m_2 - m_1}(2n_2 - 1).$$

Since the left-hand side is odd, the right-hand side must also be odd. This is possible only if $m_2 - m_1 = 0$, which implies that $m_1 = m_2$. Substituting this back into the equation, we get $2n_1 - 1 = 2n_2 - 1$, which implies that $n_1 = n_2$. This shows that f is injective. Similarly, if $m_2 \le m_1$, then it also follows that $m_1 = m_2$ and $n_1 = n_2$, and hence f is injective.

To show that f is surjective, let $k \in \mathbb{N}$ be arbitrary. We need to find $(m, n) \in \mathbb{N} \times \mathbb{N}$ such that $f(m, n) = k$. Note that every positive integer k can be uniquely expressed as $k = 2^r s$, where s is an odd positive integer and r is a nonnegative integer. Let $m = r + 1$ and $n = \frac{s+1}{2}$. Then, we have

$$f(m, n) = f\left(r + 1, \frac{s + 1}{2}\right) = 2^r \cdot s = k.$$

This shows that f is surjective.

Since f is both injective and surjective, it is bijective. □

Exercise 5.6 (*). Let F be a nonempty finite set. Does there exist a bijection between $F \times \mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$?

Exercise 5.7. Let a, b be real numbers satisfying $a < b$. Show that the map

$$f : (0, 1) \to (a, b)$$

defined by

$$f(x) = a + (b - a)x \text{ for all } x \in (0, 1)$$

is a bijection.

Using the above or otherwise, show that for any real numbers a, b, c, d satisfying $a < b$ and $c < d$, there is a bijection between the intervals (a, b) and (c, d).

Exercise 5.8. Let A be an $m \times n$ matrix with real entries. The **left multiplication by** A map

$$m_A : \mathbb{R}^n \to \mathbb{R}^m,$$

defined by

$$m_A(v) = Av \text{ for all } v \in \mathbb{R}^n$$

is

- injective if and only if the null space $\operatorname{null}(A)$ is equal to $\{0\}$ of A,
- surjective if and only if the column space $\operatorname{col}(A)$ is equal to $\mathbb{R}^m$,
- bijective if and only if it is both injective and surjective, that is, if and only if $\operatorname{null}(A) = \{0\}$ and $\operatorname{col}(A) = \mathbb{R}^m$.

Solution.

- To show that m_A is injective if and only if $\operatorname{null}(A) = \{0\}$, note that m_A is injective if and only if for any $v_1, v_2 \in \mathbb{R}^n$, the equality $Av_1 = Av_2$ implies that $v_1 = v_2$. This is equivalent[15] to saying that for any $v \in \mathbb{R}^n$, the equality $Av = 0$ implies that $v = 0$. This is precisely the condition that $\operatorname{null}(A) = \{0\}$.
- To show that m_A is surjective if and only if $\operatorname{col}(A) = \mathbb{R}^m$, note that m_A is surjective if and only if for every $w \in \mathbb{R}^m$, there exists a $v \in \mathbb{R}^n$ such that $Av = w$. This is precisely the condition that $\operatorname{col}(A) = \mathbb{R}^m$.
- The last part follows directly from the first two parts.

□

[15] Why?

If the map m_A (as above) is bijective, then $m = n$, and A is an invertible matrix.

Example

The function $f : \mathbb{Z} \to \mathbb{Z}$ defined by

$$f(n) = n + 1 \text{ for all } n \in \mathbb{Z}$$

is bijective.

It is injective because if $f(n_1) = f(n_2)$, then $n_1 + 1 = n_2 + 1$, which implies that $n_1 = n_2$.

It is surjective because for any $m \in \mathbb{Z}$, there exists an $n \in \mathbb{Z}$ such that $f(n) = m$ holds. In fact, we can take $n = m - 1$.

Example

Let A be a nonempty set. The function $\mathrm{id}_A : A \to A$ defined by

$$\mathrm{id}_A(a) = a \text{ for all } a \in A$$

is bijective. It is called the **identity map** on A.

Exercise 5.9. Let S^1 be the unit circle in the plane, defined by

$$S^1 := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

Show that the function $f : \mathbb{R} \to S^1$ defined by

$$f(t) = \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2}\right) \text{ for all } t \in \mathbb{R}$$

is injective but not surjective.

Exercise 5.10. Let $\mathbb{R}[X]_{\leq d}$ be the set of all polynomials in one variable X with real coefficients and of degree at most d. Show that the function $f : \mathbb{R}^{d+1} \to \mathbb{R}[X]_{\leq d}$ defined by

$$f(a_0, a_1, \ldots, a_d) = a_0 + a_1 X + \cdots + a_d X^d \text{ for all } (a_0, a_1, \ldots, a_d) \in \mathbb{R}^{d+1}$$

is bijective.

Exercise 5.11. Let $\mathbb{R}[X]_{\leq d}$ be the set of all polynomials in one variable X with real coefficients and of degree at most d. Consider the map sending a polynomial to its even part, that is, the map $T_e : \mathbb{R}[X]_{\leq d} \to \mathbb{R}[X]_{\leq d}$ defined by

$$T_e(p(X)) = \frac{p(X) + p(-X)}{2} \text{ for all } p(X) \in \mathbb{R}[X]_{\leq d}.$$

Show that T_e is neither injective nor surjective.

Exercise 5.12. Let $\mathbb{R}[X]_{\leq d}$ be the set of all polynomials in one variable X with real coefficients and of degree at most d. Consider the map sending a polynomial to its odd part, that is, the map $T_o : \mathbb{R}[X]_{\leq d} \to \mathbb{R}[X]_{\leq d}$ defined by

$$T_o(p(X)) = \frac{p(X) - p(-X)}{2} \text{ for all } p(X) \in \mathbb{R}[X]_{\leq d}.$$

Show that T_o is neither injective nor surjective.

Exercise 5.13. The function

$$|\cdot|:\mathbb{R}\to\mathbb{R}$$

defined by

$$|x| := \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0 \end{cases}$$

is called the **absolute value** function. Show that it is neither injective nor surjective.

Exercise 5.14. Consider the function

$$f:\mathbb{R}\to(0,1]$$

defined by

$$f(x) = \frac{1}{1+|x|} \text{ for all } x \in \mathbb{R}.$$

Show that f is not injective. Determine $\operatorname{im}(f)$.

Exercise 5.15. Let A be a nonempty set. Show that the identity map $\mathrm{id}_A : A \to A$ is bijective.

Exercise 5.16. Let A be a nonempty subset of a set B. Show that the inclusion map $i : A \to B$ defined by

$$i(a) := a \text{ for all } a \in A$$

is injective. Is it surjective? Justify your answer.

Exercise 5.17. The map

$$\mathbb{R}\times\mathbb{R}\to\mathbb{R},$$

defined by

$$(x,y)\mapsto |x-y| \text{ for all } (x,y)\in\mathbb{R}\times\mathbb{R},$$

is called the **distance function** on $\mathbb{R}$. Show that it is neither injective nor surjective.

Exercise 5.18. The map

$$d:\mathbb{R}^2\times\mathbb{R}^2\to\mathbb{R},$$

defined by

$$d((x_1,y_1),(x_2,y_2)) := \sqrt{(x_1-x_2)^2+(y_1-y_2)^2} \text{ for all } (x_1,y_1),(x_2,y_2)\in\mathbb{R}^2,$$

is called the **distance function** on $\mathbb{R}^2$. Show that it is neither injective nor surjective.

Exercise 5.19. The map

$$\|\cdot\|:\mathbb{R}^2\to\mathbb{R},$$

defined by

$$\|(x,y)\| := \sqrt{x^2+y^2} \text{ for all } (x,y)\in\mathbb{R}^2,$$

is called the **norm** on $\mathbb{R}^2$. Show that it is neither injective nor surjective.

Exercise 5.20. Let u be an element of $\mathbb{R}^2$. Let θ be a real number, and let A_θ denote the orthogonal matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Show that $\|A_\theta u\| = \|u\|$.

Exercise 5.21. The function

$$\mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R},$$

defined by

$$\langle (x_1, y_1), (x_2, y_2) \rangle := x_1 x_2 + y_1 y_2 \text{ for all } (x_1, y_1), (x_2, y_2) \in \mathbb{R}^2,$$

is called the **inner product** on $\mathbb{R}^2$. Show that the inner product of the standard basis vectors $e_1 = (1, 0)$ and $e_2 = (0, 1)$ is equal to 0.

Exercise 5.22. Does there exist a set S of three or more nonzero vectors in $\mathbb{R}^2$ such that the inner product of any two distinct vectors of S is equal to 0? Justify your answer.

Exercise 5.23. Let u, v be elements of $\mathbb{R}^2$ satisfying $\langle u, v \rangle = 0$. Let θ be a real number, and let A_θ denote the orthogonal matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Show that the vectors $A_\theta u$ and $A_\theta v$ also satisfy

$$\langle A_\theta u, A_\theta v \rangle = 0.$$

Exercise 5.24. Show that the inner product on $\mathbb{R}^2$ has the following properties: for any $u, v, w \in \mathbb{R}^2$ and any element $c \in \mathbb{R}$, we have

- $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$,
- $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$,
- $\langle cu, v \rangle = c\langle u, v \rangle$,
- $\langle u, cv \rangle = c\langle u, v \rangle$.

Exercise 5.25. For any u in $\mathbb{R}^2$, show that

$$\langle u, u \rangle = \|u\|^2.$$

Exercise 5.26. For any u, v in $\mathbb{R}^2$, show that

$$|\langle u, v \rangle| \leq \|u\| \, \|v\|.$$

Solution. Let u, v be elements of $\mathbb{R}^2$. Write $u = \left(\begin{smallmatrix} x_1 \\ y_1 \end{smallmatrix}\right)$ and $v = \left(\begin{smallmatrix} x_2 \\ y_2 \end{smallmatrix}\right)$. Note that

$$\begin{aligned} \langle u, v \rangle &= x_1 x_2 + y_1 y_2, \\ \|u\| &= \sqrt{x_1^2 + y_1^2}, \\ \|v\| &= \sqrt{x_2^2 + y_2^2}. \end{aligned}$$

This shows that

$$\begin{aligned} (\|u\| \, \|v\|)^2 - (\langle u, v \rangle)^2 &= (x_1^2 + y_1^2)(x_2^2 + y_2^2) - (x_1 x_2 + y_1 y_2)^2 \\ &= x_1^2 x_2^2 + x_1^2 y_2^2 + y_1^2 x_2^2 + y_1^2 y_2^2 - (x_1^2 x_2^2 + 2 x_1 x_2 y_1 y_2 + y_1^2 y_2^2) \\ &= x_1^2 y_2^2 + x_2^2 y_1^2 - 2 x_1 x_2 y_1 y_2 \\ &= (x_1 y_2 - x_2 y_1)^2 \\ &\geq 0. \end{aligned}$$

This implies that

$$(\|u\|\ \|v\| + |\langle u, v\rangle|)(\|u\|\ \|v\| - |\langle u, v\rangle|) \geq 0.$$

If

$$\|u\|\ \|v\| + |\langle u, v\rangle| = 0,$$

then the desired inequality holds. Otherwise, $\|u\|\ \|v\| + |\langle u, v\rangle|$ is positive, and hence

$$\|u\|\ \|v\| - |\langle u, v\rangle| \geq 0,$$

which also implies the desired inequality. □

Exercise 5.27. The map

$$\|\cdot\| : \mathbb{R}^3 \to \mathbb{R},$$

defined by

$$\|(x, y, z)\| := \sqrt{x^2 + y^2 + z^2} \text{ for all } (x, y, z) \in \mathbb{R}^3,$$

is called the **norm** on $\mathbb{R}^3$. Show that it is neither injective nor surjective.

Exercise 5.28. Show that the standard basis vectors of $\mathbb{R}^2$ satisfy $\|e_1\| = \|e_2\| = 1$.

Exercise 5.29. The map

$$d : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R},$$

defined by

$$d((x_1, y_1, z_1), (x_2, y_2, z_2)) := \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$
$$\text{for all } (x_1, y_1, z_1), (x_2, y_2, z_2) \in \mathbb{R}^3,$$

is called the **distance function** on $\mathbb{R}^3$. Show that it is neither injective nor surjective.

Exercise 5.30. The function

$$\mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R},$$

defined by

$$\langle (x_1, y_1, z_1), (x_2, y_2, z_2)\rangle := x_1x_2 + y_1y_2 + z_1z_2 \text{ for all } (x_1, y_1, z_1), (x_2, y_2, z_2) \in \mathbb{R}^3,$$

is called the **inner product** on $\mathbb{R}^3$. Show that the inner product of any two distinct standard basis vectors of $\mathbb{R}^3$ is equal to 0.

Exercise 5.31. Does there exist a set S of four or more nonzero vectors in $\mathbb{R}^3$ such that the inner product of any two distinct vectors of S is equal to 0? Justify your answer.

Exercise 5.32. Show that the inner product on $\mathbb{R}^3$ has the following properties: for any $u, v, w \in \mathbb{R}^3$ and any element $c \in \mathbb{R}$, we have

- $\langle u + v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$,
- $\langle u, v + w\rangle = \langle u, v\rangle + \langle u, w\rangle$,
- $\langle cu, v\rangle = c\langle u, v\rangle$,
- $\langle u, cv\rangle = c\langle u, v\rangle$.

Exercise 5.33. For any u in $\mathbb{R}^3$, show that

$$\langle u, u\rangle = \|u\|^2.$$

Exercise 5.34. Show that the standard basis vectors of $\mathbb{R}^3$ satisfy $\|e_1\| = \|e_2\| = \|e_3\| = 1$.

Exercise 5.35 (*). For any u, v in $\mathbb{R}^3$, show that

$$|\langle u, v\rangle| \leq \|u\| \ \|v\|.$$

Solution. Let u, v be elements of $\mathbb{R}^3$. Write $u = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$ and $v = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$. Note that

$$\begin{aligned} \langle u, v\rangle &= x_1x_2 + y_1y_2 + z_1z_2, \\ \|u\| &= \sqrt{x_1^2 + y_1^2 + z_1^2}, \\ \|v\| &= \sqrt{x_2^2 + y_2^2 + z_2^2}. \end{aligned}$$

This shows that

$$\begin{aligned} (\|u\| \ \|v\|)^2 - (\langle u, v\rangle)^2 &= (x_1^2 + y_1^2 + z_1^2)(x_2^2 + y_2^2 + z_2^2) - (x_1x_2 + y_1y_2 + z_1z_2)^2 \\ &= x_1^2x_2^2 + x_1^2y_2^2 + x_1^2z_2^2 + y_1^2x_2^2 + y_1^2y_2^2 + y_1^2z_2^2 + z_1^2x_2^2 + z_1^2y_2^2 + z_1^2z_2^2 \\ &\quad -(x_1^2x_2^2 + y_1^2y_2^2 + z_1^2z_2^2 + 2x_1x_2y_1y_2 + 2y_1y_2z_1z_2 + 2z_1z_2x_1x_2) \\ &= x_1^2y_2^2 + x_1^2z_2^2 + y_1^2x_2^2 + y_1^2z_2^2 + z_1^2x_2^2 + z_1^2y_2^2 \\ &\quad -(2x_1x_2y_1y_2 + 2y_1y_2z_1z_2 + 2z_1z_2x_1x_2) \\ &= x_1^2y_2^2 + y_1^2x_2^2 + y_1^2z_2^2 + z_1^2y_2^2 + x_1^2z_2^2 + z_1^2x_2^2 \\ &\quad -(2x_1x_2y_1y_2 + 2y_1y_2z_1z_2 + 2z_1z_2x_1x_2) \\ &= (x_1y_2 - x_2y_1)^2 + (y_1z_2 - y_2z_1)^2 + (z_1x_2 - z_2x_1)^2 \\ &\geq 0. \end{aligned}$$

This implies that

$$(\|u\| \ \|v\| + |\langle u, v\rangle|)(\|u\| \ \|v\| - |\langle u, v\rangle|) \geq 0.$$

If

$$\|u\| \ \|v\| + |\langle u, v\rangle| = 0,$$

then the desired inequality holds. Otherwise, $\|u\| \ \|v\| + |\langle u, v\rangle|$ is positive, and hence

$$\|u\| \ \|v\| - |\langle u, v\rangle| \geq 0,$$

which also implies the desired inequality. □

§5.2 Composition and inversion of functions

Definition

Let A, B, C be nonempty sets. Let $f : A \to B$ and $g : B \to C$ be functions. The **composition** of f and g, denoted by $g \circ f$, is the function from A to C defined by

$$(g \circ f)(a) := g(f(a)) \text{ for all } a \in A.$$

A function $f : A \to B$ is said to be **invertible** if there exists a function $g : B \to A$ such that

$$g \circ f = \mathrm{id}_A, f \circ g = \mathrm{id}_B,$$

where $\mathrm{id}_A : A \to A$ and $\mathrm{id}_B : B \to B$ are the identity functions on A and B, respectively.

In this case, the function g is called the **inverse** of f, and is denoted by f^{-1}.

Exercise 5.36. Let A, B be nonempty sets. Let $f : A \to B$ be a function. Show that if f is invertible, then its inverse is unique.

Exercise 5.37. Let A, B, C be nonempty sets. Let $f : A \to B$ and $g : B \to C$ be functions.
- Show that if f and g are both injective (resp. surjective, bijective), then so is $g \circ f$.
- Show that a function $f : A \to B$ is bijective if it is invertible.

Exercise 5.38. Let A, B, C be nonempty sets. Let $f : A \to B$ and $g : B \to C$ be functions.
- Show that if $g \circ f$ is injective, then so is f.
- Show that if $g \circ f$ is surjective, then so is g.

Exercise 5.39. The map

$$T : \mathbb{R}^2 \to \mathbb{R}^2$$

defined by

$$T\begin{pmatrix} x \\ y \end{pmatrix} := \begin{pmatrix} x + y \\ x - y \end{pmatrix} \text{ for all } (x, y) \in \mathbb{R}^2$$

is bijective. Find the inverse map T^{-1} of T.

§5.3 Images and inverse images under functions

Definition

Let $f : A \to B$ be a function. For any subset $C \subseteq A$, the **image** of C under f, denoted by $f(C)$, is defined as

$$f(C) := \{b \in B : b = f(a) \text{ for some } a \in C\}.$$

Note that $f(C)$ is a subset of B.

Exercise 5.40. Let $f : A \to B$ be a function. Show that

$$f(\emptyset) = \emptyset.$$

Exercise 5.41. Determine the images of the functions defined in the above exercises.

Exercise 5.42. Let $f : A \to B$ be a function. Show that for any two subsets $C_1, C_2 \subseteq A$, we have
- $f(C_1 \cup C_2) = f(C_1) \cup f(C_2)$,
- $f(C_1 \cap C_2) \subseteq f(C_1) \cap f(C_2)$.

What happens if f is injective?

Exercise 5.43. Let $f : A \to B$ be a function. Assume that for any subsets $C_1, C_2 \subseteq A$,

$$f(C_1 \cap C_2) = f(C_1) \cap f(C_2)$$

holds. Show that f is injective.

Note the the following exercise concludes the same result under a weaker assumption.

Exercise 5.44. Let $f : A \to B$ be a function. Assume that for any two subsets $C_1, C_2 \subseteq A$ satisfying $C_1 \cap C_2 = \emptyset$, we have

$$f(C_1) \cap f(C_2) = \emptyset.$$

Show that f is injective.

Definition

Let $f : A \to B$ be a function. For any subset $D \subseteq B$, the **inverse image** or **preimage** of D under f, denoted by $f^{-1}(D)$, is defined as

$$f^{-1}(D) := \{a \in A : f(a) \in D\}.$$

Note that $f^{-1}(D)$ is a subset of A.

Exercise 5.45. Determine the inverse image of the set $\{1\}$ under the norm map

$$\| \cdot \| : \mathbb{R}^2 \to \mathbb{R}.$$

Solution. The inverse image of the set $\{1\}$ under the norm map

$$\| \cdot \| : \mathbb{R}^2 \to \mathbb{R}$$

is the subset

$$\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$$

of $\mathbb{R}^2$. □

Exercise 5.46. Let $f : A \to B$ be an invertible function. Let $g : B \to A$ be its inverse. Show that for any subset $C \subseteq B$, the inverse image of C under f is equal to the image of C under g.

Exercise 5.47. Let $f : A \to B$ be a function. Show that for any two subsets $D_1, D_2 \subseteq B$,

- $f^{-1}(D_1 \cup D_2) = f^{-1}(D_1) \cup f^{-1}(D_2)$,
- $f^{-1}(D_1 \cap D_2) = f^{-1}(D_1) \cap f^{-1}(D_2)$.

Exercise 5.48. Let $f : A \to B$ be a function. For any subset $C \subseteq A$, show that

$$f^{-1}(f(C)) \supseteq C.$$

Exercise 5.49. Let $f : A \to B$ be a function. For any subset $D \subseteq B$, show that

$$f(f^{-1}(D)) \subseteq D.$$

Exercise 5.50. Let $f : A \to B$ be a function. Show that the following statements are equivalent:

- f is surjective.
- For any subset $D \subseteq B$, we have $f(f^{-1}(D)) = D$.
- For any subset $C \subseteq A$, we have $f(A \setminus C) \supseteq B \setminus f(C)$.

Exercise 5.51. Let $f : A \to B$ be a function. Show that the following statements are equivalent:

- f is injective.
- For any subset $D \subseteq B$, we have $f^{-1}(f(D)) = D$.
- For any subset $C \subseteq A$, we have $f(A \setminus C) \subseteq B \setminus f(C)$.

Exercise 5.52. Let $f : A \to B$ be an invertible function. Show that for any subset $C \subseteq A$ and any subset $D \subseteq B$, we have

- $f^{-1}(f(C)) = C$,
- $f(f^{-1}(D)) = D$.

Remark

The above exercise shows that if a function $f : A \to B$ is invertible, then the inverse image $f^{-1}(D)$ of any subset $D \subseteq B$ and the image of D under the inverse function f^{-1} are the same. Hence, no confusions arise from using the same notation $f^{-1}(D)$ for both.

Exercise 5.53. Let $f : A \to B$ be a function. For any subset $C \subseteq A$, show that $C \subseteq f^{-1}(f(C))$. What happens if f is injective?

Appendix

The content of the appendix is supplementary and not required for the main course. However, it may be useful for further studies.

Chapter 6. Sets

Proof of Fact 1.23. Note that if P is a subset of Q, then we claim that $A \cup P$ is a subset of $A \cup Q$. Indeed, for an element x of $A \cup P$, note that x lies in A, or it lies in P. If x lies in A, then it lies in $A \cup Q$. If x lies in P, it also lies in Q, and hence it lies in $A \cup Q$. This proves the claim. Consequently, $A \cup (B \cap C)$ is a subset of both of the sets

$$A \cup B, A \cup C,$$

and hence, we obtain

$$\boxed{A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)}.$$

Let y be an element of $(A \cup B) \cap (A \cup C)$. Note that y lies in $A \cup B$ and y also lies in $A \cup C$. If y lies in A, then it also lies in $A \cup (B \cap C)$. Thus, it remains to consider the case that y does not lie in A, which we assume from now on. Since y lies in $A \cup B$ and y does not lie in A, it follows that y lies in B. Similarly, it also follows that y lies in C. This shows that y lies in $B \cap C$, and hence it lies in $A \cup (B \cap C)$. This proves that

$$\boxed{(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)}.$$

This establishes that

$$\boxed{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}.$$

Let us now establish that

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Let x be an element of $A \cap (B \cup C)$. Note that x lies in A and also lies in $B \cup C$. If x lies in B, then x lies in $A \cap B$, and hence x belongs to $(A \cap B) \cup (A \cap C)$. It remains to consider the case that x does not lie in B, which we assume from now on. Since x lies in $B \cup C$ and x does not lie in B, it follows that x lies in C. Using that x lies in A, we obtain that x lies in $A \cap C$, and hence, x lies in $(A \cap B) \cup (A \cap C)$. Combining these cases, we obtain

$$\boxed{A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)}.$$

Note that if P is a subset of Q, then $A \cap P$ is contained in $A \cap Q$. It follows that $A \cap B$ is a subset of $A \cap (B \cup C)$, and $A \cap C$ is also a subset of $A \cap (B \cup C)$. This shows that

$$\boxed{(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)}.$$

This proves that

$$\boxed{A \cap (B \cup C) = (A \cap B) \cup (A \cap C)}.$$

□

Proof of Fact 1.28. Note that

$$\begin{aligned}(A \cup B)^c &= X \setminus (A \cup B) \\ &= \{x \in X : x \notin (A \cup B)\} \\ &= \{x \in X : x \notin A \text{ and } x \notin B\} \\ &= \{x \in X : x \in A^c \text{ and } x \in B^c\} \\ &= \{x \in X : x \in A^c \cap B^c\} \\ &= A^c \cap B^c\end{aligned}$$

holds for any subsets A, B of X.

As a consequence, we obtain

$$(A^c \cup B^c)^c = ((A^c)^c \cap (B^c)^c),$$

which yields

$$(A^c \cup B^c)^c = A \cap B.$$

This implies

$$((A^c \cup B^c)^c)^c = (A \cap B)^c,$$

or equivalently,

$$\boxed{(A \cap B)^c = A^c \cup B^c}\,.$$

□

Proof of Fact 1.30. Let X denote the underlying universal set. Note that

$$\begin{aligned}X \setminus (B \cup C) &= (B \cup C)^c \\ &= B^c \cap C^c \\ &= (X \setminus B) \cap (X \setminus C).\end{aligned}$$

This shows that

$$A \cap (X \setminus (B \cup C)) = A \cap ((X \setminus B) \cap (X \setminus C)),$$

which implies that

$$A \cap (X \setminus (B \cup C)) = (A \cap (X \setminus B)) \cap (A \cap (X \setminus C)).$$

Note that for any subset Y of X, we have

$$\begin{aligned}A \cap (X \setminus Y) &= \{x \in X : x \in A \text{ and } x \in X \setminus Y\} \\ &= \{x \in X : x \in A \text{ and } x \notin Y\} \\ &= A \setminus Y.\end{aligned}$$

Consequently, we obtain

$$\boxed{A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)}\,.$$

Similarly, we obtain

$$\begin{aligned}X \setminus (B \cap C) &= (B \cap C)^c \\ &= B^c \cup C^c \\ &= (X \setminus B) \cup (X \setminus C).\end{aligned}$$

This shows that

$$A \cap (X \setminus (B \cap C)) = A \cap ((X \setminus B) \cup (X \setminus C)).$$

Using the distributive property of intersection over union (see Fact 1.23), we obtain

$$A \cap (X \setminus (B \cap C)) = (A \cap (X \setminus B)) \cup (A \cap (X \setminus C)).$$

Since for any subset Y of X, the equality

$$A \cap (X \setminus Y) = A \setminus Y$$

holds, we conclude that

$$\boxed{A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)}\ .$$

□

Exercise 6.1 (*). Do there exist three finite sets A, B, C satisfying

$$|A\Delta B| = 1, |B\Delta C| = 2, |C\Delta A| = 4?$$

Solution. Note that for any three sets A, B, C,

$$\begin{aligned} C \setminus A &= (C \setminus A) \cap (B \cup B^c) \\ &= ((C \setminus A) \cap B) \cup ((C \setminus A) \cap B^c) \\ &\subseteq (A^c \cap B) \cup (C \cap B^c) \\ &= (B \setminus A) \cup (C \setminus B) \\ &\subseteq (A\Delta B) \cup (B\Delta C) \end{aligned}$$

hold. Hence, for any three sets A, B, C, we also obtain

$$\begin{aligned} A \setminus C &\subseteq (C\Delta B) \cup (B\Delta A) \\ &= (A\Delta B) \cup (B\Delta C). \end{aligned}$$

This shows that for any three sets A, B, C, the union of $A \setminus C, C \setminus A$ is a subset of $(A\Delta B) \cup (B\Delta C)$, that is, $C\Delta A$ is a subset of $(A\Delta B) \cup (B\Delta C)$. If A, B, C are finite sets, it follows that

$$\boxed{|C\Delta A| \leq |(A\Delta B) \cup (B\Delta C)| \leq |A\Delta B| + |B\Delta C|}\ .$$

This shows that there are no three finite sets A, B, C satisfying the given conditions. □

Exercise 6.2. If z, w are complex numbers, then show that

1. $|z + w| \leq |z| + |w|$ (triangle inequality).

Remark (**)

More formally, one defines $\mathbb{C}$, as the following set

$$\{(a, b) : a, b \in \mathbb{R}\},$$

equipped with addition and multiplication defined by

$$\begin{aligned} (a, b) + (c, d) &:= (a + c, b + d), \\ (a, b) \cdot (c, d) &:= (ac - bd, ad + bc), \end{aligned}$$

for any $(a, b), (c, d)$ with $a, b, c, d \in \mathbb{R}$.

Chapter 7. Induction principles

Theorem 7.1. *The well-ordering principle, the principle of mathematical induction, and the principle of strong induction are equivalent.*

Proof.

□

Exercise 7.2 (Infinitude of primes, by Saidak). Let $a_1 = 2$, and $a_{n+1} = a_n(a_n + 1)$ for any positive integer n. For any $n \in \mathbb{N}$, show that a_n has at least n distinct prime divisors.

Solution. For a positive integer n, let $P(n)$ denote the statement that $a_n \geq 1$ and a_n has at least n distinct prime divisors.

Since $a_1 = 2$, it follows that $P(1)$ is true.

Let n be a positive integer such that $P(n)$ holds. This gives that $a_n \geq 1$, and hence $a_n + 1 \geq 2$. Also note that the integers $a_n, a_n + 1$ have no common prime divisor, since any of their common divisors divides their difference, which is equal to 1. Hence, no prime divisor of a_n is a prime divisor of $a_n + 1$. By the induction hypothesis, a_n has at least n distinct prime divisors. Since $a_n + 1 \geq 2$, it admits at least one prime divisor. It follows that the product $a_n(a_n + 1)$ has at least $n + 1$ distinct prime divisors. Also note that $a_n(a_n + 1) \geq 2$ holds. This shows that $P(n + 1)$ holds.

By induction, the statement $P(n)$ holds for all $n \in \mathbb{N}$. In particular, a_n has at least n distinct prime divisors. □

Exercise 7.3 (*). Show that there are infinitely many prime numbers of the form $4m + 3$, where m is a nonnegative integer.

Solution. Le $a_1, a_2, \ldots$ be a sequence of positive integers defined by

$$a_1 := 7,$$
$$a_{n+1} := a_n(4a_n + 3)$$

for all $n \in \mathbb{N}$. For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that a_n is a positive integer, 3 does not divide a_n, and a_n has at least n distinct prime divisors of the form $4m + 3$.

Since $a_1 = 7$, it follows that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Since a_k is a positive integer, it follows that $4a_k + 3$ is an odd positive integer ≥ 3. Hence it has[16] a prime divisor of the form $4m + 3$. Since 3 does not divide a_k, it follows[17] that 3 does not divide $4a_k + 3$. Also note that any common factor of the integers a_k and $4a_k + 3$ divides

$$(4a_k + 3) - 4a_k,$$

which is equal to 3. Since 3 does not divide a_k, we obtain that the integers a_k and $4a_k + 3$ have no common prime divisor. Hence, any prime divisor of $4a_k + 3$ of the form $4m + 3$ is different from the prime divisors of a_k. This shows that a_{k+1} is a positive integer, 3 does not divide a_{k+1}, and a_{k+1} has at least $k + 1$ distinct prime divisors of the form $4m + 3$. By induction, it follows that for any $n \in \mathbb{N}$, the integer a_n has n distinct prime divisors of the form $4m + 3$.

This shows that there are infinitely many prime numbers of the form $4m + 3$. □

Exercise 7.4. Show that

[16] How does it follow? Prove it by induction that for any positive integer n, the product of any integers of the form $4m + 1$ is also of the form $4m + 1$.

[17] Why?

$$2^{2^n} - 1$$

has at least n distinct prime divisors for all $n \in \mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that $2^{2^n} - 1$ has at least n distinct prime divisors.

Since $2^{2^1} - 1 = 3$, it follows that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$2^{2^{k+1}} - 1 = \left(2^{2^k} - 1\right)\left(2^{2^k} + 1\right).$$

Also note that the integers $2^{2^k} - 1$ and $2^{2^k} + 1$ have no common prime divisors, since these are odd integers, and any of their common divisors divides their difference, which is equal to 2. By the induction hypothesis, the integer $2^{2^k} - 1$ has at least k distinct prime divisors. Since $2^{2^k} + 1 \geq 5$, it admits at least one prime divisor. It follows that the product $\left(2^{2^k} - 1\right)\left(2^{2^k} + 1\right)$ has at least $k + 1$ distinct prime divisors. This shows that $P(k+1)$ holds.

By induction, the statement $P(n)$ holds for all $n \in \mathbb{N}$. □

Remark

Note that it is false that

$$2^{2^n} + 1$$

has at least n distinct prime divisors for all $n \in \mathbb{N}$. Indeed, the integer $2^{2^2} + 1 = 17$ is a prime.

Exercise 7.5. Let S be a finite set of size n. Show that the number of subsets of S is 2^n.

Solution. For a nonnegative integer n, let $P(n)$ denote the statement that if A is a set with n elements, then the number of subsets of A is 2^n.

Since the empty set has exactly one subset, namely itself, it follows that $P(0)$ holds.

Let k be a nonnegative integer such that $P(k)$ holds, that is, for **any set** with k elements, the number of its subsets is 2^k. We show that $P(k+1)$ holds.

Let A be a set with $k + 1$ elements. Choose an element $a \in A$, and let $B = A \setminus \{a\}$. Note that B has exactly k elements. By the induction hypothesis, the number of subsets of B is 2^k. Any subset of A either contains a or does not contain a. The number of subsets of A that do not contain a is equal to the number of subsets of B, which is 2^k.

Note that any subset of A that contains a is of the form $C \cup \{a\}$, where C is a subset of B. Further, different subsets of B give rise to different subsets of A that contain a. That is, if C_1, C_2 are different subsets of B, then $C_1 \cup \{a\} \neq C_2 \cup \{a\}$. This shows that the number of subsets of A that contain a is also equal to the number of subsets of B, which is again 2^k. It follows that the total number of subsets of A is equal to

$$2^k + 2^k = 2^{k+1}.$$

This shows that $P(k+1)$ holds.

By induction, it follows that for any nonnegative integer n, if A is a set of size n, then the number of subsets of A is 2^n. □

Exercise 7.6 (*). Show that

$$\left(1 + \frac{1}{1^3}\right)\left(1 + \frac{1}{2^3}\right)\left(1 + \frac{1}{3^3}\right) \cdots \left(1 + \frac{1}{n^3}\right) < 3$$

holds for all $n \in \mathbb{N}$.

Remark (*)

Does taking $P(n)$ to be the statement that

$$\left(1+\frac{1}{1^3}\right)\left(1+\frac{1}{2^3}\right)\left(1+\frac{1}{3^3}\right)\cdots\left(1+\frac{1}{n^3}\right) < 3$$

help?

Solution. For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that

$$\left(1+\frac{1}{1^3}\right)\left(1+\frac{1}{2^3}\right)\left(1+\frac{1}{3^3}\right)\cdots\left(1+\frac{1}{n^3}\right) \leq 3-\frac{1}{n}.$$

Note that $P(1)$ holds. Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
&\left(1+\frac{1}{1^3}\right)\left(1+\frac{1}{2^3}\right)\left(1+\frac{1}{3^3}\right)\cdots\left(1+\frac{1}{k^3}\right)\left(1+\frac{1}{(k+1)^3}\right) \\
&\leq \left(3-\frac{1}{k}\right)\left(1+\frac{1}{(k+1)^3}\right) \\
&= 3-\frac{1}{k}+\frac{3}{(k+1)^3}-\frac{1}{k(k+1)^3} \\
&= 3-\frac{1}{k+1}+\frac{1}{k+1}-\frac{1}{k}+\frac{3}{(k+1)^3}-\frac{1}{k(k+1)^3} \\
&= 3-\frac{1}{k+1}-\frac{1}{k(k+1)}+\frac{3}{(k+1)^3}-\frac{1}{k(k+1)^3} \\
&= 3-\frac{1}{k+1}-\frac{(k+1)^2-3k+1}{k(k+1)^3} \\
&= 3-\frac{1}{k+1}-\frac{k^2-k+2}{k(k+1)^3} \\
&= 3-\frac{1}{k+1}-\frac{k(k-1)+2}{k(k+1)^3} \\
&< 3-\frac{1}{k+1}
\end{aligned}$$

holds. This proves that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. This implies the given inequality for all $n \in \mathbb{N}$.

□

Exercise 7.7 (*). Show that

$$1+\frac{1}{2^3}+\frac{1}{3^3}+\cdots+\frac{1}{n^3} < \frac{3}{2}$$

for all $n \in \mathbb{N}$.

Solution. For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that

$$1+\frac{1}{2^3}+\frac{1}{3^3}+\cdots+\frac{1}{n^3} \leq \frac{3}{2}-\frac{1}{2n^2}.$$

Note that $P(1)$ holds. Let k be a positive integer such that $P(k)$ holds. Note that

$$
\begin{aligned}
&1+\frac{1}{2^3}+\frac{1}{3^3}+\cdots+\frac{1}{k^3}+\frac{1}{(k+1)^3}\\
&\le \left(\frac{3}{2}-\frac{1}{2k^2}\right)+\frac{1}{(k+1)^3} && \text{(using } P(k)\text{)}\\
&=\frac{3}{2}-\frac{1}{2k^2}+\frac{1}{(k+1)^3}\\
&=\frac{3}{2}-\frac{1}{2(k+1)^2}+\frac{1}{2(k+1)^2}-\frac{1}{2k^2}+\frac{1}{(k+1)^3}\\
&=\frac{3}{2}-\frac{1}{2(k+1)^2}-\frac{(k+1)^3-k^2(k+1)-2k^2}{2k^2(k+1)^3}\\
&=\frac{3}{2}-\frac{1}{2(k+1)^2}-\frac{k^3+3k^2+3k+1-k^3-k^2-2k^2}{2k^2(k+1)^3}\\
&=\frac{3}{2}-\frac{1}{2(k+1)^2}-\frac{3k+1}{2k^2(k+1)^3}\\
&<\frac{3}{2}-\frac{1}{2(k+1)^2}.
\end{aligned}
$$

This proves that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. This implies the given inequality for all $n \in \mathbb{N}$.

□

Exercise 7.8 (*). Show that

$$\left(1+\frac{1}{2}\right)\left(1+\frac{1}{2^2}\right)\left(1+\frac{1}{2^3}\right)\cdots\left(1+\frac{1}{2^n}\right) < \frac{5}{2}$$

holds for all $n \in \mathbb{N}$.

Solution. For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that

$$\left(1+\frac{1}{2}\right)\left(1+\frac{1}{2^2}\right)\left(1+\frac{1}{2^3}\right)\cdots\left(1+\frac{1}{2^n}\right) \le \frac{5}{2}\left(1-\frac{2}{2^{n+1}+1}\right).$$

Note that $P(1)$ holds. Let k be a positive integer such that $P(k)$ holds. Note that

$$
\begin{aligned}
&\left(1+\frac{1}{2}\right)\left(1+\frac{1}{2^2}\right)\left(1+\frac{1}{2^3}\right)\cdots\left(1+\frac{1}{2^k}\right)\left(1+\frac{1}{2^{k+1}}\right)\\
&\le \frac{5}{2}\left(1-\frac{2}{2^{k+1}+1}\right)\left(1+\frac{1}{2^{k+1}}\right)\\
&\le \frac{5}{2}\times\frac{2^{k+1}-1}{2^{k+1}}\\
&\le \frac{5}{2}\left(1-\frac{1}{2^{k+1}}\right)\\
&\le \frac{5}{2}\left(1-\frac{1}{2^{k+2}+1}\right).
\end{aligned}
$$

This proves that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. This implies the given inequality for all $n \in \mathbb{N}$. □

Chapter 8. Matrices

Let us revisit Exercise 3.16, in the light of Exercise 3.19.

Exercise 8.1. Compute B^2 where B is the following $n \times n$ matrix

$$B = \begin{pmatrix} 0 & \cdots & 0 & 1 \\ 0 & \cdots & 1 & 0 \\ \vdots & ⋰ & \vdots & \vdots \\ 1 & \cdots & 0 & 0 \end{pmatrix}.$$

Solution. Let $e_1, e_2, \dots, e_n$ denote the standard basis vectors of $\mathbb{R}^n$. Note that B sends $e_1, e_2, \dots, e_{n-1}, e_n$ to the vectors $e_n, e_{n-1}, \dots, e_2, e_1$ respectively. That is,

$$Be_i = e_{n+1-i}$$

holds for all $1 \leq i \leq n$. Hence, for any integer $1 \leq i \leq n$, it follows that

$$\begin{aligned} B^2 e_i &= B(Be_i) \\ &= B(e_{n+1-i}) \\ &= e_{n+1-(n+1-i)} \\ &= e_i. \end{aligned}$$

This shows that[18]

$$B^2 = I_n.$$

□

Proof of Fact 3.20. Using $AB = I_n$, we obtain

$$CAB = CI_n = C.$$

Using $CA = I_n$, we get

$$CAB = I_n B = B.$$

This shows that $B = C$. □

Remark

Note that if A is a matrix and P is an invertible matrix such that PAP^{-1} is a diagonal matrix, then using Exercise 3.30, the powers of A can be computed. Indeed, if

$$PAP^{-1} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix},$$

then

$$A^k = P^{-1}(PA^kP^{-1})P = P^{-1}(PAP^{-1})^k P = P^{-1} \begin{pmatrix} \lambda_1^k & 0 & \cdots & 0 \\ 0 & \lambda_2^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^k \end{pmatrix} P$$

holds for any positive integer k. Thus, Exercise 3.30 can be used to compute powers of matrices which are similar to diagonal matrices.

Exercise 8.2. Does the following hold?

[18] How does it follow? Does Exercise 3.19 help?

$$\begin{pmatrix} -\frac{\sqrt{5}+1}{2} & \frac{\sqrt{5}-1}{2} \\ 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{5}+1}{2} & \frac{\sqrt{5}-1}{2} \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{-\sqrt{5}+1}{2} & 0 \\ 0 & \frac{1+\sqrt{5}}{2} \end{pmatrix}$$

? Question

What can be said about the Fibonacci numbers using Exercise 3.15 and Exercise 3.30?

Proof of Lemma 3.32. In the following, the first and the second column of A are denoted by C_1 and C_2 respectively. Write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Let us first show that the first three statements are equivalent. Assume that A is invertible. If $\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ is a solution to Equation 11, then

$$A \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This shows that

$$A^{-1} A \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

which yields

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This implies that the solution $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is the only solution to Equation 11. This shows that the first statement implies the second statement. Assume that the second statement holds, that is, Equation 11 admits no solution other than $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$. If s, t are real numbers such that

$$sC_1 + tC_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

then

$$A \begin{pmatrix} s \\ t \end{pmatrix} = sC_1 + tC_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This shows that $\begin{pmatrix} s \\ t \end{pmatrix}$ is a solution to Equation 11, and hence, $s = t = 0$. This shows that the second statement implies the third statement. Assume that the third statement holds, that is, the trivial linear combination of the columns of A is the only linear combination of the columns of A that is equal to $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$. If A is not invertible, then by Lemma 3.24, we have $ad - bc = 0$, which implies that

$$dC_1 - cC_2 = \begin{pmatrix} ad - bc \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

$$-bC_1 + aC_2 = \begin{pmatrix} 0 \\ ad - bc \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

and consequently, a, b, c, d are equal to 0, and hence, any linear combination of the columns of A is equal to $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, contradicting the third statement. This shows that if the third statement holds, then A is invertible, and hence, the third statement implies the first statement. This proves the equivalence of the first three statements.

For the equivalence of the next two statements, note that the following statements are equivalent.

- any element of $\mathbb{R}^2$ is a solution to Equation 11

- $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$ are solutions to Equation 11
- the matrix A is the zero matrix.

□

Proof of Fact 3.39. Let A be a square matrix. Note that

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T).$$

Note that the matrix $\frac{1}{2}(A + A^T)$ is symmetric, and that the matrix $\frac{1}{2}(A - A^T)$ is skew-symmetric. This shows that A can be expressed as the sum of a symmetric matrix and a skew-symmetric matrix.

Assume that

$$A = S_1 + K_1 = S_2 + K_2,$$

where S_1, S_2 are symmetric matrices, and K_1, K_2 are skew-symmetric matrices. It follows that

$$S_1 - S_2 = K_2 - K_1.$$

Note that the left-hand side of the above equality is a symmetric matrix, and that the right-hand side of the above equality is a skew-symmetric matrix. This shows that each of $S_1 - S_2, K_2 - K_1$ is both symmetric and skew-symmetric. Since the only matrix that is both symmetric and skew-symmetric is the zero matrix, we have $S_1 - S_2 = K_2 - K_1 = 0$. This yields

$$S_1 = S_2,$$
$$K_1 = K_2.$$

This implies that any square matrix A can be **uniquely** expressed as the sum of a symmetric matrix and a skew-symmetric matrix. □

Proof of Fact 3.53. Note that

$$A = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*).$$

Note that the matrix $\frac{1}{2}(A + A^*)$ is hermitian, and that the matrix $\frac{1}{2}(A - A^*)$ is skew-hermitian. This shows that A can be expressed as the sum of a hermitian matrix and a skew-hermitian matrix.

Assume that

$$A = H_1 + K_1 = H_2 + K_2,$$

where H_1, H_2 are hermitian matrices, and K_1, K_2 are skew-hermitian matrices. It follows that

$$H_1 - H_2 = K_2 - K_1.$$

Note that the left-hand side of the above equality is a hermitian matrix, and that the right-hand side of the above equality is a skew-hermitian matrix. This shows that each of $H_1 - H_2, K_2 - K_1$ is both hermitian and skew-hermitian. Since the only matrix that is both hermitian and skew-hermitian is the zero matrix, we have $H_1 - H_2 = K_2 - K_1 = 0$. This yields

$$H_1 = H_2,$$
$$K_1 = K_2.$$

This implies that any square matrix A with complex entries can be **uniquely** expressed as the sum of a hermitian matrix and a skew-hermitian matrix. □

Proof of Lemma 3.67. Let A be an $n \times m$ matrix. If

$$E = I_n + \lambda e_{ij} \text{ with } i \neq j,$$

then

$$EA = (I_n + \lambda e_{ij})A = A + \lambda e_{ij}A$$

holds, and hence, the k-th row of EA is equal to the k-th row of A if $k \neq i$, and is equal to the sum of the i-th row of A and the row vector obtained by multiplying the j-th row of A by λ if $k = i$.

Moreover, if

$$E = I_n - e_{ii} - e_{jj} + e_{ij} + e_{ji} \text{ with } i \neq j,$$

then

$$EA = (I_n - e_{ii} - e_{jj} + e_{ij} + e_{ji})A = A - e_{ii}A - e_{jj}A + e_{ij}A + e_{ji}A$$

holds, and hence, the k-th row of EA is equal to the k-th row of A if $k \neq i, j$, and the i-th row of EA is equal to the j-th row of A, and the j-th row of EA is equal to the i-th row of A.

Finally, if

$$E = I_n + (\lambda - 1)e_{ii} \text{ with } \lambda \neq 0,$$

then

$$EA = (I_n + (\lambda - 1)e_{ii})A = A + (\lambda - 1)e_{ii}A$$

holds, and hence, the k-th row of EA is equal to the k-th row of A if $k \neq i$, and is equal to the vector obtained by multiplying the i-th row of A by the scalar λ if $k = i$.

This shows that left multiplication by an elementary matrix on a matrix A performs the corresponding elementary row operation on the matrix A. □

Proof of Lemma 3.68. Let E be an elementary matrix. We consider the three types of elementary matrices separately.

Suppose E is the elementary matrix obtained by interchanging the i-th row and the j-th row of the identity matrix. Since multiplying any matrix by E from the left interchanges its i-th and j-th rows, we obtain

$$E^2 = EE = I_n.$$

This shows that E is invertible and that $E^{-1} = E$.

Now consider the case that E is the elementary matrix obtained by multiplying the i-th row of the identity matrix by a nonzero scalar λ. Since multiplying any matrix A by E from the left has the same effect of multiplying the i-th row of A by the same nonzero scalar λ, it follows that

$$E\left(I_n + \left(\frac{1}{\lambda} - 1\right)e_{ii}\right) = I_n.$$

Similarly, it follows that

$$\left(I_n + \left(\frac{1}{\lambda} - 1\right)e_{ii}\right)E = I_n.$$

This shows that E is invertible and that

$$E^{-1} = I_n + \left(\frac{1}{\lambda} - 1\right)e_{ii}.$$

Finally, consider the case that E is the elementary matrix obtained by adding a scalar multiple of the j-th row of the identity matrix to its i-th row, that is,

$$E = I_n + \lambda e_{ij} \text{ with } i \neq j,$$

where λ is a scalar. Using Exercise 3.63, it follows that

$$E(I_n - \lambda e_{ij}) = (I_n + \lambda e_{ij})(I_n - \lambda e_{ij}) = I_n,$$

and

$$(I_n - \lambda e_{ij})E = (I_n - \lambda e_{ij})(I_n + \lambda e_{ij}) = I_n,$$

which shows that E is invertible and that

$$E^{-1} = I_n - \lambda e_{ij}.$$

This shows that any elementary matrix is invertible, and its inverse is also an elementary matrix. □

Proof of Theorem 3.70. Let X_0 be a solution to the system of equations $AX = b$, that is, $AX_0 = b$ holds. It follows that

$$A'X_0 = EAX_0 = Eb = b'$$

holds, which shows that X_0 is a solution to the system of equations $A'X = b'$.

Conversely, let Y_0 be a solution to the system of equations $A'X = b'$, that is, $A'Y_0 = b'$ holds. Since the elementary matrices are invertible, and E is a product of elementary matrices, it follows that E is invertible. This yields

$$AY_0 = E^{-1}A'Y_0 = E^{-1}b' = b,$$

which shows that Y_0 is a solution to the system of equations $AX = b$. □

Proof of Fact 3.84. Let $M = (A \mid 0)$ denote the augmented matrix corresponding to the homogeneous system of equations $AX = 0$. Performing row reduction on M, we obtain a matrix $M' = (A' \mid 0)$ in row echelon form. Since A has more columns than rows, the matrix A' has more columns than rows. Hence, the matrix A' has at least one non-pivot column, which implies that the system of equations $A'X = 0$ has at least one free variable. Assigning an arbitrary nonzero value to this free variable, and then solving for the other variables, we obtain a nonzero solution to the system of equations $A'X = 0$. Since M' is obtained from M by performing a sequence of elementary row operations, it follows that the system of equations $AX = 0$ also admits a nonzero solution. Considering the scalar multiples of any such nonzero solution (cf. Exercise 3.35), we see that the homogeneous system of equations $AX = 0$ admits infinitely many solutions. □

Proof of Fact 3.85. Note that if A can be transformed into the identity matrix by performing a sequence of elementary row operations, then there are elementary matrices $E_1, E_2, \ldots, E_k$ such that

$$E_k \ldots E_2 E_1 A = I_n,$$

which implies that

$$A = (E_k \ldots E_2 E_1)^{-1} = E_1^{-1} E_2^{-1} \ldots E_k^{-1},$$

and consequently, A is a product of elementary matrices.

If A is a product of elementary matrices, then using the fact that every elementary matrix is invertible, it follows that A is invertible.

Suppose A is invertible. Let b be any column vector having n entries. Note that the vector $A^{-1}b$ is a solution to the $AX = b$. Also note that for any solution X_0 to the system of equations $AX = b$, we have $AX_0 = b$, which implies that $A^{-1}AX_0 = A^{-1}b$, which yields $X_0 = A^{-1}b$. This shows that the system of equations $AX = b$ admits a unique solution if A is invertible.

Note that if the system of equations $AX = b$ admits a unique solution for every column vector b having n entries, then in particular, the system of equations $AX = 0$ admits a unique solution, which is the zero vector.

Finally, let us assume that the zero solution is the only solution to the homogeneous system of equations $AX = 0$. Performing row reduction on the augmented matrix $(A \mid 0)$, we obtain a matrix $(A' \mid 0)$ in row echelon form. Since the homogeneous system of equations $AX = 0$ admits only one solution, each column of A' must contain a pivot of A'. This implies that the matrix A' has no non-pivot column, and hence, A' has n pivots. In other words, A' is equal to the identity matrix I_n. Since $(A' \mid 0)$ is obtained from $(A \mid 0)$ by performing a sequence of elementary row operations, it follows that A can be transformed into the identity matrix by performing a sequence of elementary row operations.

This proves the equivalence of the four statements. □

§8.1 Systems of linear equations in three variables and determinants of 3×3 matrices

Let us revisit the solution to Exercise 3.94.

? Question

Can one **come up** with a formula for the inverse of an invertible 3×3 matrix? Can one also have a notion of determinant of a 3×3 matrix, and use it to determine whether a 3×3 matrix is invertible or not? Can one have an analogue of Lemma 3.24 and Exercise 3.25 for 3×3 matrices?

Consider the system of linear equations in three variables x, y, z:

$$a_1 x + b_1 y + c_1 z = d_1, \tag{13}$$

$$a_2 x + b_2 y + c_2 z = d_2, \tag{14}$$

$$a_3 x + b_3 y + c_3 z = d_3, \tag{15}$$

where a_i, b_i, c_i, d_i are real numbers for $i = 1, 2, 3$. This system of equations can be expressed in matrix form as

$$\begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}.$$

The matrix

$$A = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}$$

is called the **coefficient matrix** of the system of equations, and the vector

$$\begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}$$

is called the **constant vector** of the system of equations. The system of equations is said to be **homogeneous** if

$$\begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Otherwise, the system of equations is said to be **non-homogeneous**.

i Remark

The discussion in Section 3.3 and Section 3.5 can be adapted to the case of systems of linear equations in three variables. In particular, if $\begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}$ is a linear combination of the columns of the coefficient matrix A, then the solutions to the system of equations

$$A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}$$

are in one-to-one correspondence with the solutions to the system of equations

$$A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Multiplying Equation 14 by c_3 and Equation 15 by $-c_2$, and adding the results, we obtain the equation

$$(a_2c_3 - a_3c_2)x + (b_2c_3 - b_3c_2)y = d_2c_3 - d_3c_2. \tag{16}$$

Similarly, multiplying Equation 14 by $-b_3$ and Equation 15 by b_2, and adding the results, we obtain the equation

$$(a_2b_3 - a_3b_2)x + (c_2b_3 - c_3b_2)z = d_2b_3 - d_3b_2. \tag{17}$$

Multiplying Equation 13 by $b_2c_3 - b_3c_2$ and Equation 16 by $-b_1$, and adding the results, we obtain the equation

$$\begin{aligned}(a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2))x + (c_1(b_2c_3 - b_3c_2))z \\ = d_1(b_2c_3 - b_3c_2) - b_1(d_2c_3 - d_3c_2).\end{aligned} \tag{18}$$

Multiplying Equation 17 by c_1 and adding to Equation 18, we obtain the equation

$$\begin{aligned}(a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2))x \\ = d_1(b_2c_3 - b_3c_2) - b_1(d_2c_3 - d_3c_2) + c_1(d_2b_3 - d_3b_2),\end{aligned} \tag{19}$$

or equivalently, we obtain

$$\begin{aligned}(a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2))x \\ = d_1(b_2c_3 - b_3c_2) - d_2(b_1c_3 - b_3c_1) + d_3(b_1c_2 - b_2c_1).\end{aligned} \tag{20}$$

Similarly[19], it follows that

$$\begin{aligned}(b_2(c_3a_1 - c_1a_3) - c_2(b_3a_1 - b_1a_3) + a_2(b_3c_1 - b_1c_3))y \\ = d_2(c_3a_1 - c_1a_3) - d_3(c_2a_1 - c_1a_2) + d_1(c_2a_3 - c_3a_2),\end{aligned} \tag{21}$$

and

$$\begin{aligned}(c_3(a_1b_2 - a_2b_1) - a_3(c_1b_2 - c_2b_1) + b_3(c_1a_2 - c_2a_1))z \\ = d_3(a_1b_2 - a_2b_1) - d_1(a_3b_2 - a_2b_3) + d_2(a_3b_1 - a_1b_3).\end{aligned} \tag{22}$$

This may indicate that it could be useful to consider the product of the matrix

$$\begin{pmatrix} b_2c_3 - b_3c_2 & -(b_1c_3 - b_3c_1) & b_1c_2 - b_2c_1 \\ c_2a_3 - c_3a_2 & c_3a_1 - c_1a_3 & -(c_2a_1 - c_1a_2) \\ -(a_3b_2 - a_2b_3) & a_3b_1 - a_1b_3 & a_1b_2 - a_2b_1 \end{pmatrix}$$

with the matrix A. Consider the product

$$M := \begin{pmatrix} b_2c_3 - b_3c_2 & -(b_1c_3 - b_3c_1) & b_1c_2 - b_2c_1 \\ -(c_3a_2 - c_2a_3) & c_3a_1 - c_1a_3 & -(c_2a_1 - c_1a_2) \\ a_2b_3 - a_3b_2 & -(a_1b_3 - a_3b_1) & a_1b_2 - a_2b_1 \end{pmatrix} \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}.$$

? Question

Write $M = (m_{ij})_{1 \le i,j \le 3}$. Does it follow that if one of m_{12}, m_{23}, m_{31} is identically zero, then the remaining two are also identically zero? Does it also follow that if one of m_{13}, m_{21}, m_{32} is identically zero, then the remaining two are also identically zero? Does it follow that M is a diagonal matrix? Does it also follow that M is a scalar matrix, that is, its diagonal entries are equal?

Fact 8.3. The product M is equal to the scalar matrix

$$\det(A)I_3,$$

where

$$\det(A) := a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2).$$

[19]How does it follow? What would be a useful **similarity** to consider?

Moreover, it follows that

$$\begin{aligned}
&\begin{pmatrix} b_2c_3-b_3c_2 & -(b_1c_3-b_3c_1) & b_1c_2-b_2c_1 \\ -(c_3a_2-c_2a_3) & c_3a_1-c_1a_3 & -(c_2a_1-c_1a_2) \\ a_2b_3-a_3b_2 & -(a_1b_3-a_3b_1) & a_1b_2-a_2b_1 \end{pmatrix}\begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix} \\
&= \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}\begin{pmatrix} b_2c_3-b_3c_2 & -(b_1c_3-b_3c_1) & b_1c_2-b_2c_1 \\ -(c_3a_2-c_2a_3) & c_3a_1-c_1a_3 & -(c_2a_1-c_1a_2) \\ a_2b_3-a_3b_2 & -(a_1b_3-a_3b_1) & a_1b_2-a_2b_1 \end{pmatrix} \\
&= \det(A)I_3.
\end{aligned}$$

Definition

Let A be a 3×3 matrix. Write $A = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}$. The **determinant** of A is denoted by $\det(A)$, and is defined to be

$$\det(A) := a_1(b_2c_3-b_3c_2) - b_1(a_2c_3-a_3c_2) + c_1(a_2b_3-a_3b_2).$$

The **adjoint** of A is denoted by $\operatorname{adj}(A)$, and is defined to be the matrix

$$\operatorname{adj}(A) := \begin{pmatrix} b_2c_3-b_3c_2 & -(b_1c_3-b_3c_1) & b_1c_2-b_2c_1 \\ -(c_3a_2-c_2a_3) & c_3a_1-c_1a_3 & -(c_2a_1-c_1a_2) \\ a_2b_3-a_3b_2 & -(a_1b_3-a_3b_1) & a_1b_2-a_2b_1 \end{pmatrix}.$$

Fact 8.4. Let E be a 3×3 elementary matrix.

1. If E is obtained by interchanging two rows of I_3, then $\det(E) = -1$.
2. If E is obtained by multiplying a row of I_3 by a nonzero scalar k, then $\det(E) = k$.
3. If E is obtained by adding a multiple of one row of I_3 to another row, then $\det(E) = 1$.

In particular, the determinant of any elementary matrix is nonzero.

Fact 8.5. If A is a 3×3 matrix, and E is a 3×3 elementary matrix, then

$$\det(EA) = \det(E)\det(A).$$

Here is a proof of Lemma 3.103 using row echelon forms, and Fact 8.5.

Proof of Lemma 3.103. Let A, B be 3×3 matrices. Let E be a product of elementary matrices such that EA is in row echelon form. If A is not invertible, then the bottom row of EA is zero, and hence, the bottom row of $E(AB)$ is also zero. Note that if P is a 3×3 matrix with a zero row, then for some elementary matrix F, the matrix FP has two equal rows, and hence, by Exercise 3.102, we obtain $\det(FP) = 0$, and using Fact 8.5 yields that $\det(P) = 0$. This shows that if A is not invertible, then $\det(E(AB)) = 0$ and $\det(EA) = 0$, and hence using Fact 8.5 and the fact that elementary matrices have nonzero determinants (by Fact 8.4), it follows that $\det(AB) = 0$ and $\det(A) = 0$, which shows that $\det(AB) = \det(A)\det(B)$ holds in this case.

Let us consider the case when A is invertible. In this case, $EA = I_3$ holds, which shows that A is invertible, and that $A^{-1} = E$ by Exercise 3.74. Write $E^{-1} = E_k \dots E_2E_1$, where each E_i is an elementary matrix. Fact 8.5 combined with induction shows that

$$\begin{aligned}
\det(A) &= \det(E^{-1}) \\
&= \det(E_k)\dots\det(E_2)\det(E_1), \\
\det(AB) &= \det(E^{-1}B) \\
&= \det(E_k)\dots\det(E_2)\det(E_1)\det(B) \\
&= \det(E^{-1})\det(B) \\
&= \det(A)\det(B).
\end{aligned}$$

□

Using Cramer's rule (Fact 3.101), the following analogue of Lemma 3.24 for 3×3 matrices can be established[20].

Lemma 8.6. *Let A be a 3×3 matrix. The following statements are equivalent.*

1. *The matrix A is invertible.*
2. *The determinant $\det(A)$ of A is nonzero.*
3. *Every vector*

$$\begin{pmatrix} e \\ f \\ g \end{pmatrix}$$

can be expressed as a linear combination of the columns of A. Moreover, if A is invertible, then

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

holds.

Note that the above lemma is same as Lemma 3.104.

Proof of Lemma 3.104. If $\det(A)$ is nonzero, then using

$$A \cdot \operatorname{adj}(A) = \operatorname{adj}(A) \cdot A = \det(A) I_3,$$

it follows that A is invertible.

If A is invertible, then every vector

$$\begin{pmatrix} e \\ f \\ g \end{pmatrix}$$

can be expressed as a linear combination of the columns of A, since the standard basis vectors can be expressed as a linear combination of the columns of A.

Moreover, if every vector can be expressed as a linear combination of the columns of A, then in particular, the standard basis vectors can be expressed as a linear combination of the columns of A. This shows that there are real numbers $x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3$ such that

$$A \begin{pmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{pmatrix} = I_3.$$

This implies that A has nonzero determinant by Lemma 3.103. □

[20] In fact, the first few steps of the proof of Lemma 3.24 is precisely the analogue of Fact 3.101 for 2×2 matrices.

Chapter 9. The space $\mathbb{R}^n$

Proof of Lemma 4.2. Let $v_1, v_2, \ldots, v_k$ be elements of $\mathbb{R}^n$. Assume that the set $\{v_1, v_2, \ldots, v_k\}$ is linearly dependent. Then there exist coefficients $\alpha_1, \alpha_2, \ldots, \alpha_k$, not all zero, such that

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k = 0.$$

Let $1 \leq i \leq k$ be such that $\alpha_i \neq 0$. Note that

$$\alpha_i v_i = -(\alpha_1 v_1 + \cdots + \alpha_{i-1} v_{i-1} + \alpha_{i+1} v_{i+1} + \cdots + \alpha_k v_k).$$

Taking the product of both sides with the scalar $\frac{1}{\alpha_i}$ gives

$$v_i = -\frac{\alpha_1}{\alpha_i} v_1 - \cdots - \frac{\alpha_{i-1}}{\alpha_i} v_{i-1} - \frac{\alpha_{i+1}}{\alpha_i} v_{i+1} - \cdots - \frac{\alpha_k}{\alpha_i} v_k.$$

Thus, the vector v_i is a linear combination of the other vectors.

Conversely, suppose one of the vectors can be expressed as a linear combination of the others. Then for some $1 \leq i \leq k$, there exist coefficients $\beta_1, \beta_2, \ldots, \beta_{i-1}, \beta_{i+1}, \ldots, \beta_k \in \mathbb{R}$ such that

$$v_i = \beta_1 v_1 + \beta_2 v_2 + \cdots + \beta_{i-1} v_{i-1} + \beta_{i+1} v_{i+1} + \cdots + \beta_k v_k.$$

Rearranging gives

$$\beta_1 v_1 + \beta_2 v_2 + \cdots + \beta_{i-1} v_{i-1} - v_i + \beta_{i+1} v_{i+1} + \cdots + \beta_k v_k = 0.$$

which is a non-trivial linear combination of the vectors

$$v_1, v_2, \ldots, v_{i-1}, v_i, v_{i+1}, \ldots, v_k$$

equaling zero, with coefficients $\beta_1, \beta_2, \ldots, \beta_{i-1}, -1, \beta_{i+1}, \ldots, \beta_k$. Hence, the vectors are linearly dependent. □

Proof of Lemma 4.47. Let V be a subspace of $\mathbb{R}^n$. If $V = \{0\}$, then V is generated by the set $\{0\}$. Suppose $V \neq \{0\}$. Then, there exists a non-zero vector $v_1 \in V$. Note that $\{v_1\}$ is a linearly independent set. If $V = \operatorname{span}(\{v_1\})$, then we are done. Otherwise, there exists a vector $v_2 \in V$ such that $v_2 \notin \operatorname{span}(\{v_1\})$. Note that the vectors v_1 and v_2 are linearly independent (by Exercise 4.33). If $V = \operatorname{span}(\{v_1, v_2\})$, then we are done. Otherwise, there exists a vector $v_3 \in V$ such that $v_3 \notin \operatorname{span}(\{v_1, v_2\})$.

Suppose for an integer $k \geq 2$, we have obtained linearly independent vectors $v_1, v_2, \ldots, v_k \in V$ such that $V \neq \operatorname{span}(\{v_1, v_2, \ldots, v_k\})$. Then, there exists a vector $v_{k+1} \in V$ such that v_{k+1} does not belong to the span of $\{v_1, v_2, \ldots, v_k\}$. Note that the vectors $v_1, v_2, \ldots, v_k, v_{k+1}$ are linearly independent (by Exercise 4.33).

Since $\mathbb{R}^n$ does not contain more than n linearly independent vectors (by Exercise 4.9), it follows that this process terminates witin n steps. That is, there exists a positive integer $m \leq n$ such that

$$V = \operatorname{span}(\{v_1, v_2, \ldots, v_m\})$$

for some linearly independent vectors $v_1, v_2, \ldots, v_m \in V$. This shows that the subspace V is generated by the set of vectors $\{v_1, v_2, \ldots, v_m\}$. □

Proof of Lemma 4.48. For any positive integer m, let $P(m)$ denote the statement that for any linear subspace V of $\mathbb{R}^n$, every generating set of V containing exactly $m + n$ vectors contains a generating set of V consisting of at most n vectors.

Let V be a linear subspace of $\mathbb{R}^n$, and S be a generating set of V containing exactly $n + 1$ vectors. By Exercise 4.9, the set S is linearly dependent. Hence, S contains a vector u which is a linear combination of the other vectors in S. It follows that the span of S is equal to the span of $S \setminus \{u\}$. This shows that $S \setminus \{u\}$ is a generating set of V consisting of n vectors. This establishes the statement $P(1)$.

Suppose for a positive integer m, the statement $P(m)$ is true. Let V be a linear subspace of $\mathbb{R}^n$, and S be a generating set of V containing exactly $m + 1 + n$ vectors. By Exercise 4.9, the set S is linearly dependent. Hence, S contains a vector u which is a linear combination of the other vectors in S. It follows that the span of S is equal to the span of $S \setminus \{u\}$. This shows that $S \setminus \{u\}$ is a generating set of V consisting of $m+n$ vectors. By the induction hypothesis, $S \setminus \{u\}$ contains a generating set of V consisting of at most n vectors. This establishes the statement $P(m+1)$.

By the principle of mathematical induction, the statement $P(m)$ is true for all positive integers m. □

Proof of Lemma 4.56. Let V be a subspace of $\mathbb{R}^n$. If $V = \{0\}$, then the set $\emptyset$ is a basis of V. It remains to consider the case where V is nonzero, which we assume from now on. Let $\mathcal{G}_0$ be a generating set of V. By Lemma 4.48, $\mathcal{G}_0$ contains a generating set $\mathcal{G}$ of V consisting of at most n vectors. It suffices to show that there exists a subset $\mathcal{B} \subseteq \mathcal{G}$ such that $\mathcal{B}$ is a basis of V.

Since V is nonzero, it follows that $\mathcal{G}$ contains a nonzero vector v_1. Note that $\{v_1\}$ is a linearly independent subset of $\mathcal{G}$. Let k be a positive integer, and suppose we have obtained a linearly independent subset $\{v_1, v_2, \ldots, v_k\}$ of $\mathcal{G}$. If V is equal to the linear span of $\{v_1, v_2, \ldots, v_k\}$, then we are done. Otherwise, there exists a vector $v_{k+1} \in V$ such that v_{k+1} does not belong to the span of $\{v_1, v_2, \ldots, v_k\}$. Note that the vectors $v_1, v_2, \ldots, v_k, v_{k+1}$ are linearly independent (by Exercise 4.33). Continuing in this manner, we obtain a linearly independent subset of $\mathcal{G}$ that generates V. Since $\mathcal{G}$ contains at most n vectors, this process terminates within n steps. Thus, we obtain a subset $\mathcal{B}$ of $\mathcal{G}$ such that $\mathcal{B}$ is linearly independent and generates V, that is, $\mathcal{B}$ is a basis of V. □

Remark

In the proof of Lemma 4.56, we have argued as in the proof of Lemma 4.47 to construct a maximal linearly independent subset of a given finite generating set of a subspace, and proved that it generates the subspace. In fact, any generating set of a subspace contains a maximal linearly independent subset, and a maximal linearly independent subset of a generating set also generates the subspace, and hence, is a basis of the subspace.

Proof of Lemma 4.58. Let V be a subspace of $\mathbb{R}^n$. If $V = \{0\}$, then the empty set is the only basis of V. It remains to consider the case where V is nonzero, which we assume from now on. Let $\{u_1, u_2, \ldots, u_m\}$ and $\{v_1, v_2, \ldots, v_k\}$ be two bases of V, where m and k are positive integers. Let A be the $n \times m$ matrix whose j-th column is the vector u_j for each $j = 1, 2, \ldots, m$. Let B be the $n \times k$ matrix whose i-th column is the vector v_i for each $i = 1, 2, \ldots, k$. Since $\{u_1, u_2, \ldots, u_m\}$ is a basis of V, it follows that each v_i can be expressed as a linear combination of $u_1, u_2, \ldots, u_m$. In other words, the columns of B lie in the span of the columns of A. Thus, there exists a $m \times k$ matrix C such that $B = AC$. Similarly, it follows that there exists a $k \times m$ matrix D such that $A = BD$. Thus, we have $A = ACD$ and $B = BDC$. This shows that CD acts as the identity on the columns of A, and DC acts as the identity on the columns of B. Since the columns of A are linearly independent, it follows that CD is the $m \times m$ identity matrix. Since the columns of B are linearly independent, it follows that DC is the $k \times k$ identity matrix. We claim that $k = m$.

If $m > k$, then the row echelon form of C has at least one zero row. This implies that the row echelon form of CD has at least one zero row. This contradicts the fact that CD is the $m \times m$ identity matrix. Thus, we must have $m \leq k$. Similarly, if $k > m$, then the row echelon form of D has at least one zero row. This implies that the row echelon form of DC has at least one zero row. This contradicts the fact that DC is the $k \times k$ identity matrix. Thus, we must have $k \leq m$. Combining these two inequalities gives $k = m$. □

Proof of Lemma 4.60. Let V be a subspace of $\mathbb{R}^n$. If $V = \{0\}$, then the empty set is the only linearly independent subset of V, and it is a basis of V. It remains to consider the case where V is nonzero, which we assume from now on. Let $\{u_1, u_2, \ldots, u_m\}$ be a basis of V, where m is a positive integer. Let $\{v_1, v_2, \ldots, v_k\}$ be a linearly independent subset of V. Let $\mathcal{G}_1$ denote the set $\{v_1, v_2, \ldots, v_k\}$ or $\{v_1, v_2, \ldots, v_k, u_1\}$ according as whether u_1 belongs to the span of $\{v_1, v_2, \ldots, v_k\}$ or not. Note that $\mathcal{G}_1$ is a linearly independent subset of V by Exercise 4.33. Moreover, $\{u_1\}$ is contained in the span of $\mathcal{G}_1$.

Let r be a positive integer, and suppose we have obtained a linearly independent subset $\mathcal{G}_r$ of V such that

- $\mathcal{G}_r$ contains $\{v_1, v_2, ..., v_k\}$,
- $\mathcal{G}_r$ is a linearly independent subset of V,
- $\{u_1, u_2, ..., u_r\}$ is contained in the span of $\mathcal{G}_r$.

If $\mathcal{G}_r$ generates V, then we are done. Otherwise, it follows that $r + 1 \leq m$. Let $\mathcal{G}_{r+1}$ denote the set $\mathcal{G}_r$ or $\mathcal{G}_r \cup \{u_{r+1}\}$ according as whether u_{r+1} belongs to the span of $\mathcal{G}_r$ or not. Note that $\mathcal{G}_{r+1}$ is a linearly independent subset of V by Exercise 4.33. Moreover, $\{u_1, u_2, ..., u_{r+1}\}$ is contained in the span of $\mathcal{G}_{r+1}$, and $\mathcal{G}_{r+1}$ contains $\{v_1, v_2, ..., v_k\}$.

Continuing in this manner, we obtain a linearly independent subset $\mathcal{G}_m$ of V which contains $\{v_1, v_2, ..., v_k\}$, and $\{u_1, u_2, ..., u_m\}$ is contained in the span of $\mathcal{G}_m$. Since $\{u_1, u_2, ..., u_m\}$ is a basis of V, it follows that V is contained in the span of $\mathcal{G}_m$. Using that $\mathcal{G}_m$ is a subset of V, we conclude that V is equal to the span of $\mathcal{G}_m$. This shows that $\mathcal{G}_m$ is a linearly independent subset of V that generates V, that is, a basis of V, which contains the linearly independent subset $\{v_1, v_2, ..., v_k\}$. This completes the proof. □

Proof of Lemma 4.61. If $U = \{0\}$, then $\dim(U) = 0 \leq \dim(V)$. Suppose $U \neq \{0\}$. By Corollary 4.57, the subspace U admits a basis $\{u_1, u_2, ..., u_m\}$, where m is a positive integer. By Lemma 4.60, the linearly independent subset $\{u_1, u_2, ..., u_m\}$ of V can be extended to a basis $\{u_1, u_2, ..., u_m\} \cup \mathcal{C}$ of V for some subset $\mathcal{C}$ of V. Let p be the number of elements of $\mathcal{C}$. Note that p is a non-negative integer. It follows that $\dim(U) = m$ and $\dim(V) = m + p$. Thus, we have $\dim(U) = m \leq m + p = \dim(V)$.

If $\dim(U) = \dim(V)$ holds, then $p = 0$, and hence, $\mathcal{C} = \emptyset$. This shows that the basis $\{u_1, u_2, ..., u_m\}$ of U is also a basis of V. Thus, we have $U = V$. □

Proof of Lemma 4.62. Let us assume that S is a basis of V. By definition of basis, the set S is a linearly independent set which generates V. By definition of dimension, the number of elements of S is equal to $\dim(V)$.

Let us assume that the number of elements of S is equal to $\dim(V)$ and that S is a linearly independent set. Using that S is a subset of V, we obtain $\operatorname{span}(S)$ is contained in V. Since S is a linearly independent subset of V, it follows that the dimension of the span of S is equal to the number of elements of S. This shows that $\dim(\operatorname{span}(S)) = \dim(V)$, and by Lemma 4.61, it follows that $\operatorname{span}(S) = V$. This shows that S is a generating set for V.

Let us assume that the number of elements of S is equal to $\dim(V)$ and that S is a generating set for V. By definition of dimension, any basis of V contains exactly $\dim(V)$ vectors. If S is linearly dependent, then it contains an element v such that v belongs to the span of $S \setminus \{v\}$, and hence $S \setminus \{v\}$ is a generating set for $\operatorname{span}(S) = V$, implying that the dimension of V is less than the number of elements of S, which contradicts the assumption that the number of elements of S is equal to $\dim(V)$. We conclude that S is linearly independent. This implies that S is a basis of V. □

Proof of Lemma 4.63. Assume that S is a basis of $\operatorname{span}(S)$. By definition of basis, the set S is a linearly independent set.

Suppose that S is a linearly independent set. By definition of linear span, the set S generates $\operatorname{span}(S)$. This shows that S is a basis of $\operatorname{span}(S)$. By definition of dimension, the dimension of $\operatorname{span}(S)$ is equal to the number of elements of S.

Assume that the number of elements of S is equal to $\dim(\operatorname{span}(S))$. If S is linearly dependent, then it contains an element v such that v belongs to the span of $S \setminus \{v\}$, and hence $S \setminus \{v\}$ is a generating set for $\operatorname{span}(S)$, and applying Lemma 4.56, it follows that the dimension of $\operatorname{span}(S)$ is less than the number of elements of S, which contradicts the assumption that the number of elements of S is equal to $\dim(\operatorname{span}(S))$. This shows that S is linearly independent. Since S generates $\operatorname{span}(S)$, it follows that S is a basis of $\operatorname{span}(S)$. □

Question

Can Lemma 4.63 be proved using Lemma 4.62?